CM-EIP — Question Bank

Contents

CM-EIP Extensive Question Bank **1**

Chapter 1 — Introduction to Investing, Investments and Financial Markets . . . 1
Chapter 2 — Risk, Return and Time Value Calculations . . . 8
Chapter 3 — Key Drivers of Market Movements and Asset Values . . . 17
Chapter 4 — Foreign Exchange . . . 28
Chapter 5 — Company Analysis and Understanding Financial Statements . . . 37
Chapter 6 — Equity Securities . . . 47
Chapter 7 — Deposits and Fixed-Income Securities . . . 63
Chapter 8 — Portfolio Management . . . 82
Chapter 9 — Exchange Traded Funds (ETFs), Unit Trusts, Real Estate Investment Trusts (REITs) and Insurance . . . 99
Chapter 10 — Warrants . . . 119
Chapter 11 — Technical Analysis and Quantitative Analysis . . . 125
Chapter 12 — Case Studies . . . 136
Appendix A — Review Questions . . . 143
Appendix B — Formulae Sheet . . . 145
Appendix C — Essential Readings . . . 146
Appendix D — Glossary . . . 147
Exam method and final revision checklist . . . 147

CM-EIP Extensive Question Bank

This question bank follows the sequence and style of the review questions in Appendix A of the IBF CM-EIP study guide, but expands them substantially. It is organised by chapter and section, covers all twelve chapters and all four appendices, and adds calculations, suitability decisions and scenario questions. Answers and rationales are hidden inside expandable sections; click **Show answer** after attempting each question.

For multiple-response questions, select every correct option. Unless stated otherwise, use a 360-day convention for all products other than Singapore Government Securities, which use a 365-day convention, and the formulae in Appendix B of the study guide. These are original revision questions, not a reproduction of the source's review-question set.

The bank is generated from `scripts/lib/questions-cmeip.mjs` by `scripts/append-questions.mjs`. Edit the source module, not this file: everything below the marker is rewritten on each run.

Chapter 1 — Introduction to Investing, Investments and Financial Markets

1. [Single choice] [CM-EIP §1.1.1] Under section 4A of the Securities and Futures Act 2001, which investor category is defined by occupation rather than by wealth?

- a. Expert investor
- b. Institutional investor
- c. Retail investor
- d. Accredited investor

Answer and explanation

Answer: A. Expert investor

Rationale: An expert investor is a person whose business involves acquiring, disposing of or holding capital markets products. The classification rests on occupational understanding of product risk, not on wealth or income.

2. [Calculation] [CM-EIP §1.1.1] An individual has net personal assets of S$2.4 million, including a primary residen million with an outstanding mortgage of S$600,000. Does the individual meet the net personal assets test for accredited investor status?

- a. Yes, because net personal assets exceed S$2 million
- b. No, because the residence is capped at S$1 million
- c. Yes, because the residence alone exceeds S$1 million
- d. No, because financial assets are not stated

Answer and explanation

Answer: B. No, because the residence is capped at S$1 million

Rationale: The residence counts net of the secured loan and is capped at S1 million : $S1.8m - S0.6m = S1.2m, capped at S1.0m. Other net personal assets are S2.4m - S1.8m = S0.6m. Total counted = S1.0m + S0.6m = S1.6m, below the S2 million threshold.

3. [Single choice] [CM-EIP §1.1.1] What is the effect of the accredited investor opt-in regime introduced by MAS?

- a. An accredited investor must re-qualify every quarter
- b. A capital markets intermediary may treat any qualifying investor as an accredited investor automatically
- c. A capital markets intermediary may not treat a qualifying investor as an accredited investor unless the investor opts in
- d. An accredited investor is barred from retail safeguards

Answer and explanation

Answer: C. A capital markets intermediary may not treat a qualifying investor as an accredited investor unless the investor opts in

Rationale: Under the opt-in regime a CMI can no longer automatically treat anyone who qualifies as an accredited investor; the investor must opt to be treated as one, and may instead elect the full range of retail safeguards.

4. [Multiple response] [CM-EIP §1.1.1] Which of the following are typical return objectives quoted for client profiling? (Select all that apply)

- a. Preserve wealth and purchasing power at 2–4%
- b. Income with some growth at 4–6%
- c. Steady balanced growth at 6–8%
- d. Guaranteed capital growth above 15%

Answer and explanation

Answer: A, B, C

Rationale: The guide quotes four bands: 2–4% for preserving wealth and purchasing power, 4–6% for income with some growth, 6–8% for steady balanced growth and 8–12% for aggressive growth. No guaranteed return above 15% is offered.

5. [Single choice] [CM-EIP §1.1.2] Which product characteristic is associated with convertible bonds but not with plain bonds in the product-benefit matrix?

- a. The ability to hedge an inverse payout
- b. A defined interest yield
- c. A regular payout stream
- d. The ability to amplify gains and losses through leverage

Answer and explanation

Answer: D. The ability to amplify gains and losses through leverage

Rationale: Convertible bonds are marked as amplifying gains and losses through leverage, because the embedded equity option gives participation in the share price. Plain bonds are marked for defined yield, capital gains, regular payout and inverse-floaters as a hedge.

6. [Single choice] [CM-EIP §1.1.3] Why must a representative be alert to a client's business assets when advising on financial assets?

- a. Business financing needs and decisions about physical assets can materially affect the client's financial assets
- b. Business assets are exempt from disclosure requirements
- c. Business assets are treated as financial assets under the SFA
- d. Business assets are always larger than financial assets

Answer and explanation

Answer: A. Business financing needs and decisions about physical assets can materially affect the client's financial assets

Rationale: For individuals the flow of funds between business, physical and financial assets is closely related. Business financing needs, and decisions about business and physical assets, can have a major impact on financial assets.

7. [Single choice] [CM-EIP §1.2.1] What is the principal opportunity cost of holding investible funds in cash over a long period?

- a. Loss of deposit insurance cover
- b. Loss of purchasing power as inflation erodes real value
- c. Exposure to currency risk
- d. Ineligibility for the CPF Investment Scheme

Answer and explanation

Answer: B. Loss of purchasing power as inflation erodes real value

Rationale: Cash pays the lowest return. In a low interest rate environment, holding investible funds in cash carries a high opportunity cost because inflation erodes purchasing power.

8. [Single choice] [CM-EIP §1.2.2] Which money market instrument is quoted on an interest-bearing basis rather than a discount basis?

- a. Commercial paper
- b. Banker's acceptance
- c. Certificate of deposit
- d. Treasury bill

Answer and explanation

Answer: C. Certificate of deposit

Rationale: Certificates of deposit carry a maturity date and a specific interest rate, generally on a compounding basis, with interest payable at maturity. Treasury bills, commercial paper and banker's acceptances are all issued at a discount to face value.

9. [Calculation] [CM-EIP §1.2.2] A treasury bill with a face value of S$100,000 is bought for S$98,500 and held to maturity in 90 days. What is the simple discount, expressed as a percentage of face value?

- a. 1.52%
- b. 1.48%
- c. 6.00%
- d. 1.50%

Answer and explanation

Answer: D. 1.50%

Rationale: The discount is $S100,000 - S98{,}500 = S1,500, which is 1.50 100{,}000$ face value. The annualised bank discount yield would be $1.50\% \times 360/90 = 6.00\%$, but the question asks for the discount as a percentage of face value.

10. [Single choice] [CM-EIP §1.2.2] A bond denominated in US dollars, sold in the United States, and issued by a Japanese corporation is a:

- a. Yankee bond
- b. Domestic bond
- c. Euroyen bond
- d. Eurodollar bond

Answer and explanation

Answer: A. Yankee bond

Rationale: A Yankee bond is sold in the United States, denominated in US dollars, but issued by a foreign corporation or government. A Eurodollar bond is denominated in USD and sold outside the United States.

11. [Single choice] [CM-EIP §1.2.2] A Japanese company issues yen-denominated bonds for sale to investors in London. These are:

- a. Domestic bonds
- b. Euroyen bonds
- c. Samurai bonds
- d. Yankee bonds

Answer and explanation

Answer: B. Euroyen bonds

Rationale: A Euroyen bond is denominated in yen and sold outside Japan. The guide gives exactly this example: a Japanese company issuing Euroyen bonds for sale in London.

12. [Single choice] [CM-EIP §1.2.3] In the order of priority of claim on a company's assets, which ranking is correct?

- a. Ordinary shareholders, preference shareholders, creditors
- b. Creditors, ordinary shareholders, preference shareholders
- c. Creditors, preference shareholders, ordinary shareholders
- d. Preference shareholders, creditors, ordinary shareholders

Answer and explanation

Answer: C. Creditors, preference shareholders, ordinary shareholders

Rationale: Creditors have the highest priority, followed by preference shareholders, then ordinary shareholders, who hold the residual claim.

13. [Single choice] [CM-EIP §1.2.4] Which derivative is described as exchange-traded with standardised terms and may be cash settled or physically settled?

- a. Convertible security
- b. Company-issued warrant
- c. Structured warrant
- d. Futures contract

Answer and explanation

Answer: D. Futures contract

Rationale: A futures contract is an agreement for future delivery of an asset at a predetermined price, traded on exchanges with standardised terms, and may be cash settled or physically settled.

14. [Single choice] [CM-EIP §1.2.4] Why does a convertible security usually pay a lower yield than the same issuer's plain corporate bond?

- a. Because the yield difference is the value of the embedded share option
- b. Because it has a shorter maturity
- c. Because it is unsecured
- d. Because it ranks below the plain bond in liquidation

Answer and explanation

Answer: A. Because the yield difference is the value of the embedded share option

Rationale: The convertible gives the holder the right to convert into ordinary shares. The price difference between the convertible's lower yield and the plain bond's yield is the value of the embedded share option.

15. [Single choice] [CM-EIP §1.2.5] Which client is professional portfolio management most clearly suited to?

- a. A client with strong views who trades daily
- b. A client who lacks the time, information, expertise, discipline or emotional temperament to invest actively
- c. A client seeking guaranteed returns
- d. A client who wishes to avoid all fees

Answer and explanation

Answer: B. A client who lacks the time, information, expertise, discipline or emotional temperament to invest actively

Rationale: Professional portfolio management benefits clients who do not have the time, appropriate information, expertise, discipline or emotional temperament to invest actively.

16. [Multiple response] [CM-EIP §1.3] Which of the following are participants in financial markets? (Select all that apply)

- a. Issuers
- b. Intermediaries such as broker-dealers and market makers
- c. Infrastructure providers such as clearing houses and central depositories
- d. Regulators

Answer and explanation

Answer: A, B, C, D

Rationale: The guide lists investors, issuers, intermediaries, regulators, electronic exchanges, infrastructure providers and financial institutions as participants.

17. [Single choice] [CM-EIP §1.3] What distinguishes high frequency trading from algorithmic trading more generally?

- a. High frequency trading is a form of fundamental analysis
- b. High frequency trading does not use computer programs
- c. High frequency trading relies on computer speed to analyse large amounts of market information and self-execute trades in fractions of a second
- d. High frequency trading is only used in the bond market

Answer and explanation

Answer: C. High frequency trading relies on computer speed to analyse large amounts of market information and self-execute trades in fractions of a second

Rationale: Algorithmic trading uses a computer program following defined instructions to execute a trade and can slice large trades to minimise market impact. High frequency trading is a form of algorithmic trading that relies on computer speed and power to analyse significant amounts of market information and self-execute trades in fractions of a second.

18. [Single choice] [CM-EIP §1.3.1] In a primary market transaction, who receives the proceeds?

- a. The exchange
- b. The clearing house
- c. The selling investor
- d. The issuer

Answer and explanation

Answer: D. The issuer

Rationale: In the primary market the issuer receives the cash from subscribers, who receive financial claims that previously did not exist.

19. [Single choice] [CM-EIP §1.3.2] What is the essential economic function of the secondary market?

- a. It provides liquidity, which is essential to the proper functioning of capital markets
- b. It sets the coupon rate on new bonds
- c. It guarantees investors against loss
- d. It raises new capital for issuers

Answer and explanation

Answer: A. It provides liquidity, which is essential to the proper functioning of capital markets

Rationale: Proceeds from a secondary market sale go to the selling investor, not the issuer. The secondary market provides liquidity, which is essential to the proper functioning of capital markets.

20. [Single choice] [CM-EIP §1.3.3] Which characteristic of a good financial market is described as internal efficiency?

- a. Availability of information
- b. Low transaction costs across reaching the market, brokerage and transfer
- c. Narrow bid-offer spreads
- d. Price discovery

Answer and explanation

Answer: B. Low transaction costs across reaching the market, brokerage and transfer

Rationale: An efficient market is characterised by low transaction costs applying to all aspects of the transaction, including the cost of reaching the market, the actual brokerage cost and the cost of transferring the security. This attribute is referred to as internal efficiency.

21. [Single choice] [CM-EIP §1.3.3] A security can be sold quickly but its price jumps substantially between transactions. Which characteristic of a good market is missing?

- a. Availability of information
- b. Marketability
- c. Price continuity
- d. Information efficiency

Answer and explanation

Answer: C. Price continuity

Rationale: Marketability is the likelihood of a quick sale and is necessary but not sufficient for liquidity. Price continuity means prices do not change much from one transaction to the next unless new information arrives, and it is the missing component here.

22. [Single choice] [CM-EIP §1.3.4] Which body's rules govern the dealing and settlement procedures for Eurobonds?

- a. The Securities and Exchange Commission
- b. The Monetary Authority of Singapore

- c. The Options Clearing Corporation
- d. The Association of International Bond Dealers

Answer and explanation

Answer: D. The Association of International Bond Dealers

Rationale: Because participation in the Eurobond market is worldwide, dealing and settlement procedures are governed by the rules of the Association of International Bond Dealers (AIBD).

23. [Single choice] [CM-EIP §1.3.5] Which market settles on a T+0 basis?

- a. China
- b. United States
- c. Singapore
- d. Hong Kong

Answer and explanation

Answer: A. China

Rationale: In China the settlement period is T+0, on the philosophy that an investor must have the funds on hand to make the purchase. The US, Singapore and Hong Kong settle at T+2.

24. [Single choice] [CM-EIP §1.3.5] How does a dealer differ from a broker in the OTC market?

- a. A dealer acts as agent for a commission; a broker has a vested interest in the transaction
- b. A dealer has a vested interest in transactions and earns the spread between bid and offer; a broker acts as agent
- c. A dealer only trades listed securities
- d. A dealer cannot hold inventory

Answer and explanation

Answer: B. A dealer has a vested interest in transactions and earns the spread between bid and offer; a broker acts as agent

Rationale: Unlike brokers, dealers have a vested interest in transactions and earn the spread between the bid and offer prices.

25. [Single choice] [CM-EIP §1.3.6] What role does the Options Clearing Corporation perform in the US options market?

- a. It regulates the exchanges
- b. It sets the strike prices of all listed options
- c. It becomes the buyer for every seller and the seller for every buyer, guaranteeing contract obligations
- d. It acts as the market maker on the exchange floor

Answer and explanation

Answer: C. It becomes the buyer for every seller and the seller for every buyer, guaranteeing contract obligations

Rationale: The OCC acts as intermediary between the brokers representing buyers and sellers, becoming the buyer for every seller and the seller for every buyer, guaranteeing that all contract obligations will be met.

26. [Single choice] [CM-EIP §1.4] Which digital development is described as reducing the daily float of cash in the banking system?

- a. Non-bank digital wallets
- b. Digital currencies
- c. Internet banking
- d. Internet business transactions clearing directly

Answer and explanation

Answer: D. Internet business transactions clearing directly

Rationale: Cash management is a traditional area of banking where transactions cleared through the banking system may take a few days, providing a daily float. That float has been reduced because of the greater efficiency of direct internet transactions.

27. [Single choice] [CM-EIP §1.4] Which statement about cryptocurrencies is correct according to the guide?

- a. They are not recognised as legal tender and make up about 1% of currency transactions
- b. They cannot be traded through futures contracts
- c. They are recognised as legal tender in Singapore
- d. They are backed by central bank reserves

Answer and explanation

Answer: A. They are not recognised as legal tender and make up about 1% of currency transactions

Rationale: Cryptocurrencies are not backed by reserves or central banks, are not recognised as legal tender, and make up only about 1% of currency transactions. Bitcoin futures trade on the CME and CBOE.

28. [Single choice] [CM-EIP §1.5] A 90-day commercial paper is best classified as which type of instrument?

- a. A capital market instrument
- b. A money market instrument
- c. A derivative
- d. A real asset

Answer and explanation

Answer: B. A money market instrument

Rationale: Money market instruments are highly liquid, short-term debt instruments with maturities of less than one year. Commercial paper maturing in 90 days is a money market instrument.

Chapter 2 — Risk, Return and Time Value Calculations

29. [Single choice] [CM-EIP §2.1] Which statement best expresses the risk-return principle?

- a. Risk and return are unrelated in efficient markets
- b. Lower risk always produces higher realised returns
- c. Rational investors demand higher potential returns when they take on more risk
- d. Investors always prefer the highest available return regardless of risk

Answer and explanation

Answer: C. Rational investors demand higher potential returns when they take on more risk

Rationale: The risk-return (or risk-reward) principle tells us that rational investors demand higher potential returns when they take on more risk, and do not demand as high a rate of return when risk is lower.

30. [Single choice] [CM-EIP §2.1] A CCC-rated bond and the stock of a AAA-rated blue-chip company with a conservative business and financial profile are compared. Which is generally considered safer?

- a. They are equally risky
- b. It cannot be determined from the information given
- c. The CCC-rated bond, because bonds rank above equities
- d. The blue-chip stock, because the CCC bond carries much higher credit risk

Answer and explanation

Answer: D. The blue-chip stock, because the CCC bond carries much higher credit risk

Rationale: The risk of a CCC-rated (junk or speculative grade) bond is much higher than that of a blue-chip corporate debt rated AAA. The stock of the blue-chip company will probably be considered safer than the CCC-rated bond if it has a conservative business and financial profile, even though equities are generally the riskier asset class.

31. [Single choice] [CM-EIP §2.1] What does the VIX measure?

- a. The expected 30-day volatility of the S&P 500
- b. The realised volatility of the STI over the past year
- c. The average credit spread on US corporate bonds
- d. The yield gap between equities and bonds

Answer and explanation

Answer: A. The expected 30-day volatility of the S&P 500

Rationale: The VIX shows the market's expectation of 30-day S&P 500 volatility and is referred to as the investor fear gauge. A higher VIX means more risk and volatility are expected.

32. [Single choice] [CM-EIP §2.1] At what level did the VIX peak during the Global Financial Crisis in October 2008?

- a. 30
- b. 89.5
- c. 20
- d. 48

Answer and explanation

Answer: B. 89.5

Rationale: At the height of the Global Financial Crisis in October 2008 the VIX shot up to a high of 89.5. During the European Debt Crisis it reached 48.

33. [Calculation] [CM-EIP §2.2.1] An investor buys 1,000 shares at $5.00 each, receives a dividend of $0.20 per share, and sells one year later at $6.00. What is the total return?

- a. 20.0%
- b. 24.0%
- c. 26.0%
- d. 30.0%

Answer and explanation

Answer: B. 24.0%

Rationale: Total return = [CF + (PE − PB)] / PB = [0.20 + (6.00 − 5.00)] / 5.00 = 1.20 / 5.00 = 0.24 = 24%.

34. [Calculation] [CM-EIP §2.2.2] An investment returns 25% with probability 0.20, −30% with probability 0.20 and 15% with probability 0.60. What is the expected rate of return?

- a. 3.33%
- b. 8.00%
- c. 10.00%
- d. 15.00%

Answer and explanation

Answer: B. 8.00%

Rationale: E(R) = 0.20(0.25) + 0.20(−0.30) + 0.60(0.15) = 0.05 − 0.06 + 0.09 = 0.08 = 8%.

35. [Calculation] [CM-EIP §2.2.3] A product is expected to generate a maximum return of 2.5% over three months. What is the annualised rate of return?

- a. 10.00%
- b. 10.38%
- c. 10.40%
- d. 12.50%

Answer and explanation

Answer: B. 10.38%

Rationale: Annualised return = (1.025)^(12/3) − 1 = (1.025)^4 − 1 = 1.10381 − 1 = 0.1038 = 10.38%.

36. [Calculation] [CM-EIP §2.2.3] A product quotes an annualised target return of 12%. What return is achievable over four months?

- a. 3.00%
- b. 4.12%
- c. 3.85%
- d. 4.00%

Answer and explanation

Answer: C. 3.85%

Rationale: (1 + r4)^(12/4) − 1 = 0.12, so (1 + r4) = (1.12)^(4/12) and r4 = (1.12)^(1/3) − 1 = 0.0385 = 3.85%.

37. [Calculation] [CM-EIP §2.2.4] An investment returns 10.4% in a year when inflation is 2%. What is the real rate of return?

- a. 8.40%
- b. 8.00%
- c. 12.61%
- d. 8.24%

Answer and explanation

Answer: D. 8.24%

Rationale: Real return = [(1 + 0.104) / (1 + 0.02)] − 1 = 1.104 / 1.02 − 1 = 0.08235 = 8.24%.

38. [Calculation] [CM-EIP §2.2.4] An investment returns 10.4% in a year when there is deflation of 1%. What is the real rate of return?

- a. 11.52%
- b. 11.40%
- c. 10.40%
- d. 9.31%

Answer and explanation

Answer: A. 11.52%

Rationale: With deflation the denominator subtracts the rate: [(1 + 0.104) / (1 − 0.01)] − 1 = 1.104 / 0.99 − 1 = 0.1152 = 11.52%.

39. [Calculation] [CM-EIP §2.2.5] A JPY portfolio returns 121.88% while SGD/JPY rises 18.89%. What is the currency-adjusted return in SGD?

- a. 140.77%
- b. 86.62%
- c. 102.99%
- d. 121.88%

Answer and explanation

Answer: B. 86.62%

Rationale: The reference currency (SGD) is the first listed currency of SGD/JPY, so divide: (1 + 1.2188) / (1 + 0.1889) − 1 = 2.2188 / 1.1889 − 1 = 0.8662 = 86.62%.

40. [Calculation] [CM-EIP §2.2.5] A USD portfolio returns 87.06% while USD/SGD rises 8.12%. What is the currency-adjusted return in SGD?

- a. 78.94%
- b. 95.18%
- c. 102.24%
- d. 87.06%

Answer and explanation

Answer: C. 102.24%

Rationale: The reference currency (SGD) is the second listed currency of USD/SGD, so multiply: $(1 + 0.8706) \times (1 + 0.0812) - 1 = 1.8706 \times 1.0812 - 1 = 1.0224 = 102.24\%$.

41. [Single choice] [CM-EIP §2.2.5] When restating a foreign portfolio return into a reference currency, what is the rule of thumb?

- a. Multiply when the reference currency is the first listed currency of the pair, divide when it is the second
- b. Always divide by (1 + currency return)
- c. Always multiply by (1 + currency return)
- d. Divide when the reference currency is the first listed currency of the pair, multiply when it is the second

Answer and explanation

Answer: D. Divide when the reference currency is the first listed currency of the pair, multiply when it is the second

Rationale: Divide (1 + portfolio return) by (1 + currency return) when the reference currency is the first listed currency of the currency pair, and multiply when the reference currency is the second listed currency.

42. [Single choice] [CM-EIP §2.2.6] Which return measure is the money-weighted return in a portfolio management context?

- a. The internal rate of return
- b. The real rate of return
- c. The currency-adjusted return
- d. The compounded annualised rate of return

Answer and explanation

Answer: A. The internal rate of return

Rationale: The IRR is the money-weighted return: it is the actual return generated for the client given the movement of their cash flows, and it is sensitive to the size and timing of those flows. The compounded annualised rate is the time-weighted return.

43. [Calculation] [CM-EIP §2.2.6] A fund returns 2% in year 1, 10% in year 2 and −4% in year 3. What is the annualised compounded (time-weighted) return?

- a. 2.57%
- b. 2.51%
- c. 2.67%
- d. 7.71%

Answer and explanation

Answer: B. 2.51%

Rationale: Cumulative return $= (1.02 \times 1.10 \times 0.96) - 1 = 1.07712 - 1 = 7.712\%$. Annualised $= (1.07712)\hat{}(1/3) - 1 = 0.0251 = 2.51\%$.

44. [Calculation] [CM-EIP §2.3.1] Using returns of 25% (p = 0.20), −30% (p = 0.20) and 15% (p = 0.60), with an expected return of 8%, what is the standard deviation?

- a. 37.60%
- b. 3.76%
- c. 19.39%
- d. 13.30%

Answer and explanation

Answer: C. 19.39%

Rationale: Variance = $0.20(0.25 - 0.08)^2 + 0.20(-0.30 - 0.08)^2 + 0.60(0.15 - 0.08)^2 = 0.00578 + 0.02888 + 0.00294 = 0.03760$. Standard deviation = $\sqrt{0.03760} = 0.1939 = 19.39\%$.

45. [Single choice] [CM-EIP §2.3.1] What is the difference between historical volatility and implied volatility?

- a. Historical volatility is forward-looking; implied volatility is backward-looking
- b. They are the same measure
- c. Implied volatility is calculated from bond prices
- d. Historical volatility is calculated from past prices; implied volatility is derived from the price of a share option

Answer and explanation

Answer: D. Historical volatility is calculated from past prices; implied volatility is derived from the price of a share option

Rationale: Volatility is the standard deviation of the security. Historical volatility is calculated from past price changes; implied volatility is the volatility implied by the price of a share option.

46. [Calculation] [CM-EIP §2.3.2] Investment A has an expected return of 5% and a standard deviation of 8%. Investment B has an expected return of 15% and a standard deviation of 20%. Which is more attractive on a risk-per-unit-of-return basis?

- a. Investment B, with a coefficient of variation of 1.33
- b. They are equally attractive
- c. Investment A, because its standard deviation is lower
- d. Investment A, with a coefficient of variation of 1.60

Answer and explanation

Answer: A. Investment B, with a coefficient of variation of 1.33

Rationale: CV_A = 0.08 / 0.05 = 1.60; CV_B = 0.20 / 0.15 = 1.33. On a risk per unit of return basis, Investment B is more attractive because it has the lower coefficient of variation.

47. [Calculation] [CM-EIP §2.3.3] A portfolio has an expected return of 6% and a standard deviation of 12%. What is the Z-score for a return of 0%?

- a. −2.0
- b. −0.5
- c. +0.5
- d. −1.0

Answer and explanation

Answer: B. −0.5

Rationale: Z = (X − E(r)) / σ = (0% − 6%) / 12% = −0.5. From the standard normal table this implies a 31% chance of a negative return.

48. [Single choice] [CM-EIP §2.3.3] For a normal distribution, approximately what proportion of outcomes falls within plus or minus two standard deviations of the mean?

- a. 68%

- b. 90%
- c. 95.4%
- d. 99.7%

Answer and explanation

Answer: C. 95.4%

Rationale: A Z-score of 2.00 corresponds to 0.9772, or about 97.7% on one side and 2.3% on the other, so about 95.4% of returns fall within plus or minus two standard deviations. Plus or minus one standard deviation covers 68%.

49. [Calculation] [CM-EIP §2.3.4] A balanced portfolio has an expected return of 6% and a standard deviation of 12%. Using a Z-score of −1.96, what is the value at risk at the 2.5% tail?

- a. −12%
- b. −6%
- c. −18%
- d. −25%

Answer and explanation

Answer: C. −18%

Rationale: X = Z + E(r) = −1.96(0.12) + 0.06 = −0.2352 + 0.06 = −0.1752, approximately −18%. There is a 2.5% chance of the portfolio losing at least 18% in a given year.

50. [Calculation] [CM-EIP §2.3.4] A defensive portfolio has an expected return of 4% and a standard deviation of 4%. Using a Z-score of −2.58, what is the value at risk at the 0.5% tail?

- a. −10%
- b. −25%
- c. −4%
- d. −7%

Answer and explanation

Answer: D. −7%

Rationale: X = Z + E(r) = −2.58(0.04) + 0.04 = −0.1032 + 0.04 = −0.0632, approximately −7%. There is a 0.5% chance of losing at least 7% in a given year.

51. [Single choice] [CM-EIP §2.3.4] What is the principal limitation of using historical data to estimate the probability of future losses?

- a. Factors and conditions influencing returns over the next few years could be quite different from those influencing past returns
- b. Historical data cannot be used with the normal distribution
- c. Value at risk cannot be computed from historical data
- d. Historical data is always unavailable

Answer and explanation

Answer: A. Factors and conditions influencing returns over the next few years could be quite different from those influencing past returns

Rationale: The main problem with using historical data as the basis for predicting the probability of returns is that factors and conditions influencing returns for the next few years could be quite different from those influencing past returns. The financial community therefore also uses Monte Carlo simulation.

52. [Single choice] [CM-EIP §2.4.1] Which category of risk includes business risk, credit risk and fraud risk?

- a. Product-specific risk
- b. Issuer and counterparty risk
- c. Market risk
- d. Management risk

Answer and explanation

Answer: B. Issuer and counterparty risk

Rationale: Issuer and counterparty risks relate mostly to business risk and include business risk, credit risk and fraud risk.

53. [Single choice] [CM-EIP §2.4.1] In the Lehman Brothers case, what was the approximate ratio of assets to the firm's own capital?

- a. USD680 million of assets supported by USD22.5 million of capital
- b. USD68 billion of assets supported by USD2.25 billion of capital
- c. USD680 billion of assets supported by USD22.5 billion of capital
- d. USD22.5 billion of assets supported by USD680 billion of capital

Answer and explanation

Answer: C. USD680 billion of assets supported by USD22.5 billion of capital

Rationale: At one point Lehman had close to USD680 billion of assets supported only by USD22.5 billion of the firm's capital.

54. [Single choice] [CM-EIP §2.4.1] Which risk cannot be fully diversified away?

- a. Concentration risk
- b. Business risk
- c. Fraud risk
- d. Market risk

Answer and explanation

Answer: D. Market risk

Rationale: Market risk refers to risk that affects most assets in the market. Hence such risk cannot be fully diversified. All asset classes are affected by market risks in one way or another.

55. [Single choice] [CM-EIP §2.4.1] During the Global Financial Crisis of 2008–2009, which asset classes did not fall in tandem with the rest?

- a. Cash and government bonds
- b. Real estate and commodities
- c. High-yield bonds and hedge funds
- d. Equities and corporate bonds

Answer and explanation

Answer: A. Cash and government bonds

Rationale: During the Global Financial Crisis almost all classes of financial assets except cash and government bonds fell in tandem. The correlation of asset classes tends toward 1 during periods of crisis.

56. [Single choice] [CM-EIP §2.4.1] On SGX, at what price movement is a circuit breaker triggered?

- a. A move of more than 5% from the reference price
- b. A move of more than 10% from the traded price 5 minutes prior
- c. A move of more than 20% from the previous close
- d. A move of more than 3% from the opening price

Answer and explanation

Answer: B. A move of more than 10% from the traded price 5 minutes prior

Rationale: On SGX, circuit breakers are triggered if the price for a stock would move by more than 10% of the traded price 5 minutes prior (the reference price). Once triggered, a 5-minute cooling-off period follows during which trading can only take place plus or minus 10% from the reference price.

57. [Single choice] [CM-EIP §2.4.1] In the 2015 Mizuho Securities incident, what was the cause of the loss of at least 27 billion yen?

- a. A fraudulent trade
- b. A system failure
- c. A typing error in the trade size and price
- d. A flash crash

Answer and explanation

Answer: C. A typing error in the trade size and price

Rationale: A typing error caused Mizuho Securities to lose at least 27 billion yen (about $225 million) when a trader tried to sell 610,000 shares at 1 yen each, having intended to sell 1 share at 610,000 yen.

58. [Single choice] [CM-EIP §2.4.1] Which of the following is a management risk rather than a market risk?

- a. A rise in factor costs
- b. A natural disaster
- c. A sudden market-wide shock
- d. A risk profile mismatch between the client and the recommended product

Answer and explanation

Answer: D. A risk profile mismatch between the client and the recommended product

Rationale: Management risk relates to how investments are potentially mismanaged. Risk profile mismatch and concentration risk are the two forms. Market shocks, factor costs and natural disasters are market risks.

59. [Single choice] [CM-EIP §2.4.1] What is a short squeeze?

- a. A situation where an absence of sellers forces short positions to cover at substantially higher prices
- b. A regulatory restriction on short selling
- c. A narrowing of the bid-offer spread
- d. A sudden fall in price forcing long positions to liquidate

Answer and explanation

Answer: A. A situation where an absence of sellers forces short positions to cover at substantially higher prices

Rationale: Those with a short position on a derivative instrument may find the price of the underlying asset suddenly rising on good news. If there is an absence of sellers, in this short squeeze situation, traders may be forced to cover their short positions at substantially higher prices.

60. [Multiple response] [CM-EIP §2.4.2] Which of the following are elements of risk management described in the guide? (Select all that apply)

- a. Due diligence checks on fund managers
- b. Timely research and risk management processes
- c. Client profile and product suitability assessment
- d. Process and system risk management

Answer and explanation

Answer: A, B, C, D

Rationale: Risk management involves due diligence, timely research and risk management processes, assessing client profile and product suitability, risk diversification, and process and system risk management.

61. [Calculation] [CM-EIP §2.5.1] A SGD 1 million deposit earns 8% simple interest for 91 days. What is the interest earned?

- a. SGD 40,288.23
- b. SGD 19,945.21
- c. SGD 20,000.00

- d. SGD 20,343.02

Answer and explanation

Answer: B. SGD 19,945.21

Rationale: Simple interest = (1,000,000 × 8 × 91) / 36,500 = SGD 19,945.21. The denominator is 36,500 because Singapore dollar interest uses a 365-day basis.

62. [Single choice] [CM-EIP §2.5.1] Why does Singapore dollar interest use a denominator of 36,500 while US dollar interest uses 36,000?

- a. Because US deposits are always shorter term
- b. Because Singapore has higher interest rates
- c. Because the day basis differs: 365 days for Singapore dollars and 360 days for US dollars
- d. Because Singapore uses simple rather than compound interest

Answer and explanation

Answer: C. Because the day basis differs: 365 days for Singapore dollars and 360 days for US dollars

Rationale: The denominator depends on the currency's day basis. Singapore dollar interest is based on a 365-day basis; US dollar interest is based on a 360-day basis.

63. [Calculation] [CM-EIP §2.5.2] What is the future value of $1,000 invested for 10 years at 5% compounded annually?

- a. $1,500.00
- b. $1,628.89
- c. $1,643.62
- d. $1,647.01

Answer and explanation

Answer: B. $1,628.89

Rationale: FV = PV(1 + k)^n = 1,000(1.05)^10 = 1,000(1.62889) = $1,628.89.

64. [Calculation] [CM-EIP §2.5.4] What is the future value of $1,000 invested for 10 years at 5% compounded quarterly?

- a. $1,647.01
- b. $1,500.00
- c. $1,628.89
- d. $1,643.62

Answer and explanation

Answer: D. $1,643.62

Rationale: FV = PV(1 + k/m)^(mn) = 1,000(1 + 0.05/4)^(4×10) = 1,000(1.0125)^40 = 1,000(1.64362) = $1,643.62.

65. [Calculation] [CM-EIP §2.5.4] An investment grows from $1,000 to $1,628.89 over 10 years. What is the compounded annual growth rate?

- a. 5.00%
- b. 6.29%
- c. 4.88%
- d. 62.89%

Answer and explanation

Answer: A. 5.00%

Rationale: CAGR = (FV/PV)^(1/n) − 1 = (1,628.89/1,000)^(1/10) − 1 = 1.62889^0.1 − 1 = 0.05 = 5%.

66. [Calculation] [CM-EIP §2.5.5] At a 6% discount rate, what is the present value of $1,000 receivable in five years?

- a. $792.09
- b. $747.27
- c. $747.26
- d. $1,338.23

Answer and explanation

Answer: B. $747.27

Rationale: PV = FV / (1 + k)^n = 1,000 / (1.06)^5 = 1,000 / 1.33823 = $747.27.

67. [Single choice] [CM-EIP §2.5.5] What is the minimum appropriate discount rate for a present value calculation?

- a. The average market return
- b. The issuer's coupon rate
- c. The risk-free government bond yield of the same maturity as the intended holding period
- d. The current inflation rate

Answer and explanation

Answer: C. The risk-free government bond yield of the same maturity as the intended holding period

Rationale: The actual discount rate is at least the risk-free government bond yield that has the same maturity as the intended holding period of the investment.

68. [Single choice] [CM-EIP §2.5.3] What is the effect of negative interest rates on the present value of short-end fixed income investments?

- a. It lowers the present value
- b. It has no effect
- c. It makes the present value indeterminate
- d. It raises the present value

Answer and explanation

Answer: D. It raises the present value

Rationale: Negative central bank rates drive short-term rates down, and investors initially tried to lock in yields at the short end, driving short-term fixed income values up. In time value terms, negative interest rates raise the present value of fixed income investments at the short end of the yield curve.

69. [Single choice] [CM-EIP §2.6] Which pair of risks make opposite predictions about the impact of a rise in interest rates on a bond investor's return?

- a. Interest rate risk and reinvestment risk
- b. Inflation risk and political risk
- c. Event risk and sector risk
- d. Credit risk and liquidity risk

Answer and explanation

Answer: A. Interest rate risk and reinvestment risk

Rationale: A rise in interest rates pushes the bond price down (interest rate risk) while raising the rate at which coupons can be reinvested (reinvestment risk). These are not fully offsetting: generally the reinvestment effect only slightly mitigates the interest rate effect.

Chapter 3 — Key Drivers of Market Movements and Asset Values

70. [Single choice] [CM-EIP §3.1] Which three primary factors influence asset values and market movement?

- a. Earnings, dividends and growth
- b. Fundamental factors, market-driven factors and investment behaviour

- c. Interest rates, inflation and exchange rates
- d. Supply, demand and regulation

Answer and explanation

Answer: B. Fundamental factors, market-driven factors and investment behaviour

Rationale: There are three primary factors: fundamental factors, market-driven factors and investment behaviour. Fundamental and market-driven factors provide the rational explanation; investor behaviour reflects how emotion and sentiment affect price.

71. [Single choice] [CM-EIP §3.1] What does the Efficient Market Hypothesis contend?

- a. Prices never reflect available information
- b. Markets are always irrational
- c. Asset prices adjust rapidly to reflect new publicly available information, so price is fair value and no individual can consistently outperform
- d. Fundamental analysis always beats technical analysis

Answer and explanation

Answer: C. Asset prices adjust rapidly to reflect new publicly available information, so price is fair value and no individual can consistently outperform

Rationale: The EMH postulated by Eugene Fama contends that markets are efficient and asset prices adjust rapidly to reflect any new publicly available information. Therefore an asset's price is its fair value and no individual may consistently outperform the market.

72. [Single choice] [CM-EIP §3.1.1] Which factor group is described as reflecting how emotional and market sentiment affects price movement?

- a. Regulatory factors
- b. Fundamental factors
- c. Market-driven factors
- d. Investment behaviour

Answer and explanation

Answer: D. Investment behaviour

Rationale: Fundamental and market-driven factors provide the rational and logical explanation for how an asset's worth is reflected in its price, while investor behaviour reflects how emotional and market sentiment affects the price movement of the asset.

73. [Single choice] [CM-EIP §3.2.1] What is top-down asset allocation?

- a. Allocating a larger proportion of funds to developed economies, growing sectors and leading corporate performers
- b. Allocating equally across all asset classes
- c. Allocating only to index funds
- d. Selecting individual stocks first, then sectors

Answer and explanation

Answer: A. Allocating a larger proportion of funds to developed economies, growing sectors and leading corporate performers

Rationale: In top-down asset allocation, a larger proportion of funds is allocated to the developed economies, growing sectors and industries, and the leading corporate performers within these industries.

74. [Single choice] [CM-EIP §3.2.2] What is the multiplier effect of fiscal policy?

- a. The tendency of tax cuts to increase the budget deficit
- b. The process by which an initial round of government spending ripples through industries, raising incomes and further consumption
- c. The multiplication of money supply through bank lending
- d. The compounding of investment returns over time

Answer and explanation

Answer: B. The process by which an initial round of government spending ripples through industries, raising incomes and further consumption

Rationale: Money from fiscal expenditure passes through the economy and eventually flows to consumers, producing a multiplier effect. The initial round of spending ripples through different industries, raising wages and profits, creating higher disposable incomes and triggering further consumption.

75. [Calculation] [CM-EIP §3.2.3] Using GDP = C + I + G + NE, which component is directly raised by a government programme of infrastructure development?

- a. C
- b. I
- c. G
- d. NE

Answer and explanation

Answer: C. G

Rationale: Government spending includes both investment and consumption, covering education, defence, capital investment, infrastructure development and health services. Infrastructure projects therefore raise G.

76. [Single choice] [CM-EIP §3.2.3] Which fiscal measure is intended to improve net exports?

- a. Increasing infrastructure spending
- b. Offering tax incentives for companies to relocate factories home
- c. Lowering the personal income tax rate
- d. Imposing tariffs on selected foreign goods

Answer and explanation

Answer: D. Imposing tariffs on selected foreign goods

Rationale: Imposing tariffs on selected foreign goods makes the prices of some foreign goods less attractive relative to domestically made goods, which is intended to improve net exports, traditionally negative.

77. [Single choice] [CM-EIP §3.2.4] Which of the Fed's three tools has it relied on most heavily as its key method for influencing market liquidity?

- a. Open market operations
- b. Reserve requirement changes
- c. The discount window
- d. The Fed Funds Rate

Answer and explanation

Answer: A. Open market operations

Rationale: Of the three, the Fed has relied heavily on open market operations as its key method for influencing market liquidity. It was used extensively to pump liquidity into the market in 2008.

78. [Single choice] [CM-EIP §3.2.4] In what order does a change in liquidity from open market operations affect the economy?

- a. The aggregate economy, then common stocks, then bond prices
- b. Bond prices, then common stocks, then the aggregate economy
- c. Common stocks, then bond prices, then the aggregate economy
- d. All three simultaneously

Answer and explanation

Answer: B. Bond prices, then common stocks, then the aggregate economy

Rationale: This type of change in liquidity first affects bond prices, then common stocks and, finally, the aggregate economy.

79. [Single choice] [CM-EIP §3.2.4] What is the primary intermediate target of Singapore's monetary policy?

- a. The money supply
- b. Domestic interest rates
- c. The trade-weighted exchange rate
- d. The inflation rate directly

Answer and explanation

Answer: C. The trade-weighted exchange rate

Rationale: Because of its open economy, Singapore's monetary regime centres around the management of its exchange rate. MAS conducts policy by managing a trade-weighted exchange rate index. The choice of the exchange rate as intermediate target implies MAS gives up control over domestic interest rates and money supply.

80. [Single choice] [CM-EIP §3.2.4] How often is Singapore's exchange rate policy band reviewed, and what is released after each review?

- a. Annually, with an Annual Report
- b. Monthly, with a press release
- c. Quarterly, with a Budget Statement
- d. About every six months, with a Monetary Policy Statement

Answer and explanation

Answer: D. About every six months, with a Monetary Policy Statement

Rationale: The length of the policy review cycle is typically six months. A Monetary Policy Statement is released after each review, accompanied by a Macroeconomic Review.

81. [Calculation] [CM-EIP §3.2.4] With a 10% reserve requirement, a new deposit of 10,000 passes through the banking system. What is the total money supply created after all rounds?

- a. 10,000
- b. 90,000
- c. 100,000
- d. 1,000,000

Answer and explanation

Answer: C. 100,000

Rationale: A bank takes in a new deposit and lends out 90%, keeping 10% as statutory reserves. The process repeats through other banks, leading to a 10-fold increase in money supply: 10,000 becomes 100,000.

82. [Single choice] [CM-EIP §3.2.4] According to Fisher's equation of exchange, what happens when the central bank increases the money supply and the equation is fully operative?

- a. The price level rises, and velocity determines output
- b. Velocity falls proportionally
- c. Nothing changes
- d. Output rises proportionally

Answer and explanation

Answer: A. The price level rises, and velocity determines output

Rationale: When the equation is fully operative, the central bank cannot increase output by increasing money supply because as money M increases, the price level P increases, and the velocity of money V, which is socially determined, then determines output Q.

83. [Single choice] [CM-EIP §3.2.4] In the positive scenario for stock prices when interest rates rise, why may stock prices remain fairly stable?

- a. Because earnings fall as fast as the required return rises

- b. Because the increase in the required rate of return is offset by the increase in earnings growth
- c. Because dividends are cut
- d. Because the market ignores interest rates

Answer and explanation

Answer: B. Because the increase in the required rate of return is offset by the increase in earnings growth

Rationale: In the positive scenario, rates rise with inflation and strong corporate earnings, and firms pass cost increases on to consumers. Stock prices may remain fairly stable because the increase in required rate of return k is offset by the increase in earnings growth g.

84. [Single choice] [CM-EIP §3.2.4] Why do banks typically benefit from a gradual increase in interest rates during the initial expansion?

- a. Because loan defaults fall to zero
- b. Because their bond portfolios appreciate
- c. Because they can widen the net interest spread by raising lending rates faster than deposit rates while still enjoying loan growth
- d. Because deposit rates rise faster than lending rates

Answer and explanation

Answer: C. Because they can widen the net interest spread by raising lending rates faster than deposit rates while still enjoying loan growth

Rationale: During the initial expansion, the banking sector benefits from its ability to widen the net interest spread, raising lending cost faster than deposit rates, while still enjoying the demand in loan growth, creating net interest income expansion.

85. [Single choice] [CM-EIP §3.2.5] What is a technical recession?

- a. A rise in unemployment above 5%
- b. A decline in industrial production for one month
- c. A fall in the stock market of more than 20%
- d. Two quarters of contraction in GDP growth

Answer and explanation

Answer: D. Two quarters of contraction in GDP growth

Rationale: Before GDP growth numbers reflecting two quarters of contraction, that is a technical recession, are announced, the equity markets would have typically corrected ahead of the announcements.

86. [Single choice] [CM-EIP §3.2.5] Which investment strategy is appropriate in the late bear to early bull phase of the business cycle?

- a. Invest in cyclical and high-beta equities and reduce cash holdings
- b. Buy growth stocks and high-yield bonds
- c. Hold defensive high-dividend stocks and investment grade bonds
- d. Keep cash, investment grade bonds and precious metals

Answer and explanation

Answer: A. Invest in cyclical and high-beta equities and reduce cash holdings

Rationale: In the late bear to early bull phase, markets perceive the worst is over and asset values begin to rebound sharply. The appropriate strategy is to invest in cyclical and high-beta equities and reduce cash holdings.

87. [Single choice] [CM-EIP §3.2.5] What does a flattening yield curve suggest?

- a. An imminent economic boom
- b. One of the early predictors of an upcoming recession
- c. Rising inflation expectations
- d. A widening credit spread

Answer and explanation

Answer: B. One of the early predictors of an upcoming recession

Rationale: When longer-term credit demand appears weak, long-maturity bond rates may not rise as much even with Fed fund rate hikes. This results in a flattening of the yield curve, which is considered one of the early predictors of an upcoming recession.

88. [Single choice] [CM-EIP §3.2.6] Why do investors monitor economic calendars closely?

- a. Because calendars predict the exact market direction
- b. To avoid trading altogether
- c. Because price volatility of affected financial assets tends to increase immediately before and after an information release
- d. Because releases are always accurate

Answer and explanation

Answer: C. Because price volatility of affected financial assets tends to increase immediately before and after an information release

Rationale: Knowing the dates and times of the releases of key information is useful because the price volatility of affected financial assets tends to increase during the period immediately before and after the information release.

89. [Single choice] [CM-EIP §3.2.7] How many components make up the US Composite Leading Economic Index published by The Conference Board?

- a. 4
- b. 7
- c. 10
- d. 12

Answer and explanation

Answer: C. 10

Rationale: The CLEI comprises 10 indicators used to predict the direction of the economy. The Coincident Economic Indicators comprise 4 series and the lagging index comprises 7 components.

90. [Single choice] [CM-EIP §3.2.7] What does a Purchasing Managers' Index reading of 50 signify?

- a. Severe contraction
- b. No data available
- c. Strong expansion
- d. The dividing line between expansion and contraction

Answer and explanation

Answer: D. The dividing line between expansion and contraction

Rationale: In general, a reading of 50 or higher suggests expansion in the manufacturing sector whilst a reading below 50 indicates a contraction.

91. [Single choice] [CM-EIP §3.2.7] Which of the following is a coincident indicator?

- a. Nonfarm payroll employment
- b. Average prime rate charged by banks
- c. Housing starts and building permits
- d. Average weekly hours in manufacturing

Answer and explanation

Answer: A. Nonfarm payroll employment

Rationale: Nonfarm payroll employment is one of the four Coincident Economic Indicators. Average weekly hours and housing starts are leading indicators; the average prime rate is a lagging indicator.

92. [Single choice] [CM-EIP §3.2.7] What is the principal limitation of composite leading indicators?

- a. They are published too infrequently to be useful
- b. They signal turning points but say nothing about the magnitude of the shifts
- c. They only apply to developing economies
- d. They are always wrong

Answer and explanation

Answer: B. They signal turning points but say nothing about the magnitude of the shifts

Rationale: While such composite indicators are useful as short-term indicators of turning points in the economy, they do not tell investors anything about the magnitude of these shifts.

93. [Single choice] [CM-EIP §3.2.8] Why can a country with a widening trade deficit become a target of speculative attack?

- a. Because its inflation rate is too low
- b. Because it has too much foreign currency reserves
- c. Because the deficit makes the country a debtor nation
- d. Because its interest rates are too high

Answer and explanation

Answer: C. Because the deficit makes the country a debtor nation

Rationale: A country with a widening trade deficit can become the target of speculative attacks because this deficit makes the country a debtor nation. Such weaknesses are generally believed to have led to the Asian Currency Crisis in 1997.

94. [Single choice] [CM-EIP §3.2.8] Which inflation measure does the US Fed use as its primary gauge?

- a. The producer price index
- b. Average hourly earnings
- c. The headline Consumer Price Index
- d. Core Personal Consumption Expenditure

Answer and explanation

Answer: D. Core Personal Consumption Expenditure

Rationale: The US Fed uses Core Personal Consumption Expenditure data, or core inflation, as its primary measure and has traditionally set 2% as the long-run inflation rate.

95. [Single choice] [CM-EIP §3.2.9] What happened to the pound sterling after the Brexit referendum in June 2016?

- a. It fell 10.5% in four days and was down 18% by October
- b. It was unchanged
- c. It fell 51.9% immediately
- d. It rose 10.5% in four days

Answer and explanation

Answer: A. It fell 10.5% in four days and was down 18% by October

Rationale: The surprise 51.9% affirmation vote for Britain to leave the European Union resulted in panic selling of the GBP. It fell 10.5% in 4 days and continued to drift downwards to a low of GBP/USD 1.21714, down 18%, by October.

96. [Single choice] [CM-EIP §3.2.10] In which stage of the industry life cycle are profit margins very high and profits able to more than double from a low earnings base?

- a. Early development
- b. Rapid growth

- c. Mature growth
- d. Decline

Answer and explanation

Answer: B. Rapid growth

Rationale: In rapid growth, rising sales growth coupled with greater efficiency results in accelerating profit growth. During this phase profits can more than double from the low earnings base, and profit margins are also very high.

97. [Single choice] [CM-EIP §3.2.10] Which stage of the industry life cycle is described as probably the longest phase?

- a. Early development
- b. Rapid growth
- c. Stabilisation
- d. Decline

Answer and explanation

Answer: C. Stabilisation

Rationale: Stabilisation is probably the longest phase of the life cycle. The industry growth rate tends to be in line with the growth rate of the overall economy.

98. [Single choice] [CM-EIP §3.2.10] Which of Porter's five forces is illustrated by the bankruptcy of Kodak following the emergence of digital photography?

- a. Bargaining power of suppliers
- b. Rivalry among existing competitors
- c. Threat of new entrants
- d. Threat of substitute products

Answer and explanation

Answer: D. Threat of substitute products

Rationale: Substitute products limit the profit potential of an industry because they limit the prices that firms can charge. The emergence of digital photography and imaging resulted in the demise of camera film, bankrupting film-producing giant Kodak.

99. [Single choice] [CM-EIP §3.2.10] Which industry type is least affected by recession and economic adversity?

- a. Defensive
- b. Growth
- c. Interest-sensitive
- d. Cyclical

Answer and explanation

Answer: A. Defensive

Rationale: Least affected by recession and economic adversity are the defensive industries. The food and public utilities industries are considered to be defensive.

100. [Single choice] [CM-EIP §3.2.11] Which approach to stock-picking focuses on firms with specific competitive advantages that distinguish them from others in the same industry?

- a. Top-down
- b. Bottom-up
- c. Sector rotation
- d. Market timing

Answer and explanation

Answer: B. Bottom-up

Rationale: By applying a bottom-up approach to stock-picking, firms with specific competitive advantages distinguish themselves from others.

101. [Single choice] [CM-EIP §3.3.1] What typically happens when the bulk of available funds is almost fully invested in the market?

- a. New IPOs are cancelled
- b. The market accelerates upward
- c. The market plateaus and funds rotate to lagging sectors, with parts of the market becoming overvalued
- d. Interest rates fall sharply

Answer and explanation

Answer: C. The market plateaus and funds rotate to lagging sectors, with parts of the market becoming overvalued

Rationale: When there are insufficient funds to sustain the upward drive, the market will start to plateau, with funds rotating to industry sectors and assets that have been lagging in relative performance. Typically this is when parts of the market become overvalued.

102. [Single choice] [CM-EIP §3.3.2] Why can a cluster of large IPOs depress the prices of similar stocks already trading?

- a. Because IPOs reduce market capitalisation
- b. Because IPOs raise interest rates
- c. Because IPOs increase the money supply
- d. Because funds are absorbed by the IPOs, and some investors take profit on existing stocks to participate

Answer and explanation

Answer: D. Because funds are absorbed by the IPOs, and some investors take profit on existing stocks to participate

Rationale: Funds will be absorbed by IPOs, in some cases deliberately priced at a discount to ensure demand. This flow of monies to the IPOs can come at the expense of similar stocks already trading, and some investors will take profit on existing stocks to participate.

103. [Single choice] [CM-EIP §3.3.3] Why can futures and options amplify a market decline?

- a. Because leverage magnifies losses, forcing investors to reduce holdings to meet margin requirements and driving the market down further
- b. Because they are only traded by retail investors
- c. Because they settle in cash
- d. Because they are unleveraged

Answer and explanation

Answer: A. Because leverage magnifies losses, forcing investors to reduce holdings to meet margin requirements and driving the market down further

Rationale: Because of the leverage, returns to a derivative investor are magnified. As underlying assets fall, they cause further reduction to futures and options prices with magnified losses, and derivative investors may be forced to reduce holdings to meet margin requirements, further driving down the market.

104. [Single choice] [CM-EIP §3.3.3] When does the triple witching hour occur in the US market?

- a. The first Friday of each quarter, at the open
- b. The third Friday of March, June, September and December, in the last trading hour
- c. The last trading day of each month, at midday
- d. Every Friday at the close

Answer and explanation

Answer: B. The third Friday of March, June, September and December, in the last trading hour

Rationale: On the third Friday of every March, June, September and December, the last hour of the stock market trading session, 3.00 to 4.00pm New York local time, is called the triple witching hour. Index futures, index options and stock options all mature that day.

105. [Single choice] [CM-EIP §3.3.4] What is hot money?

- a. Money raised through an IPO
- b. Money held in central bank reserves
- c. Global money from hedge and other large funds seeking temporary inefficiencies and arbitrage opportunities
- d. Money deposited in savings accounts

Answer and explanation

Answer: C. Global money from hedge and other large funds seeking temporary inefficiencies and arbitrage opportunities

Rationale: Hot money is global money from hedge and other large funds that seek to exploit temporary inefficiencies and arbitrage opportunities. It may flow into a market, sector or set of stocks and temporarily push up volume, interest and prices.

106. [Single choice] [CM-EIP §3.3.5] In a call market such as some French bourses, what is prioritised?

- a. Both price and timing equally
- b. Neither price nor timing
- c. Timing, but not price
- d. Price, but not timing

Answer and explanation

Answer: D. Price, but not timing

Rationale: In a call market all the bids are submitted and batched for simultaneous execution once or twice a day when the market is called. Price, but not timing, prioritisation is practised.

107. [Single choice] [CM-EIP §3.3.5] What is the principal concern about dark pools for less sophisticated participants?

- a. Internally agreed trade prices, together with the size and lack of transparency of participant identities, put them at a disadvantage
- b. They only trade government bonds
- c. They guarantee better prices
- d. They are illegal in all jurisdictions

Answer and explanation

Answer: A. Internally agreed trade prices, together with the size and lack of transparency of participant identities, put them at a disadvantage

Rationale: In the case of dark pools, internally agreed trade prices, along with the size and lack of transparency of identities of dark pool participants, put less sophisticated market participants at a disadvantage.

108. [Single choice] [CM-EIP §3.4.1] When does the herd instinct kick in most strongly?

- a. When information is abundant and markets are calm
- b. When information is scarce and threats seem imminent
- c. When interest rates are falling
- d. When volatility is low

Answer and explanation

Answer: B. When information is scarce and threats seem imminent

Rationale: Behavioural studies show that the herd instinct kicks in especially when information is scarce, and threats seem imminent.

109. [Single choice] [CM-EIP §3.4.1] What was the market price-earnings ratio at the peak of the US Internet Dot.com Bubble?

- a. 20x
- b. 50x
- c. 200x
- d. 500x

Answer and explanation

Answer: C. 200x

Rationale: NASDAQ rose sharply from 1,000 to 5,000, with a market price-earnings ratio of 200x. Alan Greenspan coined the term irrational exuberance in December 1996, but the bubble continued until 2001.

110. [Single choice] [CM-EIP §3.4.2] How does loss aversion affect portfolio behaviour?

- a. Investors buy more of their losers immediately
- b. Investors sell losing positions first to realise losses
- c. Investors sell profitable holdings to lock in gains while holding on to book losses in the hope of a rebound
- d. Investors hold all positions equally

Answer and explanation

Answer: C. Investors sell profitable holdings to lock in gains while holding on to book losses in the hope of a rebound

Rationale: Investors and investment managers may be unconsciously influenced by loss aversion to sell off their profitable holdings to lock in positive returns while still holding on to book losses on poorer performing securities in the hope that the securities rebound. This potentially has a double negative impact.

111. [Single choice] [CM-EIP §3.4.3] An investor who has committed funds to an investment only reads research supporting that investment. Which bias is this?

- a. Anchoring
- b. Herd instinct
- c. Loss aversion
- d. Confirmation bias

Answer and explanation

Answer: D. Confirmation bias

Rationale: Confirmation bias refers to how we, as investors, tend to look for views and recommendations that are aligned with our own. We then reinforce the view by committing even more funds to the same investment.

112. [Single choice] [CM-EIP §3.5] Which of the following is a market-driven factor rather than a fundamental factor?

- a. Liquidity and funds flow
- b. Firm-specific competitive advantage
- c. Monetary policy
- d. Industry life cycle

Answer and explanation

Answer: A. Liquidity and funds flow

Rationale: Market-driven factors refer to physical market conditions that affect short-term demand and supply. Major market-driven factors include liquidity flows, new issues of securities, trading influences from derivatives, hot-money flows and computer-driven momentum trading.

Chapter 4 — Foreign Exchange

113. [Single choice] [CM-EIP §4.1.1] In the quotation USD/SGD 1.2520, which currency is the base currency and what does the rate mean?

- a. SGD is the base; SGD 1 = USD 1.2520
- b. USD is the base; USD 1 = SGD 1.2520
- c. USD is the counter currency; SGD 1 = USD 1.2520
- d. Both are base currencies

Answer and explanation

Answer: B. USD is the base; USD 1 = SGD 1.2520

Rationale: The first listed currency is the base currency and the second is the counter or term currency. The quotation refers to the number of units of counter currency that equals one unit of base currency, so USD 1 = SGD 1.2520.

114. [Single choice] [CM-EIP §4.1.1] An investor borrowed USD to fund a SGD investment. How is the resulting foreign exchange position described?

- a. Short both currencies
- b. Long USD and short SGD
- c. Short USD and long SGD
- d. Long both currencies

Answer and explanation

Answer: C. Short USD and long SGD

Rationale: The investor sold USD spot for SGD, so the position is short USD and long SGD. To close it, the investor enters the opposite transaction, selling spot SGD and buying USD, and is said to have squared off the position.

115. [Single choice] [CM-EIP §4.1.1] What is the standard settlement period for a spot foreign exchange transaction?

- a. Five working days
- b. Same day
- c. Next working day
- d. Two working days

Answer and explanation

Answer: D. Two working days

Rationale: Usually foreign exchange transactions are spot deals, made for delivery in two working days (value spot). There is also a market for same day delivery (value today) and next day delivery (value tomorrow).

116. [Single choice] [CM-EIP §4.2.1] What is the effect of higher interest rates on a currency, all else equal?

- a. They attract foreign capital inflow, causing the exchange rate to rise
- b. They cause capital outflow, weakening the currency
- c. They have no effect on the exchange rate
- d. They always cause immediate depreciation

Answer and explanation

Answer: A. They attract foreign capital inflow, causing the exchange rate to rise

Rationale: Higher interest rates offer lenders in an economy higher returns relative to other countries. Consequently, higher interest rates attract inflow of foreign capital, which causes the exchange rate to rise. The impact may be moderated by higher inflation or other factors.

117. [Single choice] [CM-EIP §4.2.2] What happened to the New Venezuelan Bolivar on the day of its launch in August 2018?

- a. It appreciated 95%
- b. It fell 95% in value
- c. It was unchanged
- d. It was pegged to the US dollar

Answer and explanation

Answer: B. It fell 95% in value

Rationale: Despite launching the New Venezuelan Bolivar in late August 2018, the new currency fell by 95% in value the very day of the launch. The IMF indicated Venezuela's inflation would have hit 1,000,000% in 2018.

118. [Single choice] [CM-EIP §4.2.4] Which countries successfully defended their currencies during the Asian Currency Crisis despite speculative attack?

- a. India, Philippines and Vietnam
- b. Indonesia, Thailand and Malaysia
- c. Hong Kong, Singapore and Taiwan
- d. Japan, Korea and China

Answer and explanation

Answer: C. Hong Kong, Singapore and Taiwan

Rationale: Hedge funds and other speculators were able to exploit the vulnerability of some Asian currencies that had weaker foreign currency reserves, but countries with strong currency reserves such as Hong Kong, Singapore and Taiwan successfully defended their currencies.

119. [Single choice] [CM-EIP §4.2.5] What does a trade deficit imply for demand for the domestic currency, all else equal?

- a. No change in demand
- b. Immediate appreciation
- c. Higher demand for the domestic currency
- d. Weaker demand for the local currency

Answer and explanation

Answer: D. Weaker demand for the local currency

Rationale: When a country has a trade deficit it needs to acquire more foreign currency to pay for imports, resulting in higher demand for foreign currency. All other things held constant, a trade deficit should lead to weaker demand for local currency.

120. [Single choice] [CM-EIP §4.2.5] What does an improvement in a country's terms of trade imply?

- a. Export prices rise by a greater rate than import prices, implying higher demand for the currency and appreciation
- b. Import prices rise faster than export prices
- c. The trade balance is unchanged
- d. The currency must depreciate

Answer and explanation

Answer: A. Export prices rise by a greater rate than import prices, implying higher demand for the currency and appreciation

Rationale: The terms of trade improve when a country's export prices rise by a greater rate than its import prices. This in turn implies higher demand for the country's currency and accompanying currency appreciation.

121. [Single choice] [CM-EIP §4.2.6] Why could Greece not inflate its way out of its public debt crisis?

- a. Because its debt was denominated in gold
- b. Because as a Eurozone member it had no monetary policy independence
- c. Because it had no central bank
- d. Because its debt was all short-term

Answer and explanation

Answer: B. Because as a Eurozone member it had no monetary policy independence

Rationale: As Greece did not have monetary policy independence because it is a member of the Eurozone, it could not simply inflate its way out of debt. Greece has needed three bail-out programmes totalling 300 billion euros.

122. [Single choice] [CM-EIP §4.3] In what order do foreign exchange markets open through the trading day?

- a. New York, London, Tokyo, Sydney
- b. London, New York, Tokyo, Sydney
- c. Sydney, Tokyo, Singapore/Hong Kong, London, New York, San Francisco
- d. Tokyo, London, Sydney, New York

Answer and explanation

Answer: C. Sydney, Tokyo, Singapore/Hong Kong, London, New York, San Francisco

Rationale: The foreign exchange markets work round the clock, starting the day in Sydney, then Tokyo, Singapore/Hong Kong, London, New York and San Francisco.

123. [Single choice] [CM-EIP §4.3] According to the BIS Triennial Survey, what was average daily foreign exchange turnover in April 2019?

- a. USD2.0 trillion
- b. USD3.2 trillion
- c. USD5.1 trillion
- d. USD6.6 trillion

Answer and explanation

Answer: D. USD6.6 trillion

Rationale: Trading in FX markets averaged USD6.6 trillion per day in April 2019, up from USD5.1 trillion in April 2016. FX swaps rose to USD3.2 trillion per day, almost half of global FX trading.

124. [Single choice] [CM-EIP §4.3] Which currency was on one side of 88% of all foreign exchange trades in April 2019?

- a. The US dollar
- b. The Japanese yen
- c. The Chinese renminbi
- d. The euro

Answer and explanation

Answer: A. The US dollar

Rationale: The US dollar remained the world's dominant vehicle currency, on one side of 88% of all trades in April 2019. The euro remained second at 32%.

125. [Single choice] [CM-EIP §4.3] Which currencies are described as commodity currencies?

- a. CHF and JPY
- b. AUD and ZAR
- c. SGD and HKD
- d. EUR and GBP

Answer and explanation

Answer: B. AUD and ZAR

Rationale: The Australian dollar and South African rand are considered commodity currencies and tend to be particularly strong during periods of higher global economic growth or fiscal stimulus, when demand for metals and raw materials drives demand for both.

126. [Single choice] [CM-EIP §4.4] What is a pip in foreign exchange?

- a. 1% of the exchange rate
- b. 0.01, the second decimal place
- c. 0.0001, the fourth decimal place
- d. 0.001, the third decimal place

Answer and explanation

Answer: C. 0.0001, the fourth decimal place

Rationale: FX rates are usually quoted to the fourth decimal place. The last digit of the quote is called a pip, equal to 0.0001, the smallest unit of price movement or quotation for foreign exchange.

127. [Calculation] [CM-EIP §4.4.1] A dealer quotes USD/SGD 1.2520-25. What is the spread in basis points?

- a. 0.5 basis points
- b. 50 basis points
- c. 25 basis points
- d. 5 basis points

Answer and explanation

Answer: D. 5 basis points

Rationale: The difference between 1.2525 and 1.2520 is 0.0005, also known as 5 basis points, which is the spread or profit earned by the bank.

128. [Single choice] [CM-EIP §4.4.1] When a client asks a bank for a quote, at which prices does the client transact?

- a. Buys at the offer and sells at the bid
- b. Buys and sells at the mid rate
- c. Buys and sells at the bid
- d. Buys at the bid and sells at the offer

Answer and explanation

Answer: A. Buys at the offer and sells at the bid

Rationale: The party asking for a quote will always buy at the offer rate and sell at the bid rate. In market parlance, the price is always against the price taker.

129. [Single choice] [CM-EIP §4.4.2] Which of the following is a direct quote in Singapore?

- a. GBP/USD 1.6816
- b. USD/SGD 1.2520
- c. AUD/USD 0.9235
- d. EUR/USD 1.3622

Answer and explanation

Answer: B. USD/SGD 1.2520

Rationale: A direct quote expresses the price of one unit of foreign currency in units of domestic currency, that is foreign/domestic. In Singapore, USD/SGD is a direct quote. GBP/USD, AUD/USD and EUR/USD are indirect quotes in Singapore.

130. [Calculation] [CM-EIP §4.4.3] In Singapore the direct quote is USD/SGD 1.2520. What is the corresponding indirect quote SGD/USD?

- a. 1.0000
- b. 0.8000
- c. 0.7987
- d. 1.2520

Answer and explanation

Answer: C. 0.7987

Rationale: The indirect quote is the reciprocal of the direct quote: 1 / 1.2520 = 0.7987.

131. [Single choice] [CM-EIP §4.4.3] Which currencies are quoted as the base currency against the US dollar, contrary to the usual convention?

- a. CHF, CAD and SEK
- b. INR, IDR and THB
- c. JPY, CNY and SGD
- d. GBP, AUD, NZD and EUR

Answer and explanation

Answer: D. GBP, AUD, NZD and EUR

Rationale: Currencies in the colonial or old sterling bloc, comprising the UK pound sterling, the Australian dollar and the New Zealand dollar, are the base currency when quoted against the USD. The EUR is also quoted in this manner.

132. [Calculation] [CM-EIP §4.4.4] USD/SGD is 1.3362/1.3382 and USD/JPY is 112.6530/112.7330. What is the SGD/JPY bid rate?

- a. 84.1825
- b. 84.3684
- c. 84.2755
- d. 150.6100

Answer and explanation

Answer: A. 84.1825

Rationale: To get the SGD/JPY bid, divide the USD/JPY bid by the USD/SGD ask: 112.6530 / 1.3382 = 84.1825. To get the ask, divide the USD/JPY ask by the USD/SGD bid: 112.7330 / 1.3362 = 84.3684.

133. [Single choice] [CM-EIP §4.5] What does an investor forfeit by covering a foreign currency exposure with a forward contract?

- a. The right to receive dividends
- b. Potential gains from favourable movements in market rates
- c. The ability to sell the underlying security
- d. The interest earned on the foreign currency

Answer and explanation

Answer: B. Potential gains from favourable movements in market rates

Rationale: By covering the exposure through locking in a known forward FX rate, the investor also forfeits potential gains from favourable movements in market rates.

134. [Single choice] [CM-EIP §4.5.1] Which currency generally trades at a forward discount?

- a. Both currencies equally
- b. Neither, since forward rates equal spot rates
- c. The currency of the country with the higher interest rate
- d. The currency of the country with the lower interest rate

Answer and explanation

Answer: C. The currency of the country with the higher interest rate

Rationale: Generally, the currency of the country with the higher interest rate will trade at a forward discount, meaning the forward rate is cheaper than the spot rate. The currency with the lower interest rate trades at a forward premium.

135. [Calculation] [CM-EIP §4.5.2] Spot USD/SGD is 1.2520, the USD rate is 1.75% and the SGD rate is 0.75%, for 30 days. What are the forward points?

- a. +10 points
- b. −100 points
- c. +100 points
- d. −10 points

Answer and explanation

Answer: D. −10 points

Rationale: Forward points = S(Rc − Rb)(n/360) / [1 + Rb(n/360)] = 1.2520(0.0075 − 0.0175)(30/360) / [1 + 0.0175(30/360)] = −0.0010 / 1.0015 = −SGD 0.0010, or −10 points. The forward rate is 1.2510.

136. [Single choice] [CM-EIP §4.5.2] What do forward points reflect?

- a. The relative interest rates of the two currencies only
- b. The credit risk of the counterparties
- c. The expected inflation differential
- d. A forecast of the future spot rate

Answer and explanation

Answer: A. The relative interest rates of the two currencies only

Rationale: Forward points are only a reflection of the relative interest rates of the currencies. They are not an indication of what the future spot rate of the currencies will actually be.

137. [Calculation] [CM-EIP §4.5.3] The USD interest rate is 6%, the GBP rate is 5%, and GBP spot is USD 1.50. What is the one-year forward rate?

- a. USD 1.5000
- b. USD 1.5143
- c. USD 1.4857
- d. USD 1.5100

Answer and explanation

Answer: B. USD 1.5143

Rationale: By interest rate parity, F = S[(1 + Rc n/360)/(1 + Rb n/360)] = 1.50(1.06/1.05) = USD 1.5143 per GBP.

138. [Calculation] [CM-EIP §4.5.3] With the parity forward at USD 1.5143 but the market quoting USD 1.5100, an arbitrageur borrows GBP 1,000,000 at 5%, sells spot at 1.50, invests USD at 6% and buys GBP forward at 1.5100. What is the profit?

- a. USD 1,500
- b. USD 4,050
- c. USD 4,500
- d. USD 14,300

Answer and explanation

Answer: C. USD 4,500

Rationale: Repayment due is GBP 1,050,000. USD received = 1,000,000 × 1.50 = 1,500,000, growing at 6% to 1,590,000. Forward purchase of GBP 1,050,000 at 1.5100 costs 1,585,500. Profit = 1,590,000 − 1,585,500 = USD 4,500.

139. [Calculation] [CM-EIP §4.5.4] USD/JPY spot is 119.22/25 and one-month forward points are 52/51. What is the outright one-month rate?

- a. 119.2252/119.2551
- b. 119.2720/119.2760
- c. 119.1680/119.1980
- d. 119.2148/119.2449

Answer and explanation

Answer: D. 119.2148/119.2449

Rationale: Descending points (52/51) are negative, so subtract: $119.22 - 0.52 = 119.2148$ and $119.25 - 0.51 = 119.2449$. The bid is higher than the offer in point terms because the currency is at a forward discount.

140. [Calculation] [CM-EIP §4.5.4] AUD/USD spot is 0.6475/80 and one-month forward points are 17/21. What is the outright one-month rate?

- a. 0.6492/0.6501
- b. 0.6458/0.6459
- c. 0.6475/0.6480
- d. 0.6492/0.6480

Answer and explanation

Answer: A. 0.6492/0.6501

Rationale: Ascending points (17/21) are positive, so add: $0.6475 + 0.0017 = 0.6492$ and $0.6480 + 0.0021 = 0.6501$.

141. [Single choice] [CM-EIP §4.5.5] What is the key difference between a non-deliverable forward and a regular FX forward?

- a. An NDF has a longer tenor
- b. There is no physical delivery or exchange of the domestic currency in the offshore market
- c. An NDF is exchange-traded
- d. An NDF requires delivery of the notional amount

Answer and explanation

Answer: B. There is no physical delivery or exchange of the domestic currency in the offshore market

Rationale: The concept of an NDF is relatively similar to that of a regular FX forward. The key difference is that there is no physical delivery or exchange of the domestic currency in the offshore market. An NDF is a cash-settled, short-term forward contract.

142. [Single choice] [CM-EIP §4.5.5] What is the normal tenor of a non-deliverable forward contract?

- a. Five years or more
- b. One week to one month
- c. One month to one year
- d. One year to five years

Answer and explanation

Answer: C. One month to one year

Rationale: The normal tenor of the NDF contract ranges from 1 month to a year.

143. [Calculation] [CM-EIP §4.5.5] A client must pay CNY 10 million in three months. USD/CNY spot is 6.5845. If USD/CNY falls 5% to 6.2553, what is the additional USD cost compared with the current rate?

- a. USD 72,314
- b. USD 1,598,644
- c. USD 1,518,718

- d. USD 79,926

Answer and explanation

Answer: D. USD 79,926

Rationale: At 6.5845 the cost is 10,000,000 / 6.5845 = USD 1,518,718. At 6.2553 it is 10,000,000 / 6.2553 = USD 1,598,644. The difference is USD 79,926, which is the loss if the client does nothing.

144. [Single choice] [CM-EIP §4.6] Why is credit risk in a foreign exchange transaction only a fraction of the total transaction amount?

- a. Because the transaction is an exchange of buy and sell, not an outright extension of credit
- b. Because FX is exchange-traded
- c. Because settlement is immediate
- d. Because FX transactions are collateralised

Answer and explanation

Answer: A. Because the transaction is an exchange of buy and sell, not an outright extension of credit

Rationale: In forex transactions, credit risk is but a fraction of the total transaction amount because the transaction is an exchange, buy and sell, and not an outright extension of credit.

145. [Single choice] [CM-EIP §4.6] When does settlement or delivery risk arise in a foreign exchange transaction?

- a. At the moment the trade is agreed
- b. In the period just before settlement, which in FX is two days before payment
- c. Only if the counterparty becomes insolvent
- d. After settlement is complete

Answer and explanation

Answer: B. In the period just before settlement, which in FX is two days before payment

Rationale: Settlement risk is a specific counterparty credit risk which takes place in the period just before settlement. In FX, this is 2 days before payment. Continuous Linked Settlement has reduced delivery risk significantly.

146. [Single choice] [CM-EIP §4.7.1] Which exchange rate regime involves a country adopting another nation's currency as legal tender?

- a. Managed soft peg
- b. Currency board
- c. Currency substitution or dollarisation
- d. Monetary union

Answer and explanation

Answer: C. Currency substitution or dollarisation

Rationale: A country may choose not to manage its own currency but to simply adopt another nation's currency as legal tender. This exchange rate regime is known as currency substitution or dollarization. The adopting country inherits the credibility of that currency, but not that other country's creditworthiness.

147. [Single choice] [CM-EIP §4.7.1] Under the Currency Interchangeability Agreement of 1967, how is the Brunei dollar related to the Singapore dollar?

- a. It is convertible only for trade payments
- b. It is not related
- c. It is pegged at a 5% discount
- d. It is interchangeable at par, and each is legal tender in the other country

Answer and explanation

Answer: D. It is interchangeable at par, and each is legal tender in the other country

Rationale: Under a Currency Interchangeability Agreement in 1967, the Brunei Dollar is interchangeable with the Singapore Dollar at par. The Brunei Dollar is legal tender in Singapore and vice versa.

148. [Single choice] [CM-EIP §4.7.1] What did China's defence of the CNY between July 2014 and December 2016 cost in foreign reserves?

- a. Almost USD 1 trillion, about 25%
- b. Almost USD 100 billion, about 5%
- c. Almost USD 500 billion, about 15%
- d. Nothing

Answer and explanation

Answer: A. Almost USD 1 trillion, about 25%

Rationale: From July 2014 to December 2016, USD/CNY rose only about 11% from CNY6.20 to CNY6.90, but China ran down its foreign currency reserves by almost USD 1 trillion, or 25%.

149. [Calculation] [CM-EIP §4.7.2] A Singapore firm buys a GBP 1 million UK property, borrowing 80% of the value. GBP/SGD is 1.8060 and the GBP depreciates 5% to 1.7157 in a year. If the firm borrows in SGD, what is the loss?

- a. −SGD 1,444,800
- b. −SGD 90,300
- c. −SGD 18,060
- d. −SGD 361,200

Answer and explanation

Answer: B. −SGD 90,300

Rationale: Down payment = $0.2 \times 1{,}000{,}000 \times 1.8060$ = SGD 361,200. SGD loan = $0.8 \times 1{,}000{,}000 \times 1.8060$ = SGD 1,444,800. Sale proceeds = $1{,}000{,}000 \times 1.7157$ = SGD 1,715,700. Loss = 1,715,700 − 361,200 − 1,444,800 = −SGD 90,300.

150. [Calculation] [CM-EIP §4.7.2] Using the same property example, if the firm borrows in GBP instead, what is the loss?

- a. −SGD 343,140
- b. −SGD 361,200
- c. −SGD 18,060
- d. −SGD 90,300

Answer and explanation

Answer: C. −SGD 18,060

Rationale: Down payment = SGD 361,200. Net proceeds after repaying the GBP 800,000 loan = $(1{,}000{,}000 - 800{,}000) \times 1.7157$ = SGD 343,140. Loss = 343,140 − 361,200 = −SGD 18,060. Borrowing in the investment currency is a natural hedge.

151. [Single choice] [CM-EIP §4.7.3] What does a currency carry trade rely on to be profitable?

- a. Falling interest rates in the invested currency
- b. A strengthening borrowing currency
- c. Rising volatility in the invested currency
- d. Relative stability in currency prices, so the interest differential is not wiped out by an adverse FX move

Answer and explanation

Answer: D. Relative stability in currency prices, so the interest differential is not wiped out by an adverse FX move

Rationale: This strategy relies on relative stability in currency prices as an adverse exchange rate movement can wipe out the returns from the underlying interest rate differential.

152. [Single choice] [CM-EIP §4.7.3] **In the JPY/AUD carry trade from 2008 to 2016, in which years did the strategy fail?**

- a. 2008, 2011 and 2015
- b. 2010, 2013 and 2016
- c. Every year
- d. 2009, 2012 and 2014

Answer and explanation

Answer: A. 2008, 2011 and 2015

Rationale: For most years the strategy worked, but for 2008, 2011 and 2015 the strategy failed because the AUD, the invested currency, fell sharply against the borrowed currency, the JPY.

153. [Calculation] [CM-EIP §4.7.3] **In 2008 the AUD fell 34.2% against the JPY and 19% against the SGD. What was the net loss on a JPY 100 million borrowing invested in AUD at 7%?**

- a. −SGD 6,240
- b. −SGD 503,946
- c. +SGD 526,960
- d. −SGD 80,223

Answer and explanation

Answer: B. −SGD 503,946

Rationale: JPY 100 million was borrowed at 2.25%, converted into AUD 1.020 million (SGD 1.286 million) and invested at 7%. Total Australian proceeds converted back to SGD were SGD 1.111 million, lower than the SGD 1.615 million equivalent of the JPY principal and interest, producing a loss of SGD 503,946.

154. [Single choice] [CM-EIP §4.8] **Which of the following is NOT one of the leading factors explaining exchange rate fluctuations?**

- a. Inflation rates
- b. Political stability
- c. The level of the stock market index
- d. Interest rates

Answer and explanation

Answer: C. The level of the stock market index

Rationale: The leading factors are interest rates, inflation rates, political stability, foreign currency reserves, trade account balances and terms of trade, and government or public debt. The stock market index level is not among them.

Chapter 5 — Company Analysis and Understanding Financial Statements

155. [Single choice] [CM-EIP §5.1] **Why is company analysis relevant beyond equity analysis?**

- a. Because bonds are legally equities
- b. Because regulators require it
- c. Because equities are the only asset class
- d. Because it is also relevant for credit analysis of bond investments and other securities whose payments depend on the issuer's viability

Answer and explanation

Answer: D. Because it is also relevant for credit analysis of bond investments and other securities whose payments depend on the issuer's viability

Rationale: Company analysis does not only relate to equity analysis, but is also relevant for credit analysis of bond investments and other securities whose cashflow payments are dependent on the ongoing viability of the issuing company.

156. [Single choice] [CM-EIP §5.1] Why does the guide argue that qualitative factors matter more than ratios?

- a. Because qualitative factors drive and precede the quantitative outcomes
- b. Because ratios cannot be calculated for most companies
- c. Because regulators forbid ratio analysis
- d. Because ratios are always wrong

Answer and explanation

Answer: A. Because qualitative factors drive and precede the quantitative outcomes

Rationale: It is more important to focus on the qualitative factors and conditions that allow for the company to be successful than to simply rely on ratios and numbers, as these qualitative factors drive and precede the quantitative outcomes.

157. [Single choice] [CM-EIP §5.2] Which competitive advantage is illustrated by De Beers, which still controls about 30–35% of world diamond supply?

- a. Superior brand differentiation
- b. Control over key resource supply
- c. Cost efficiency
- d. Control of distribution channels

Answer and explanation

Answer: B. Control over key resource supply

Rationale: Companies that have substantial control over the raw materials and resources required for production have an advantage. Despite giving up its almost monopolistic control of diamond supply decades ago, De Beers continues to control about 30–35% of the world's diamond supply.

158. [Single choice] [CM-EIP §5.2] How many core businesses do diversified companies with shared costs typically hold?

- a. More than 10
- b. 1–2
- c. 3–4
- d. 5–6

Answer and explanation

Answer: C. 3–4

Rationale: Such companies will typically have 3–4 core businesses where they have a competitive advantage and lead in market share. Resources are mainly focused on their core businesses, with a smaller portion allocated to developing new businesses.

159. [Single choice] [CM-EIP §5.3.2] Which accounting standard applies to listed companies in Singapore?

- a. Neither
- b. Both, at the company's choice
- c. US GAAP
- d. IFRS

Answer and explanation

Answer: D. IFRS

Rationale: Listed US companies follow US GAAP. All listed European companies follow IFRS, as does Singapore. As there are variations between US GAAP and IFRS, adjustments and restatements must be made to compare companies across markets.

160. [Single choice] [CM-EIP §5.3.2] What is a qualified audit report?

- a. A report issued when the auditors are not completely satisfied with how the financial statements are prepared
- b. A report issued only for banks
- c. A report confirming the accounts are perfect
- d. A report required only for IPOs

Answer and explanation

Answer: A. A report issued when the auditors are not completely satisfied with how the financial statements are prepared

Rationale: The investment community expects auditors to make the necessary qualifying comments if financial statements are not well-prepared. Qualified reports are provided when the auditors are not completely satisfied with how the company's financial statements are prepared.

161. [Single choice] [CM-EIP §5.4] Which balance sheet equation is used in British practice?

- a. Assets = Liabilities + Shareholders' Equity
- b. Assets − Liabilities = Shareholders' Equity
- c. Assets + Liabilities = Shareholders' Equity
- d. Assets = Shareholders' Equity − Liabilities

Answer and explanation

Answer: B. Assets − Liabilities = Shareholders' Equity

Rationale: For British companies, the practice is to present assets net of liabilities, that is net assets on one side, which then equates to shareholders' equity on the other side. US companies present assets on one side with liabilities and shareholders' equity summing to the same total.

162. [Single choice] [CM-EIP §5.4] What does a common size balance sheet express each item as a percentage of?

- a. Total liabilities
- b. Sales revenue
- c. Total assets
- d. Shareholders' equity

Answer and explanation

Answer: C. Total assets

Rationale: The items on the common size statement are reflected as a percentage of total assets, which is 100%. The common size statement allows for an easier comparison of balance sheet items across different companies.

163. [Single choice] [CM-EIP §5.5.1] What is the principal disadvantage of holding a perpetually large cash balance?

- a. It reduces the current ratio
- b. It increases leverage
- c. It increases the tax liability
- d. It represents underutilised assets and sacrifices potential returns

Answer and explanation

Answer: D. It represents underutilised assets and sacrifices potential returns

Rationale: When not invested in the business, cash balances represent underutilized assets. Keeping a perpetually large cash balance would also mean sacrificing potential returns that could have been generated from investing in other financial assets.

164. [Single choice] [CM-EIP §5.5.2] Which of the following is a warning signal for a retailer?

- a. A sudden build-up in inventories amid falling sales
- b. A steady inventory level with rising sales
- c. A fall in inventory with rising sales
- d. An increase in finished goods ahead of expected demand

Answer and explanation

Answer: A. A sudden build-up in inventories amid falling sales

Rationale: When carrying out quarterly reviews of inventory balances for retailers, one warning signal to watch for is the sudden build-up in inventories amidst falling sales numbers. An increase in finished goods in anticipation of greater demand is a positive reason.

165. [Single choice] [CM-EIP §5.5.3] Approximately what proportion of an airline's assets are accounts receivables, and why?

- a. About 50%, because airlines sell on credit
- b. Less than 5%, because airline ticket sales are largely cash-based
- c. About 25%, because of travel agent credit
- d. About 75%, because of corporate accounts

Answer and explanation

Answer: B. Less than 5%, because airline ticket sales are largely cash-based

Rationale: Airline passenger ticket sales are largely a cash-based business, usually 95% or more, hence accounts receivables usually make up less than 5% of an airline's assets.

166. [Single choice] [CM-EIP §5.5.5] Why was Apple's plant, property and equipment only 7–8% of overall assets in its 2015/16 accounts?

- a. Because Apple leases all its equipment
- b. Because Apple sold its factories
- c. Because Apple outsourced manufacturing to subcontractors while retaining software, intellectual property and research
- d. Because Apple is a service company

Answer and explanation

Answer: C. Because Apple outsourced manufacturing to subcontractors while retaining software, intellectual property and research

Rationale: Some companies have chosen to pass on some of the obsolescence risk to subcontractors. In Apple's 2015/16 accounts, PPE accounts for only 7–8% of overall assets.

167. [Single choice] [CM-EIP §5.5.6] What proportion of total assets do property holdings typically represent for Asian property companies, including holdings through associates and joint ventures?

- a. 95–100%
- b. 20–30%
- c. 40–50%
- d. 70–85%

Answer and explanation

Answer: D. 70–85%

Rationale: Property holdings form the bulk of the value of these companies, typically 55–70%, and once holdings through associates and joint ventures are factored in, they account for 70–85% of overall asset value.

168. [Single choice] [CM-EIP §5.5.7] Why does a goodwill write-down improve reported ROE in subsequent years?

- a. Because it reduces equity, the denominator, without any operational change

- b. Because it increases total assets
- c. Because it reduces the tax rate
- d. Because it increases net income

Answer and explanation

Answer: A. Because it reduces equity, the denominator, without any operational change

Rationale: A goodwill write-down is typically accompanied by a direct charge against equity reserves. As unproductive asset value is removed from total assets but nothing changes operationally, ROA and ROE in subsequent years are higher as a result of the lower denominators.

169. [Single choice] [CM-EIP §5.6.1] What happened to Southeast Asian companies that financed operations with USD-denominated Eurobonds during the 1997 crisis?

- a. Their debt fell in local currency terms
- b. Their USD borrowings quadrupled in local currency terms when the Rupiah and Baht fell more than 75%
- c. They were unaffected
- d. They refinanced at lower rates

Answer and explanation

Answer: B. Their USD borrowings quadrupled in local currency terms when the Rupiah and Baht fell more than 75%

Rationale: When the Rupiah and the Baht fell by more than 75% during the Asian Currency Crisis, the USD Eurobond borrowings suddenly quadrupled in value in local currency terms, and several companies had to declare bankruptcy.

170. [Single choice] [CM-EIP §5.6.2] What does a build-up of accounts payables each quarter accompanied by falling sales suggest?

- a. An increase in credit sales
- b. Strong bargaining power with suppliers
- c. The company may be facing difficulty in making payments
- d. Improved operating cash flow

Answer and explanation

Answer: C. The company may be facing difficulty in making payments

Rationale: If there are signs in the quarterly statements that a company is stretching payables as indicated by a build-up of payables, it could suggest that the company is trying to increase liquidity. A negative scenario could be that the company is facing difficulty in making payments, signalled by a build-up of payables each quarter accompanied by falling sales.

171. [Single choice] [CM-EIP §5.7.3] How are hidden reserves created in a bank?

- a. By issuing new shares
- b. By paying high dividends
- c. By overstating loan losses
- d. By making large provisions in a downturn and being slow to reverse them when customers repay

Answer and explanation

Answer: D. By making large provisions in a downturn and being slow to reverse them when customers repay

Rationale: Banks make huge provisions for loan losses during economic downturns, but the actual losses may not materialize. When economies recover and customers start repaying, some banks may be slow in reversing their provisioning. This creates hidden reserves.

172. [Calculation] [CM-EIP §5.8] A company's revenue falls 76% while operating expenses fall only 60%. What does this illustrate?

- a. That costs are more inelastic than revenue, so a revenue fall produces an operating loss
- b. That the company is overstaffed

- c. That margins improve in a downturn
- d. That costs are more elastic than revenue

Answer and explanation

Answer: A. That costs are more inelastic than revenue, so a revenue fall produces an operating loss

Rationale: Given the more inelastic nature of costs, the cuts to operating expenses were not enough to offset the loss in revenue. This is exactly the SIA case, where a S$9.5 *billion cost cut did not offset a* S$12 billion revenue loss, producing an operating loss of about S$2.5 billion.

173. [Single choice] [CM-EIP §5.8.1] Into which two components should sales revenue growth be broken down?

- a. Domestic and international sales
- b. Unit sales growth and pricing changes
- c. Recurring and non-recurring revenue
- d. Cash and credit sales

Answer and explanation

Answer: B. Unit sales growth and pricing changes

Rationale: When analysing the key drivers of sales growth, we should break the sales revenue component into two parts: unit sales growth and pricing changes. Whenever a company raises either of the two, sales revenue will increase.

174. [Single choice] [CM-EIP §5.8.3] How can depreciation expense be used to cosmetically improve reported profit?

- a. By increasing the asset base
- b. By shortening the depreciation period
- c. By stretching depreciation over more years, which lowers the annual charge without changing cash flow
- d. By capitalising all costs

Answer and explanation

Answer: C. By stretching depreciation over more years, which lowers the annual charge without changing cash flow

Rationale: When a company is not doing well, one cosmetic approach is to change its accounting practice by stretching depreciation expense over more years. A taxi fleet operator depreciating taxis over 10 years instead of 5 dramatically lowers depreciation expense and improves reported net profit, while nothing changes in cash flow.

175. [Single choice] [CM-EIP §5.9] Why is the cash flow statement particularly important for capital-intensive industries?

- a. Because it replaces the income statement
- b. Because it is audited
- c. Because it shows dividends paid
- d. Because net earnings may not accurately show the ability to support liabilities, and the statement adds back depreciation and other non-cash items

Answer and explanation

Answer: D. Because net earnings may not accurately show the ability to support liabilities, and the statement adds back depreciation and other non-cash items

Rationale: For capital-intensive industries, a company's net earnings may not provide an accurate picture about its ability to support short-term liabilities and borrowings. The cash flow statement adds back depreciation and other non-cash items to assess the ability of the company to support financing and investing activities.

176. [Single choice] [CM-EIP §5.9] What is the bondholder's central concern in the cash flow statement?

- a. Consistently negative cash flow from operations
- b. High capital expenditure
- c. Large share buybacks
- d. Rising dividends

Answer and explanation

Answer: A. Consistently negative cash flow from operations

Rationale: For most investors, especially bond holders, a major concern is when the cashflows from operations are consistently negative. A poor overall net cashflow position can be temporarily masked by the selling of assets and businesses.

177. [Single choice] [CM-EIP §5.10] Which share count is used to calculate book value per share?

- a. The weighted average number of shares
- b. The total number of shares outstanding at the financial year-end
- c. The fully diluted number of shares
- d. The shares outstanding at the ex-dividend date

Answer and explanation

Answer: B. The total number of shares outstanding at the financial year-end

Rationale: For per-share computations using numbers on the balance sheet, we use the number of outstanding shares at the end of the financial year, because the balance sheet reflects a snapshot of the financial position.

178. [Single choice] [CM-EIP §5.10] Which share count is used to calculate dividend per share?

- a. The fully diluted number of shares
- b. The weighted average number of shares
- c. The total number of shares outstanding as at the ex-dividend date
- d. The year-end number of shares

Answer and explanation

Answer: C. The total number of shares outstanding as at the ex-dividend date

Rationale: When we consider dividends paid to shareholders, we use the total number of outstanding shares as at the ex-dividend date in order to calculate DPS.

179. [Calculation] [CM-EIP §5.11.1] A company has current assets of $600,000 including inventory of $250,000, and current liabilities of $300,000. What is the quick ratio?

- a. 0.83
- b. 1.50
- c. 2.00
- d. 1.17

Answer and explanation

Answer: D. 1.17

Rationale: The quick ratio excludes inventory: (600,000 − 250,000) / 300,000 = 350,000 / 300,000 = 1.17. The current ratio would be 2.00.

180. [Single choice] [CM-EIP §5.11.1] Why might a supermarket have a quick ratio as low as 0.2 without a liquidity problem?

- a. Because its current assets are fast-moving consumer goods financed by short-term supplier credit
- b. Because it holds no inventory
- c. Because it is loss-making

- d. Because it has no current liabilities

Answer and explanation

Answer: A. Because its current assets are fast-moving consumer goods financed by short-term supplier credit

Rationale: The quick ratio of a supermarket may be as low as 0.2 because it has substantial inventory, but it may not be a liquidity concern because a supermarket's current assets are fast-moving consumer goods that are financed by short-term supplier credit.

181. [Single choice] [CM-EIP §5.11.1] What is window dressing in the context of liquidity ratios?

- a. Reporting results more frequently
- b. Collecting receivables early, reducing inventory below normal and delaying purchases near year-end to flatter the ratios
- c. Increasing the dividend
- d. Issuing new shares

Answer and explanation

Answer: B. Collecting receivables early, reducing inventory below normal and delaying purchases near year-end to flatter the ratios

Rationale: Current and quick ratios at financial reporting dates are susceptible to manipulation, otherwise known as window dressing. Toward the financial year-end, companies can collect receivables earlier, reduce inventory below normal levels and delay purchases.

182. [Single choice] [CM-EIP §5.11.1] A company has a current ratio of 1.5. It then increases both current assets and current liabilities by an equal dollar amount. What happens to the current ratio?

- a. It becomes negative
- b. It rises
- c. It falls
- d. It is unchanged

Answer and explanation

Answer: C. It falls

Rationale: An equal dollar increase in both current assets and current liabilities does not leave the current ratio unchanged. It leads to a lower ratio if it was previously more than 1, and to a higher ratio if it was previously less than 1.

183. [Single choice] [CM-EIP §5.11.2] Which ratio is of primary interest to bondholders?

- a. Inventory turnover
- b. Dividend yield
- c. Gross profit margin
- d. Leverage ratios, because their main risk is default on principal and interest

Answer and explanation

Answer: D. Leverage ratios, because their main risk is default on principal and interest

Rationale: Leverage ratios help determine a firm's amount of financial risk and measure its capacity to meet principal and interest repayments. These ratios are of primary interest for bondholders, whose main risk is default on principal and interest payments.

184. [Single choice] [CM-EIP §5.11.2] How are debts carried on the balance sheet for the purpose of the debt-to-total assets ratio?

- a. At face value, not revalued
- b. At the present value of future cash flows
- c. At the original issue discount

- d. At market value

Answer and explanation

Answer: A. At face value, not revalued

Rationale: Note that debts are carried at their face values on the balance sheet and are not revalued. This is why foreign currency-denominated debt must be checked separately, because appreciation of that currency makes the market value of the debt much larger.

185. [Single choice] [CM-EIP §5.11.3] A firm has an interest coverage ratio of −1 and another has −3, both having made the same trading loss. Which has higher financial risk?

- a. The firm with −3
- b. The firm with −1, because it has higher financing charges
- c. They are equal
- d. It cannot be determined

Answer and explanation

Answer: B. The firm with −1, because it has higher financing charges

Rationale: The size of a negative coverage ratio is also determined by the absolute amount of interest charges. A firm with a ratio of −1 does not necessarily indicate lower financial risk than a firm with −3: both are incurring trading losses, but the lower ratio of −1 is due to higher financing charges, which is not a good thing.

186. [Single choice] [CM-EIP §5.11.4] Why do efficiency ratios use average balances in the denominator?

- a. Because year-end balances are unavailable
- b. Because averages are easier to compute
- c. Because the numerator covers a period, so the denominator should be an average of the opening and closing balances
- d. Because accounting standards require it

Answer and explanation

Answer: C. Because the numerator covers a period, so the denominator should be an average of the opening and closing balances

Rationale: An average is calculated for the denominator because the numerator is measured over the period and, so, the denominator should be an average of the two year-end balances.

187. [Calculation] [CM-EIP §5.11.4] Sales are $3,816 and average total assets are $35,647. What is the total asset turnover?

- a. 0.11
- b. 0.50
- c. 0.58
- d. 9.34

Answer and explanation

Answer: A. 0.11

Rationale: Total asset turnover = Sales / Average total assets = 3,816 / 35,647 = 0.107, approximately 0.11. This is the SIA FY2020/2021 figure, depressed by the pandemic.

188. [Single choice] [CM-EIP §5.11.4] What range of total asset turnover is typical across industries?

- a. Always between 0.5 and 1.5
- b. Always above 5
- c. Always below 1
- d. From about 1 for capital-intensive industries to over 10 for some retailing operations

Answer and explanation

Answer: D. From about 1 for capital-intensive industries to over 10 for some retailing operations

Rationale: Total assets turnover ratios range from 1 for capital-intensive industries to over 10 for some retailing operations. This ratio should be compared with that of other firms in the industry because it varies substantially across industries.

189. [Single choice] [CM-EIP §5.11.5] What does the variability of the operating profit margin over time indicate?

- a. The firm's business risk
- b. The firm's liquidity
- c. The firm's dividend policy
- d. The firm's financial risk

Answer and explanation

Answer: A. The firm's business risk

Rationale: The variability of the operating profit margin over time is an indicator of the firm's business risk. Business risk is the uncertainty of income that is caused by the firm's industry.

190. [Single choice] [CM-EIP §5.11.5] Why is ROA normally expressed before interest?

- a. Because interest is not an expense
- b. To separate operating decisions from financing decisions and enable comparison of companies with different capital structures
- c. Because interest is paid after tax
- d. Because it simplifies the calculation

Answer and explanation

Answer: B. To separate operating decisions from financing decisions and enable comparison of companies with different capital structures

Rationale: Profit before interest is used to separate operating decisions from financing decisions and to enable comparisons of efficiency for companies with different capital structures.

191. [Single choice] [CM-EIP §5.11.5] Why did IAG and United Airlines show high ROE in 2019 before the pandemic?

- a. Because they paid no dividends
- b. Because their earnings were exceptionally strong
- c. Because their equity base had been substantially reduced by losses and charges in prior years
- d. Because they had no debt

Answer and explanation

Answer: C. Because their equity base had been substantially reduced by losses and charges in prior years

Rationale: IAG charged losses of EUR0.9 billion to reserves in 2012, EUR2 billion in 2013 and EUR1.2 billion in 2014. United Airlines charged its losses to retained earnings in 2011 and 2012. Subsequent years of ROE look high due to the reduction of the value of shareholders' equity with losses in prior years.

192. [Multiple response] [CM-EIP §5.11.6] Which of the following are signs of strong and high-quality financial statements? (Select all that apply)

- a. Limited or lower use of debt relative to the industry
- b. Assets generally carried below market values
- c. Absence of off-balance sheet activities
- d. Aggressive revenue recognition that reflects higher sales

Answer and explanation

Answer: A, B, C

Rationale: Signs of quality include limited or lower use of debt, higher liquidity, assets carried below market values, a preference not to reflect intangible assets, funds set aside for replacement, conservative revenue recognition, absence of off-balance-sheet activities, discipline in core business, realistic bad-debt recognition and steady sustainable growth. Aggressive revenue recognition is the opposite of quality.

193. [Single choice] [CM-EIP §5.11.7] Why must trend analysis cover at least one full business cycle?

- a. Because cycles are always five years long
- b. Because regulators require it
- c. Because accounting standards require five years of data
- d. Because otherwise the analysis could mislead, showing a company doing well when it is simply at the crest of a cycle about to turn down

Answer and explanation

Answer: D. Because otherwise the analysis could mislead, showing a company doing well when it is simply at the crest of a cycle about to turn down

Rationale: Business cycles affect the performance of all companies. To have a better understanding, it is necessary to cover at least one cycle and track overall performance over a cycle. Otherwise the analysis could be misleading and indicate that a company is doing well when it is simply at the crest of a cycle that is about to turn down.

194. [Single choice] [CM-EIP §5.11.8] Which of the following is a limitation of ratio analysis?

- a. Companies in emerging sectors with little comparable or reliable financial information do not lend themselves easily to ratio analysis
- b. Ratios are always manipulated
- c. Ratios ignore the balance sheet
- d. Ratios cannot be calculated for listed companies

Answer and explanation

Answer: A. Companies in emerging sectors with little comparable or reliable financial information do not lend themselves easily to ratio analysis

Rationale: Companies in emerging sectors, such as emerging technology or business models, do not lend themselves easily to ratio or financial analysis. The lack of comparable and reliable financial information greatly limits ratio analysis.

Chapter 6 — Equity Securities

195. [Single choice] [CM-EIP §6.2.1] Why might the owners of a successful private company list it on an exchange?

- a. To avoid all regulation
- b. To share the risk of ownership, monetise part of the value and redeploy funds
- c. To reduce the number of shareholders
- d. To eliminate the need for management

Answer and explanation

Answer: B. To share the risk of ownership, monetise part of the value and redeploy funds

Rationale: Original owners may want to share their risk of ownership, realise and monetise the value of the company, and redeploy funds to other businesses for even greater gains. Through a well-planned IPO, ownership control can still be retained while the business value is partially monetised.

196. [Single choice] [CM-EIP §6.2.2] What is the maximum number of votes a multiple-vote share may carry under SGX dual class listing rules?

- a. 2
- b. 5
- c. 10
- d. Unlimited

Answer and explanation

Answer: C. 10

Rationale: To prevent founder entrenchment, each multiple vote share can only carry a maximum of 10 votes. Extra votes also cannot be exercised on important matters where there may be conflicts of interest between founders and minority shareholders.

197. [Single choice] [CM-EIP §6.2.2] How many Class B shares of Berkshire Hathaway can one Class A share be converted into?

- a. 100
- b. 500
- c. 1,500
- d. 2,000

Answer and explanation

Answer: C. 1,500

Rationale: Each share of Class A common stock is convertible, at the option of the holder, into 1,500 shares of Class B common stock. Class B shares are non-convertible and have 1/200th the voting power of Class A shares.

198. [Single choice] [CM-EIP §6.2.3] Which right do ordinary shareholders NOT have?

- a. A residual claim on remaining assets
- b. The right to vote on major policy decisions
- c. The right to be informed of major events
- d. A guaranteed right to dividends

Answer and explanation

Answer: D. A guaranteed right to dividends

Rationale: Ordinary share dividends are decided at the discretion of management and depend on the performance of the company. Common shareholders do not have a right to dividend payouts and such pay-outs are not guaranteed.

199. [Single choice] [CM-EIP §6.2.4] Why do non-callable preference shares carry a lower rate of return than callable ones?

- a. Because the issuer has no option to redeem them
- b. Because they rank above bonds
- c. Because they have a shorter life
- d. Because they have voting rights

Answer and explanation

Answer: A. Because the issuer has no option to redeem them

Rationale: Non-callable preference shares carry a lower rate of return than the callable ones because the issuer has no option to redeem the shares.

200. [Single choice] [CM-EIP §6.2.4] What happens if an issuer fails to pay a preference dividend?

- a. It constitutes default
- b. It does not constitute default, unlike a missed bond coupon
- c. The shares convert to ordinary shares
- d. The issuer must be liquidated

Answer and explanation

Answer: B. It does not constitute default, unlike a missed bond coupon

Rationale: Unlike a bond, the issuer can decide not to pay the preference dividend if earnings are insufficient and failure to pay the specified preference dividend does not result in a default.

201. [Calculation] [CM-EIP §6.2.5] **A company has 300 million shares at $2.00 and announces a 1-for-3 rights issue at $1.90. What is the theoretical ex-rights price?**

- a. $1.90
- b. $1.50
- c. $1.975
- d. $2.00

Answer and explanation

Answer: C. $1.975

Rationale: Before the issue, market capitalisation is 300m × $2.00 = $600m. After a 1-for-3 issue there are 400m shares and new capital of 100m × $1.90 = $190m, so the new market capitalisation is $790m. Theoretical ex-rights price = $790m / 400m = $1.975.

202. [Calculation] [CM-EIP §6.2.5] **Using the same company, what is the theoretical ex-bonus price for a 1-for-3 bonus issue?**

- a. $1.975
- b. $2.00
- c. $1.90
- d. $1.50

Answer and explanation

Answer: D. $1.50

Rationale: A bonus issue raises no capital, so the market capitalisation stays at $600m while the share count rises to 400m. Theoretical ex-bonus price = $600m / 400m = $1.50, sharply lower than the rights-issue price because no new capital is injected.

203. [Calculation] [CM-EIP §6.2.5] **A company has 300 million shares for 9 months and 400 million for 3 months after a rights issue. What is the weighted-average number of shares?**

- a. 325m
- b. 350m
- c. 400m
- d. 300m

Answer and explanation

Answer: A. 325m

Rationale: Weighted-average shares = (300m × 9 + 400m × 3) / 12 = (2,700m + 1,200m) / 12 = 3,900m / 12 = 325m.

204. [Single choice] [CM-EIP §6.2.5] **What is the maximum pro-rata renounceable rights issue a company may now make, as a proportion of share capital?**

- a. 25%
- b. 50%
- c. 75%
- d. 100%

Answer and explanation

Answer: D. 100%

Rationale: Companies are now allowed to issue pro-rata renounceable rights shares of up to 100% of share capital, which is an increase from the previous limit of 50%.

205. [Single choice] [CM-EIP §6.2.5] **Why did Citigroup carry out a 1-for-10 reverse stock split in 2011?**

- a. To raise new capital
- b. Because institutional investors are discouraged from holding penny stocks priced below $5

- c. To reduce the number of shareholders
- d. To increase its market capitalisation

Answer and explanation

Answer: B. Because institutional investors are discouraged from holding penny stocks priced below $5

Rationale: Certain institutional investors are governed by rules which discourage them from investing in penny stocks. The SEC defines penny stocks as stocks priced below $5 per share, which made it difficult for institutional investors to own Citigroup shares. The reverse split took the shares from $4.52 to $44.16.

206. [Single choice] [CM-EIP §6.3.1] Which of the following is a characteristic of blue-chip companies?

- a. No dividend payments
- b. High earnings growth in the current period
- c. A long and consistent record of earnings, allowing steady dividends through good and bad years
- d. High beta and high volatility

Answer and explanation

Answer: C. A long and consistent record of earnings, allowing steady dividends through good and bad years

Rationale: As these companies have a long and consistent record of earnings, they generally can pay steady dividends over a long period of time during both good and bad years. Growth would have slowed over time due to their sheer size.

207. [Single choice] [CM-EIP §6.3.2] What growth rate is typically meant by a high-growth company?

- a. Above 50% for one year
- b. Above 5% compounded annually
- c. Above 10% compounded annually
- d. Above 20% compounded annually over several years

Answer and explanation

Answer: D. Above 20% compounded annually over several years

Rationale: When the investment community talks about growth companies, they are usually referring to companies that can generate high double-digit sales and earnings growth over a certain number of years, for example more than 20% growth compounded annually over several years.

208. [Single choice] [CM-EIP §6.3.3] If the fixed deposit rate is 3%, at what dividend yield are dependable high-yield stocks likely to find price support?

- a. 4–5%
- b. 8–10%
- c. Above 15%
- d. 1–2%

Answer and explanation

Answer: A. 4–5%

Rationale: If the fixed deposit rate is 3% for annual deposits, stocks with dependable dividend payouts are unlikely to fall below a dividend yield of 4–5% before tax. The price is supported at the point where dividend/price = 4%.

209. [Single choice] [CM-EIP §6.3.4] Which sectors are described as cyclical in Singapore and Hong Kong?

- a. Technology and electronics
- b. Property and banking
- c. Utilities and food
- d. Healthcare and pharmaceuticals

Answer and explanation

Answer: B. Property and banking

Rationale: For Singapore and Hong Kong, the cyclical stocks will also come from the property and banking sectors because they are usually amongst the first sectors to be affected in an economic downturn. For Taiwan and South Korea, technology and electronic manufacturing are cyclical.

210. [Single choice] [CM-EIP §6.3.5] Which of the following is a typical defensive stock?

- a. An automobile manufacturer
- b. A semiconductor company
- c. A food and beverage company such as Nestlé
- d. A property developer

Answer and explanation

Answer: C. A food and beverage company such as Nestlé

Rationale: Defensive companies tend to be those that provide consumers with necessities, such as pharmaceuticals, food, utilities and transportation. Shares of companies like Nestlé and Coca-Cola would be typical defensive stocks.

211. [Calculation] [CM-EIP §6.4.1] A value-weighted index has a base market value of 64,000,000 set at an index level of 100. The current market value is 67,000,000. What is the new index value?

- a. 100
- b. 104.69
- c. 105
- d. 107

Answer and explanation

Answer: C. 105

Rationale: New index value = (Current market value / Base value) × Beginning index value = (67,000,000 / 64,000,000) × 100 = 1.05 × 100 = 105.

212. [Single choice] [CM-EIP §6.4.1] Why does a value-weighted index adjust automatically for bonus and rights issues?

- a. Because the index excludes such companies
- b. Because the index is rebalanced annually
- c. Because the divisor is recalculated daily
- d. Because these corporate actions typically do not affect market capitalisation, as a price fall is offset by more shares

Answer and explanation

Answer: D. Because these corporate actions typically do not affect market capitalisation, as a price fall is offset by more shares

Rationale: An advantage of the value-weighted index is that it automatically adjusts for bonus and rights issues, as well as other capital changes, because these corporate actions typically do not affect market capitalisation: a decrease in share price is offset by an increase in the number of shares outstanding.

213. [Single choice] [CM-EIP §6.4.2] Why has the Dow Jones Industrial Average been criticised?

- a. Because it uses only 30 non-randomly selected blue-chip shares that are not representative of the NYSE
- b. Because it includes too many technology companies
- c. Because it is rebalanced monthly
- d. Because it is value-weighted

Answer and explanation

Answer: A. Because it uses only 30 non-randomly selected blue-chip shares that are not representative of the NYSE

Rationale: The DJIA has been criticized because it uses only 30 non-randomly selected blue-chip shares, which are not considered to be representative of the shares listed on the New York Stock Exchange.

214. [Calculation] [CM-EIP §6.4.3] Three shares change by factors of 1.04, 1.03 and 1.05. What is the equal-weighted geometric average change?

- a. 1.060
- b. 1.040
- c. 1.0400
- d. 1.120

Answer and explanation

Answer: B. 1.040

Rationale: Geometric average = $(1.04 \times 1.03 \times 1.05)^{(1/3)} = (1.12476)^{(1/3)} = 1.040$. The index value on day 2 equals the day 1 value multiplied by 1.040.

215. [Single choice] [CM-EIP §6.5] Which of the following is an absolute valuation approach?

- a. Dividend yield
- b. Price-to-earnings ratio
- c. Discounted cash flow
- d. Price-to-book value

Answer and explanation

Answer: C. Discounted cash flow

Rationale: There are essentially three ways of valuing a listed company: net asset valuation, discounting of future cash flows, which is also called the absolute valuation approach, and relative valuation, where comparisons are made against historical trends and other companies in the industry.

216. [Single choice] [CM-EIP §6.6.1] When does book value often represent the floor value of a company?

- a. When the share price is at a record high
- b. When the company has large intangible assets
- c. When the company is growing rapidly
- d. During bad times, if shareholders' equity has not been wiped out by losses

Answer and explanation

Answer: D. During bad times, if shareholders' equity has not been wiped out by losses

Rationale: The book value also represents the accumulation of retained earnings over the years. During bad times, if shareholder's equity has not been wiped out by losses, the book value often represents the floor value.

217. [Single choice] [CM-EIP §6.6.3] When revising asset values to market for a revised net asset value, what else must be checked?

- a. Whether debt and other liabilities also need revising to market value
- b. Whether the dividend has been declared
- c. Whether the company has changed auditors
- d. Whether the share count has changed

Answer and explanation

Answer: A. Whether debt and other liabilities also need revising to market value

Rationale: When we revise the asset values to their respective market values, we should also check whether there is a need to revise debt and other liabilities to market value as well. For example, the market value of USD-denominated debt may have appreciated in the interim.

218. [Single choice] [CM-EIP §6.7] What is the examination expectation for the DCF approach in CM-EIP?

- a. Candidates must compute and derive fair value
- b. Candidates must explain the process and the variables used, but will not be asked to compute fair value
- c. Candidates must memorise the terminal value formula only
- d. DCF is not examinable

Answer and explanation

Answer: B. Candidates must explain the process and the variables used, but will not be asked to compute fair value

Rationale: For the CM-EIP examination, investment representatives are only expected to know how to explain the process of valuing stocks and the variables used but will not be asked to compute and derive the fair value.

219. [Single choice] [CM-EIP §6.7.1] Which DCF variant does the CM-EIP module focus on?

- a. Replacement cost model
- b. Dividend discount model
- c. Free cash flow to firm discounted at the WACC
- d. Residual income model

Answer and explanation

Answer: C. Free cash flow to firm discounted at the WACC

Rationale: For this module, we focus on the Free Cash Flow to Firm approach to DCF valuation, using the Weighted Average Cost of Capital as the discount rate.

220. [Calculation] [CM-EIP §6.7.2] The risk-free rate is 2.33%, the market return is 9.28% and beta is 0.68. What is the cost of equity by CAPM?

- a. 6.31%
- b. 8.65%
- c. 9.28%
- d. 7.056%

Answer and explanation

Answer: D. 7.056%

Rationale: $\mathrm{R} = \mathrm{Rf} + \ (\mathrm{Rm} - \mathrm{Rf}) = 2.33 + 0.68(9.28 - 2.33) = 2.33 + 0.68(6.95) = 2.33 + 4.726 = 7.056\%$.

221. [Calculation] [CM-EIP §6.7.2] Debt is 30.414, equity is 224.17, the after-tax cost of debt is 1.08% and the cost of equity is 7%. What is the WACC?

- a. 6.29%
- b. 5.50%
- c. 7.00%
- d. 6.80%

Answer and explanation

Answer: A. 6.29%

Rationale: Total capital = 30.414 + 224.17 = 254.584. WACC = $0.0108 \times (30.414/254.584) + 0.07 \times (224.17/254.584) = 0.0108 \times 0.1195 + 0.07 \times 0.8805 = 0.00129 + 0.06164 = 0.0629 = 6.29\%$.

222. [Single choice] [CM-EIP §6.7.4] How is the terminal value calculated in the FCFF approach?

- a. The year-10 FCFF multiplied by 10
- b. The year-11 FCFF divided by (WACC − terminal growth rate)
- c. The year-10 FCFF divided by the WACC
- d. The sum of all projected FCFFs

Answer and explanation

Answer: B. The year-11 FCFF divided by (WACC − terminal growth rate)

Rationale: The TV is calculated using the estimated FCFF in the 11th year, divided by (WACC − gc), where gc refers to the terminal growth rate, the rate at which the company's FCFF are expected to grow indefinitely. This lump sum is then discounted back over 10 years.

223. [Single choice] [CM-EIP §6.7.4] What constraint must hold for the terminal value calculation to be valid?

- a. The WACC must equal the risk-free rate
- b. The growth rate must exceed the WACC
- c. The discount rate must be higher than the growth rate
- d. The growth rate must be zero

Answer and explanation

Answer: C. The discount rate must be higher than the growth rate

Rationale: The calculation of terminal value always requires the discount rate to be higher than the growth rate. If the growth rate exceeded the discount rate the terminal value would be negative or infinite.

224. [Calculation] [CM-EIP §6.7.5] Enterprise value is 260,391, debt is 30,514 and preference shares are 1,038. With 2,557.6 million shares, what is the intrinsic share value?

- a. USD 101.80
- b. USD 87.65
- c. USD 91.20
- d. USD 89.47

Answer and explanation

Answer: D. USD 89.47

Rationale: Estimated market value of share capital = 260,391 − 30,514 − 1,038 = 228,839. Intrinsic share value = 228,839 / 2,557.6 = USD 89.47. Against a price of USD 87.65 this implies about 2% upside.

225. [Single choice] [CM-EIP §6.7.6] What is the main criticism of the DCF model?

- a. It depends heavily on the terminal value, which can vary significantly with different assumptions
- b. It cannot be applied to listed companies
- c. It ignores the time value of money
- d. It ignores cash flows

Answer and explanation

Answer: A. It depends heavily on the terminal value, which can vary significantly with different assumptions

Rationale: The DCF model is still heavily dependent on the calculation of the terminal value, which can vary significantly if different assumptions are made. The process of determining fair value is not an exact science but only serves to provide an estimate based on certain assumptions.

226. [Single choice] [CM-EIP §6.8.1] What is the difference between a trailing and a leading P/E ratio?

- a. Trailing uses forecast earnings; leading uses historical earnings
- b. Trailing uses the most recent four quarters of EPS; leading uses consensus forecast earnings for the next year

- c. Trailing is used for bonds; leading for equities
- d. There is no difference

Answer and explanation

Answer: B. Trailing uses the most recent four quarters of EPS; leading uses consensus forecast earnings for the next year

Rationale: A trailing P/E, also known as the current P/E, is the current market price divided by the most recent four quarters of EPS. A leading or prospective P/E divides the current market price by the consensus forecasted earnings for the next year.

227. [Calculation] [CM-EIP §6.8.1] A stock trades at USD 34.68 with FY2016 EPS of USD 2.72. What is the trailing P/E?

- a. 13.5x
- b. 12.1x
- c. 12.8x
- d. 11.7x

Answer and explanation

Answer: C. 12.8x

Rationale: Trailing P/E = 34.68 / 2.72 = 12.8x. The prospective P/E for 2017 using forecast EPS of 2.86 would be 12.1x, and for 2018 using 2.97 would be 11.7x.

228. [Single choice] [CM-EIP §6.8.1] Which factor justifies a lower P/E multiple?

- a. A dominant market position
- b. Strong brand value
- c. High earnings growth
- d. Low transparency and predictability of earnings

Answer and explanation

Answer: D. Low transparency and predictability of earnings

Rationale: When the risks are high, and transparency and predictability of earnings is low, even with high earnings growth, the lower predictability increases risks for investors; hence a lower P/E multiple is justified.

229. [Calculation] [CM-EIP §6.8.2] A stock has a prospective P/E of 44.5 and expected annual EPS growth of 28%. What is its PEG ratio?

- a. 1.6
- b. 0.63
- c. 2.8
- d. 1.59

Answer and explanation

Answer: A. 1.6

Rationale: PEG = P/E / % growth in EPS = 44.5 / 28 = 1.59, approximately 1.6. This is Alibaba's PEG, among the lowest despite having the highest prospective P/E, reflecting its high projected growth.

230. [Single choice] [CM-EIP §6.8.2] What does a PEG ratio below 1 indicate?

- a. The share is overvalued
- b. The share is undervalued, though such stocks are typically hard to find and may carry higher risk
- c. The company is loss-making
- d. The P/E is meaningless

Answer and explanation

Answer: B. The share is undervalued, though such stocks are typically hard to find and may carry higher risk

Rationale: A low PEG ratio indicates an undervalued share. As a rule of thumb, a PEG ratio below 1 would be attractive but typically very hard to find. Such stocks could be relatively new and therefore not actively covered and may also carry a higher level of risk.

231. [Single choice] [CM-EIP §6.8.3] For which type of company is the P/BV ratio best suited?

- a. A company with negative book value
- b. A software company with large intangible assets
- c. A bank, insurer or property company whose value is tied to tangible asset holdings
- d. A start-up with no assets

Answer and explanation

Answer: C. A bank, insurer or property company whose value is tied to tangible asset holdings

Rationale: The P/BV ratio is best suited for companies whose value is tied directly to their asset holdings. Book value is more important for investment companies, banks, insurance companies and property companies, as many of the assets held are reflected in the books and can be liquidated.

232. [Single choice] [CM-EIP §6.8.3] How do hidden reserves affect the calculated P/BV ratio?

- a. They have no effect
- b. They make the ratio meaningless
- c. They make book value overstated, so the true P/BV is higher
- d. They make book value understated, so the true P/BV is lower than calculated

Answer and explanation

Answer: D. They make book value understated, so the true P/BV is lower than calculated

Rationale: In such situations, the book value is understated, and the true P/BV ratio would be lower than what is calculated, making it more likely that the company is undervalued.

233. [Single choice] [CM-EIP §6.8.4] Why is the P/Sales ratio conceptually flawed?

- a. Because the numerator relates to shareholders only, while sales are generated from funds supplied by both shareholders and debtholders
- b. Because it ignores the share price
- c. Because it cannot be computed for listed companies
- d. Because sales are always negative

Answer and explanation

Answer: A. Because the numerator relates to shareholders only, while sales are generated from funds supplied by both shareholders and debtholders

Rationale: Conceptually, the P/Sales ratio is flawed. The numerator, price, relates to shareholders only, even though sales are generated from funds supplied by both shareholders and debtholders. Hence the numerator and denominator are inconsistent.

234. [Calculation] [CM-EIP §6.8.5] Market capitalisation is 18,000, debt is 400, preferred shares are 10, cash is 8,000 and operating cash flow is 2,210. What is the EV/CF ratio?

- a. 3.62
- b. 4.71
- c. 4.53
- d. 8.24

Answer and explanation

Answer: B. 4.71

Rationale: EV/CF = (Market cap + Debt + Preferred − Cash) / Cash flow from operations = (18,000 + 400 + 10 − 8,000) / 2,210 = 10,410 / 2,210 = 4.71. A lower ratio is desirable.

235. [Single choice] [CM-EIP §6.8.6] Why is EBITDA preferred over operating cash flow in valuation?

- a. Because EBITDA is a shareholder concept
- b. Because it is easier to audit
- c. Because operating cash flow can swing widely with working capital changes that may not reflect business performance
- d. Because EBITDA includes interest

Answer and explanation

Answer: C. Because operating cash flow can swing widely with working capital changes that may not reflect business performance

Rationale: EBITDA is preferred over operating cashflows because the latter can experience wide swings due to changes in working capital. Depending on the industry, working capital adjustments may not really reflect business performance.

236. [Single choice] [CM-EIP §6.8.6] Why is EV/EBITDA preferred over P/E?

- a. Because it ignores debt
- b. Because it uses forecast earnings
- c. Because it is easier to calculate
- d. Because it cannot be manipulated by changes in capital structure, allowing fair comparison of companies with different capital structures

Answer and explanation

Answer: D. Because it cannot be manipulated by changes in capital structure, allowing fair comparison of companies with different capital structures

Rationale: Unlike the P/E, the EV/EBITDA ratio cannot be manipulated by changes in capital structure, which makes possible fair comparison of companies with different capital structures. EV considers the impact of debt and other capital on the total value of the firm while price relates only to equity value.

237. [Calculation] [CM-EIP §6.8.8] A share priced at $2.00 pays an annual gross dividend of $0.10. What is the dividend yield?

- a. 2.0%
- b. 5.0%
- c. 10.0%
- d. 20.0%

Answer and explanation

Answer: B. 5.0%

Rationale: Dividend yield = (Dividend per share / Market price per share) × 100 = (0.10 / 2.00) × 100 = 5%.

238. [Calculation] [CM-EIP §6.8.8] A company earns $0.40 per share and pays a dividend of $0.10 per share. What are the payout ratio and retention ratio?

- a. 25% payout, 75% retention
- b. 75% payout, 25% retention
- c. 40% payout, 60% retention
- d. 10% payout, 90% retention

Answer and explanation

Answer: A. 25% payout, 75% retention

Rationale: Dividend payout ratio = 0.10 / 0.40 = 25%. The retention ratio is 1 minus the dividend payout ratio = 75%, indicating the percentage of current earnings retained for reinvestment.

239. [Calculation] [CM-EIP §6.8.8] An analyst expects a 3-cent dividend and requires a 7.5% dividend yield. What is the implied share price?

- a. 31 cents
- b. 40 cents
- c. 22.5 cents
- d. 4 cents

Answer and explanation

Answer: B. 40 cents

Rationale: Implied share price = Dividend / Required yield = 3 cents / 7.5% = 40 cents. This illustrates yield compression: as the price rises, the dividend yield falls.

240. [Single choice] [CM-EIP §6.8.8] Why can REITs be valued using a dividend discount model?

- a. Because they pay no dividends
- b. Because they have no debt
- c. Because they are required to pay out 90% of distributable income and their earnings are relatively predictable
- d. Because they are unlisted

Answer and explanation

Answer: C. Because they are required to pay out 90% of distributable income and their earnings are relatively predictable

Rationale: Real estate investment trusts can be valued using a dividend discount model as they are required to pay out 90% of their distributable income and their earnings are relatively predictable.

241. [Calculation] [CM-EIP §6.8.9] Company A trades at 2.50 with EPS of 0.25 and net cash per share of 0.50. Company B trades at 3.00 with EPS of 0.30 and net debt. Both have a P/E of 10x. Which is undervalued?

- a. Company B, because it has higher EPS
- b. Neither, since both have the same P/E
- c. Company A, because it has a lower share price
- d. Company A, because after netting idle cash the effective P/E is 8x

Answer and explanation

Answer: D. Company A, because after netting idle cash the effective P/E is 8x

Rationale: If Company A is acquired at 2.50 and the idle cash of $0.50 per share is netted out, the buyer is buying the company at a P/E of only 2.00/0.25 = 8x, which is lower than the 10x P/E of Company B. Company A is therefore undervalued.

242. [Single choice] [CM-EIP §6.8.10] What is break-up value in an acquisition?

- a. The expectation that the sum of the parts sold exceeds the target's value as an ongoing concern
- b. The liquidation value of the acquirer
- c. The book value of the target
- d. The value of the target as a going concern

Answer and explanation

Answer: A. The expectation that the sum of the parts sold exceeds the target's value as an ongoing concern

Rationale: There are also acquisitions in which the sum of the parts sold is expected to be higher than the target company's value as an ongoing concern, that is the company is currently undervalued because of how it is managed.

243. [Single choice] [CM-EIP §6.8.11] What is the yield gap?

- a. The difference between a bond's coupon and its yield
- b. The difference between the earnings yield and the bond yield
- c. The difference between the bid and offer price
- d. The difference between two companies' dividend yields

Answer and explanation

Answer: B. The difference between the earnings yield and the bond yield

Rationale: Earnings yields are usually compared against bond yields to assess whether share prices are over- or under-valued. The difference between these yields is referred to as the yield gap.

244. [Single choice] [CM-EIP §6.9] How does listing on SGX Catalist differ from the Mainboard?

- a. Catalist has no listing rules at all
- b. Catalist requires higher quantitative standards
- c. Catalist is supervised by a sponsor and no quantitative requirements are needed
- d. Catalist is only for foreign companies

Answer and explanation

Answer: C. Catalist is supervised by a sponsor and no quantitative requirements are needed

Rationale: Listing on the SGX Mainboard requires the fulfilment of listing requirements spelt out by the exchange. However, listing on Catalist is supervised by a sponsor and no quantitative requirements are needed. Sponsors decide on the suitability of the candidate company.

245. [Single choice] [CM-EIP §6.9] What is the Greenshoe option?

- a. A claw-back of shares from retail investors
- b. A discount to the offer price
- c. A lock-up on pre-IPO shareholders
- d. An over-allotment option letting the issuer increase the offering, used by underwriters to stabilise the price

Answer and explanation

Answer: D. An over-allotment option letting the issuer increase the offering, used by underwriters to stabilise the price

Rationale: The over-allotment option, also known as the Greenshoe option, gives the issuer the right to increase its offering through an over-allotment of shares. IPO underwriters can use this option to stabilise the price of the IPO shares if they perform poorly on initial trading by buying back the additional shares sold from the market.

246. [Single choice] [CM-EIP §6.9] What is a claw-back provision in an IPO?

- a. The re-allocation of shares between institutional and retail investors
- b. A penalty for early selling
- c. A reduction in the offer price
- d. A right to buy back shares from the market

Answer and explanation

Answer: A. The re-allocation of shares between institutional and retail investors

Rationale: The issuer's right to claw back refers to the re-allocation of shares between institutional and retail investors. If institutional take-up is poorer than expected, the claw-back provision may be invoked and retail investors will get a higher allocation of a share that is probably avoided by institutional investors for good reason.

247. [Single choice] [CM-EIP §6.9] **What document does a company lodge with MAS for a rights issue after listing, rather than a prospectus?**

- a. A Product Highlights Sheet
- b. An Offer Information Statement
- c. A trust deed
- d. A circular to shareholders

Answer and explanation

Answer: B. An Offer Information Statement

Rationale: In a rights issue, the company will lodge an Offer Information Statement instead of a prospectus with the MAS.

248. [Calculation] [CM-EIP §6.10] **Q-Star has sales of USD 7.65bn and the discounted EV/Sales multiple is 3.53. With debt of USD 11bn and 750m shares, what is the target price?**

- a. USD 12.67
- b. USD 27.00
- c. USD 21.33
- d. USD 14.25

Answer and explanation

Answer: C. USD 21.33

Rationale: Firm value = 7.65 × 3.53 = USD 27.00bn. Market capitalisation = 27.00 − 11.00 = USD 16.00bn. Price = 16.00bn / 750m = USD 21.33.

249. [Single choice] [CM-EIP §6.11.1] **What is the main difference between a non-voting depositary receipt and direct investment in the issuer's shares?**

- a. NVDRs are not transferable
- b. NVDRs are only available to institutions
- c. NVDRs pay no dividends
- d. NVDRs carry no voting rights, so foreign investors cannot participate in corporate decisions

Answer and explanation

Answer: D. NVDRs carry no voting rights, so foreign investors cannot participate in corporate decisions

Rationale: The main difference between investing indirectly through an NVDR and directly in the issuer's shares is that NVDRs do not come with voting rights and thus do not allow foreign investors to participate in corporate decisions. This feature enables foreign investment in firms that would otherwise be prohibited.

250. [Single choice] [CM-EIP §6.11.2] **Which level of ADR allows the issuer to raise new capital?**

- a. Level 3
- b. All levels equally
- c. Level 1
- d. Level 2

Answer and explanation

Answer: A. Level 3

Rationale: Level 1 ADRs trade in the traditional US OTC market and are not listed. Level 2 ADRs are listed on a US securities exchange but do not involve capital raising. Level 3 ADRs use a listed ADR programme to offer new shares to US investors in a capital-raising exercise.

251. [Single choice] [CM-EIP §6.11.3] **Why might an ADR trade at a premium to its underlying shares?**

- a. Because ADRs pay higher dividends
- b. Because of regulatory restrictions in the issuer's home market on foreign holdings of the shares

- c. Because ADRs are unregulated
- d. Because the depositary bank guarantees the price

Answer and explanation

Answer: B. Because of regulatory restrictions in the issuer's home market on foreign holdings of the shares

Rationale: ADRs often trade at a premium to their underlying shares because of regulatory restrictions in the issuer's home market on foreign holdings of the shares.

252. [Single choice] [CM-EIP §6.13.1] What is the main practical difficulty with the general dividend discount model?

- a. It requires the share price
- b. It ignores the time value of money
- c. Investors must forecast and discount an infinite stream of dividends
- d. It cannot handle growth

Answer and explanation

Answer: C. Investors must forecast and discount an infinite stream of dividends

Rationale: The main difficulty with implementing the equation as a valuation model is that investors must forecast and discount an infinite stream of dividends, both insurmountable tasks. The model can be modified to account for a subsequent sale of the shares.

253. [Calculation] [CM-EIP §6.13.1] Dividends of $1.00, $1.05 and $1.10 are expected over three years, with a sale price of $10 at the end of year 3. At a 12% required return, what is the share value?

- a. $10.00
- b. $8.75
- c. $11.20
- d. $9.63

Answer and explanation

Answer: D. $9.63

Rationale: $P0 = 1.00/1.12 + 1.05/1.12^2 + 1.10/1.12^3 + 10.00/1.12^3 = 0.8929 + 0.8371 + 0.7830 + 7.1178 = \$9.63$.

254. [Calculation] [CM-EIP §6.13.2] A company pays a dividend of $0.20, expected to grow at 5% a year. Investors require 12%. What is the estimated price?

- a. $3.00
- b. $2.86
- c. $4.00
- d. $1.67

Answer and explanation

Answer: A. $3.00

Rationale: $P0 = D1 / (R - g) = (0.20 \times 1.05) / (0.12 - 0.05) = 0.21 / 0.07 = \$3.00$. The current dividend must be compounded for one period because the formula uses the dividend expected one period from now.

255. [Calculation] [CM-EIP §6.13.2] Using the same company, if the required return rises from 12% to 13%, what is the new estimated price?

- a. $2.10
- b. $2.63
- c. $3.00
- d. $3.53

Answer and explanation

Answer: B. $2.63

Rationale: P0 = 0.21 / (0.13 − 0.05) = 0.21 / 0.08 = $2.63. A 1% change in R produces a large difference in estimated value, which is the model's central weakness.

256. [Calculation] [CM-EIP §6.13.3] A firm retains 60% of earnings and has an ROE of 15%. What is its sustainable growth rate?

- a. 6%
- b. 9%
- c. 15%
- d. 25%

Answer and explanation

Answer: B. 9%

Rationale: g = Retention Rate × ROE = 0.6 × 0.15 = 0.09 = 9%. A firm can increase its sustainable growth rate by increasing its retention rate or by increasing its ROE.

257. [Calculation] [CM-EIP §6.13.3] P&G has an ROE of 17.36% and a payout ratio of 0.70. What is its estimated growth rate?

- a. 7.00%
- b. 5.21%
- c. 12.15%
- d. 17.36%

Answer and explanation

Answer: B. 5.21%

Rationale: g = ROE × (1 − payout) = 0.1736 × (1 − 0.70) = 0.1736 × 0.30 = 0.0521 = 5.21%.

258. [Single choice] [CM-EIP §6.13.4] In the two-stage DDM, what does the second term represent?

- a. The current dividend
- b. The present value of dividends in the high-growth period
- c. The constant-growth value at the end of the abnormal growth period, discounted back to today
- d. The terminal dividend

Answer and explanation

Answer: C. The constant-growth value at the end of the abnormal growth period, discounted back to today

Rationale: The second term is the constant-growth version, which divides the dividend expected at the end of period n+1 by the difference between R and g. Since the resulting figure is a present value at the end of period n, it must be discounted back to time period zero.

259. [Calculation] [CM-EIP §6.13.4] In the P&G two-stage example, the terminal value contributes 71.5797 of a total value of 84.15. What proportion of value comes from the terminal value?

- a. About 95%
- b. About 50%
- c. About 65%
- d. About 85%

Answer and explanation

Answer: D. About 85%

Rationale: 71.5797 / 84.15 = 0.8506, about 85%. Because 85% of the value is determined by the terminal value, changes that affect the terminal value still affect the model significantly.

260. [Single choice] [CM-EIP §6.13.4] For which companies is the multi-stage DDM most useful?

- a. Mature companies with very high dividend payouts of 70–120% of EPS
- b. Companies with negative earnings
- c. Companies with no dividend history
- d. Early-stage technology companies with no dividends

Answer and explanation

Answer: A. Mature companies with very high dividend payouts of 70–120% of EPS

Rationale: The accuracy and effectiveness of the dividend valuation model is better for stocks of companies that are in the mature phase of their life cycles and that have very high dividend payouts, for example 70–120% of EPS. Hence the DDM tends to be more useful for gaming companies, tobacco companies and other select mature companies.

261. [Calculation] [CM-EIP §6.13.5] A preference share has a $100 par value and pays $7 a year. At an 8% required return, what is its intrinsic value?

- a. $56.00
- b. $87.50
- c. $100.00
- d. $114.29

Answer and explanation

Answer: B. $87.50

Rationale: Because the dividend is a perpetuity, P0 = Dividend / Discount Rate = $7 / 0.08 = $87.50.

Chapter 7 — Deposits and Fixed-Income Securities

262. [Single choice] [CM-EIP §7.1] Why was keeping excessive cash from 2008 to 2017 non-optimal?

- a. Because interest rates were high
- b. Because cash was illegal to hold
- c. Because of opportunity costs and loss of purchasing power
- d. Because deposit insurance was withdrawn

Answer and explanation

Answer: C. Because of opportunity costs and loss of purchasing power

Rationale: Cash is often classified as a non-volatile, safe asset class. However, for that reason, cash pays the lowest return. Keeping excessive cash from 2008 to 2017 could be non-optimal because of the opportunity costs and loss of purchasing power.

263. [Single choice] [CM-EIP §7.2.1] When was the Singapore Deposit Insurance Scheme introduced, and what is the coverage limit from 1 April 2024?

- a. 1 April 2019; S$75,000
- b. 1 April 2024; S$75,000
- c. 1 April 2006; S$50,000
- d. 1 April 2006; S$100,000

Answer and explanation

Answer: D. 1 April 2006; S$100,000

Rationale: The DI Scheme was introduced in Singapore on 1 April 2006. The coverage limit was increased from $S50,000 to S75,000 on 1 April 2019, and further revised to S$100,000 effective 1 April 2024.

264. [Multiple response] [CM-EIP §7.2.1] Which of the following are NOT insured by the Singapore Deposit Insurance Scheme? (Select all that apply)

- a. Foreign currency deposits

- b. Structured deposits
- c. Unit trusts and shares
- d. Singapore dollar savings accounts

Answer and explanation

Answer: A, B, C

Rationale: The scheme is only valid for Singapore dollar-denominated deposits placed with a DI member. Foreign currency deposits, structured deposits and investment products such as unit trusts, shares and other securities are not insured.

265. [Calculation] [CM-EIP §7.2.1] An individual has S$300,000 of Singapore dollar deposits and S$100,000 of CPFIS deposits with one bank, plus USD 250,000 in a currency investment. How much is insured?

- a. S$200,000
- b. S$400,000
- c. S$650,000
- d. S$100,000

Answer and explanation

Answer: A. S$200,000

Rationale: For accounts with a single bank, the scheme covers only up to S$100,000 of the aggregate S$ deposit balance. CPF deposits invested with the bank receive separate cover of up to S$100,000. The USD currency investment is not in S$200,000.

266. [Single choice] [CM-EIP §7.3] Which of the following is a money market instrument rather than a capital market instrument?

- a. A 10-year government bond
- b. A repurchase agreement
- c. A convertible bond
- d. An asset-backed security

Answer and explanation

Answer: B. A repurchase agreement

Rationale: Money market instruments have maturities of one year or less and include Treasury bills, commercial paper, negotiable certificates of deposit, banker's acceptances, repurchase agreements and trade bills.

267. [Single choice] [CM-EIP §7.4] What is the typical minimum denomination of a money market instrument?

- a. S$1,000 and above
- b. S$10,000 and above
- c. S$250,000 and above
- d. S$1 million and above

Answer and explanation

Answer: C. S$250,000 and above

Rationale: Owing to their relatively short maturity, money market instruments have relatively low risks. However, the minimum denomination of money market instruments is usually relatively large, at S$250,000 and above.

268. [Single choice] [CM-EIP §7.4] Why do money market instruments carry more reinvestment risk than longer-dated bonds?

- a. Because they are unsecured
- b. Because they are illiquid
- c. Because they pay higher coupons

- d. Because they are short-term, requiring the investor to reinvest proceeds each time the instrument matures

Answer and explanation

Answer: D. Because they are short-term, requiring the investor to reinvest proceeds each time the instrument matures

Rationale: Money market instruments are subject to more reinvestment risk because they are short-term instruments, requiring the investor to reinvest his investment proceeds each time the investment matures.

269. [Single choice] [CM-EIP §7.4.1] Why is the T-bill yield used as a benchmark for the risk-free rate?

- a. Because they are obligations of the government and considered free of default risk
- b. Because they are issued by banks
- c. Because they have the longest maturity
- d. Because T-bills pay the highest yield

Answer and explanation

Answer: A. Because they are obligations of the government and considered free of default risk

Rationale: T-bills issued by a creditworthy government are the safest investments and are generally considered to be risk-free. Risk-free here means that there is no risk of defaulting on interest or principal payments when due. The yield on T-bills is usually used as a benchmark for risk-free rates.

270. [Single choice] [CM-EIP §7.4.3] Why can only corporations with very strong credit ratings issue commercial paper?

- a. Because commercial paper is secured
- b. Because commercial paper is an unsecured promissory note
- c. Because commercial paper is exchange-traded
- d. Because commercial paper has a long maturity

Answer and explanation

Answer: B. Because commercial paper is an unsecured promissory note

Rationale: Commercial papers are short-dated unsecured promissory notes issued by a corporation. Due to their unsecured nature, only corporations with very strong credit ratings are able to issue commercial papers.

271. [Single choice] [CM-EIP §7.4.4] What is a repurchase agreement economically?

- a. A forward contract
- b. An unsecured loan
- c. A collateralised short-term loan, with the money market instrument as collateral
- d. An equity investment

Answer and explanation

Answer: C. A collateralised short-term loan, with the money market instrument as collateral

Rationale: A repo is the sale of a money market instrument, with a commitment by the seller to buy the security back at a specified price on a future date. Effectively it is a collateralised short-term loan, where the collateral is the money market instrument.

272. [Single choice] [CM-EIP §7.5] What maturity range defines medium-term notes?

- a. More than 10 years
- b. More than 30 years
- c. Up to one year
- d. More than one year and up to 10 years

Answer and explanation

Answer: D. More than one year and up to 10 years

Rationale: The debt securities market may be divided into three maturity segments: money market instruments maturing within one year, medium-term notes with maturities of more than one year and up to 10 years, and long-term bonds with maturities longer than 10 years.

273. [Single choice] [CM-EIP §7.5.1] What is the major disadvantage of fixed-coupon bonds?

- a. They do not adjust for inflation, which erodes purchasing power and can cause price fluctuations
- b. They are always illiquid
- c. They cannot be sold before maturity
- d. They pay no income

Answer and explanation

Answer: A. They do not adjust for inflation, which erodes purchasing power and can cause price fluctuations

Rationale: A major disadvantage of investing in fixed-income securities with fixed coupons and principal repayment is that these securities do not adjust for inflation. Inflation can erode the purchasing power of cash flows and lead to swings in interest rates, producing price fluctuations and capital losses if not held to maturity.

274. [Single choice] [CM-EIP §7.5.2] What is the difference between a term bond and a serial bond?

- a. A term bond is exchange-traded; a serial bond is OTC
- b. A term bond has a single maturity date; a serial bond has a series of maturity dates
- c. A term bond is unsecured; a serial bond is secured
- d. A term bond is floating rate; a serial bond is fixed rate

Answer and explanation

Answer: B. A term bond has a single maturity date; a serial bond has a series of maturity dates

Rationale: Most corporate bonds are typically term bonds, which have a single maturity date. In contrast, serial bonds have a series of maturity dates and are essentially bundles of bonds issued at the same time, but with different maturities.

275. [Calculation] [CM-EIP §7.6.1] A T-bill with 90 days to maturity, face value $100,000, sells for $98,000. What is the bank discount yield?

- a. 7.35%
- b. 2.04%
- c. 8.00%
- d. 8.16%

Answer and explanation

Answer: C. 8.00%

Rationale: D = 100,000 − 98,000 = 2,000. Yd = (D/F) × (360/t) = (2,000/100,000) × (360/90) = 0.02 × 4 = 0.08 = 8.0%.

276. [Single choice] [CM-EIP §7.6.1] Why is the bank discount yield not a meaningful measure of return?

- a. Because it uses a 365-day year
- b. Because it ignores the coupon
- c. Because it ignores the maturity date
- d. Because it is calculated based on face value rather than the actual dollar amount invested

Answer and explanation

Answer: D. Because it is calculated based on face value rather than the actual dollar amount invested

Rationale: Though it is widely used, the quoted yield on a bank discount basis is not a meaningful measure of return from holding a T-bill because it is calculated based on the face value rather than on the actual dollar amount invested.

277. [Calculation] [CM-EIP §7.6.2] A corporate bond is quoted at 95¾ with a par value of $1,000. What is the price?

- a. $957.50
- b. $955.75
- c. $975.00
- d. $950.00

Answer and explanation

Answer: A. $957.50

Rationale: Prices are quoted as a percentage of par value. A quote of 95¾ represents 95.75% of $1,000 = $957.50. Corporate bonds are quoted in eighths of a percentage point.

278. [Calculation] [CM-EIP §7.6.2] An 8% bond pays semi-annual interest on 1 June and 1 December and is sold on 1 September. What accrued interest is the seller entitled to?

- a. $20
- b. $40
- c. $60
- d. $80

Answer and explanation

Answer: A. $20

Rationale: The annual coupon is $80, paid semi-annually as $40. The seller has held the bond for three months since the last coupon, so accrued interest = $80 × 3/12 = $20.

279. [Single choice] [CM-EIP §7.7.1] Why does the Singapore Government issue Singapore Government Securities if it runs a balanced budget?

- a. To fund infrastructure spending
- b. To build a liquid market and establish a benchmark yield curve for pricing Singapore dollar private debt
- c. To finance the CPF
- d. To control the money supply

Answer and explanation

Answer: B. To build a liquid market and establish a benchmark yield curve for pricing Singapore dollar private debt

Rationale: The Singapore Government does not need to finance its expenditures through the issuance of government bonds as it operates on a balanced budget policy. SGS issuance serves primarily to build a liquid SGS market to establish a benchmark yield curve for the pricing of Singapore dollar private debt securities, encourage an active secondary market, and attract issuers and investors.

280. [Single choice] [CM-EIP §7.7.2] What are the three defining features of a Eurobond?

- a. Issued by a European company, sold in Europe, settled in euros
- b. Rated AAA, unsecured, long-dated
- c. Underwritten by an international syndicate, offered simultaneously in several countries, issued outside any single jurisdiction
- d. Denominated in euros, sold in Europe, listed in London

Answer and explanation

Answer: C. Underwritten by an international syndicate, offered simultaneously in several countries, issued outside any single jurisdiction

Rationale: A Eurobond is a bond that is underwritten by an international syndicate, offered at issuance simultaneously to investors in a number of countries, and issued outside of the jurisdiction of any single country.

281. [Single choice] [CM-EIP §7.7.3] What important function does the secondary bond market perform beyond trading?

- a. It guarantees bond payments
- b. It regulates issuers
- c. It sets the coupon on new issues
- d. It benchmarks the market values and yields of traded bonds, which is needed for valuation and for establishing benchmark yield curves

Answer and explanation

Answer: D. It benchmarks the market values and yields of traded bonds, which is needed for valuation and for establishing benchmark yield curves

Rationale: Other than being a place where bonds can be bought and sold, the secondary market also serves the important market function of benchmarking the market values and yields of the traded bonds. Such information is necessary for the valuation of bond portfolios and for establishing benchmark yield curves.

282. [Single choice] [CM-EIP §7.8.1] Why are floating rate bonds attractive to some institutional investors?

- a. Because the income stream more closely matches the floating nature of some of their liabilities
- b. Because they have no credit risk
- c. Because they are tax-exempt
- d. Because they pay a fixed coupon

Answer and explanation

Answer: A. Because the income stream more closely matches the floating nature of some of their liabilities

Rationale: Floating rate securities are attractive to some institutional investors because the securities allow investors to buy an asset with an income stream that more closely matches the floating nature of some of their liabilities.

283. [Single choice] [CM-EIP §7.8.2] What is the most attractive feature of a zero-coupon bond?

- a. High current income
- b. No reinvestment risk within the life of the bond, so the return is locked in if held to maturity
- c. Low price volatility
- d. Call protection

Answer and explanation

Answer: B. No reinvestment risk within the life of the bond, so the return is locked in if held to maturity

Rationale: Perhaps the most attractive feature of zero-coupon bonds is that there is no reinvestment risk within the life of the bond because there is no coupon to reinvest, so the rate of return is locked in when they are held to maturity.

284. [Single choice] [CM-EIP §7.8.2] Which is a disadvantage of zero-coupon bonds?

- a. They have no maturity value
- b. They pay coupons that must be reinvested
- c. They experience high price volatility, and a rise in interest rates affects them more than a coupon bond
- d. They cannot be sold before maturity

Answer and explanation

Answer: C. They experience high price volatility, and a rise in interest rates affects them more than a coupon bond

Rationale: Zero-coupon bonds experience high price volatility: a rise in interest rates will negatively impact the price of a zero-coupon bond much more than that of a coupon bond. Liquidity may also be limited, and call features can limit appreciation.

285. [Single choice] [CM-EIP §7.8.3] Why are convertible bond yields lower than comparable straight bond yields?

- a. Because convertibles have shorter maturities
- b. Because convertibles are tax-exempt
- c. Because convertibles are unsecured
- d. Because the bondholder effectively pays for the conversion privilege through the lower yield

Answer and explanation

Answer: D. Because the bondholder effectively pays for the conversion privilege through the lower yield

Rationale: Yields on convertible bonds are generally lower than those on comparable straight bonds because of the conversion privilege, which the bondholder effectively pays for through the lower yield.

286. [Single choice] [CM-EIP §7.8.4] What is a covered bond?

- a. A highly rated paper backed by bank assets, typically costing less to issue than unsecured bonds
- b. A bond with an insurance wrapper
- c. A bond convertible into shares
- d. A bond secured by a government guarantee

Answer and explanation

Answer: A. A highly rated paper backed by bank assets, typically costing less to issue than unsecured bonds

Rationale: Generally, a covered bond is a highly rated paper, backed by bank assets, and typically costs less to issue than unsecured bonds. Covered bonds have a long history in Europe dating to 1769 and have never experienced a default.

287. [Single choice] [CM-EIP §7.8.5] What rating threshold separates high-yield bonds from investment grade?

- a. Below A
- b. Below BBB
- c. Below BB
- d. Below CCC

Answer and explanation

Answer: B. Below BBB

Rationale: Bonds with a credit rating below triple B, BBB, are called high-yield bonds or junk bonds. High-yield bonds are securities that trade primarily on their creditworthiness as opposed to the level of interest rates.

288. [Single choice] [CM-EIP §7.8.6] Which deferred-coupon structure gives the issuer the option to pay cash or issue a similar bond?

- a. Deferred-interest bond
- b. Step-up bond
- c. Payment-in-kind bond
- d. Perpetual bond

Answer and explanation

Answer: C. Payment-in-kind bond

Rationale: Payment-in-kind bonds give the issuer an option to pay cash at a coupon payment date or give the bondholder a similar bond, with the same coupon and a par value equal to the amount of the coupon payment that would have been paid. The choice period varies from five to ten years.

289. [Single choice] [CM-EIP §7.8.6] How do perpetual bonds rank in the order of discharge on default?

- a. Equal to equities
- b. After preference shares but before ordinary shares

- c. After equities
- d. Before equities

Answer and explanation

Answer: D. Before equities

Rationale: Perpetual bonds rank before equities in the order of discharge in the event of default. They pay a higher coupon rate to compensate the investor for an indefinite holding period and the correspondingly higher interest rate risk.

290. [Single choice] [CM-EIP §7.9.1] Why do callable bonds carry higher yields than equivalent non-callable bonds?

- a. Because the investor risks losing a high-coupon bond when rates decline and the upside price potential is limited
- b. Because they are unsecured
- c. Because they are less liquid
- d. Because they have longer maturities

Answer and explanation

Answer: A. Because the investor risks losing a high-coupon bond when rates decline and the upside price potential is limited

Rationale: The call provision is unattractive to investors who run the risk of losing a high-coupon bond when interest rates decline. The prospect of a call also limits the upside price potential of the bond in a falling interest rate environment. Due to these disadvantages, bonds with call features carry higher yields.

291. [Single choice] [CM-EIP §7.9.1] How does the call premium behave over the life of a bond?

- a. It rises towards maturity
- b. It tends to be higher in the early life of the bond and declines gradually towards par as maturity approaches
- c. It is constant
- d. It is zero throughout

Answer and explanation

Answer: B. It tends to be higher in the early life of the bond and declines gradually towards par as maturity approaches

Rationale: The call price is normally higher than the par value of the bond. The difference between the call price and par value is the call premium, which tends to be higher in the early life of the bond and declines gradually towards par as the bond approaches maturity.

292. [Single choice] [CM-EIP §7.9.2] What is the purpose of a sinking-fund provision?

- a. To convert the bond into shares
- b. To increase the coupon rate
- c. To reduce credit risk by requiring the issuer to retire a certain amount of debt each year
- d. To allow the holder to sell back at par

Answer and explanation

Answer: C. To reduce credit risk by requiring the issuer to retire a certain amount of debt each year

Rationale: Corporate bond indentures may require the issuer to retire a certain amount of the outstanding debt each year. The purpose of a sinking-fund provision is to reduce credit risk.

293. [Single choice] [CM-EIP §7.9.3] Why does a put provision lower a bond's yield?

- a. Because it extends the maturity
- b. Because it removes the coupon
- c. Because it increases credit risk

- d. Because the investor has effectively bought a put option from the issuer and pays for it with a lower yield

Answer and explanation

Answer: D. Because the investor has effectively bought a put option from the issuer and pays for it with a lower yield

Rationale: Puttable bonds give the bondholder the right to sell the bonds back to the issuer at par value on designated dates. As with a convertible bond, the investor has effectively bought a put option from the issuer and will pay for it with a lower yield.

294. [Single choice] [CM-EIP §7.9.4] Which bondholders have a claim subordinated to secured debt and after the main creditors?

- a. Subordinated (junior) bondholders
- b. Income bondholders
- c. Debenture holders
- d. Mortgage bondholders

Answer and explanation

Answer: A. Subordinated (junior) bondholders

Rationale: Subordinated (junior) bondholders have a claim on the income and assets of the issuer that is subordinated to secured debt and after the main creditors have made their claims. Income bonds are junior debt as the issuing firm can omit or delay interest if earnings are too low.

295. [Single choice] [CM-EIP §7.10] What is the primary difference between asset-backed securities and mortgage-backed securities?

- a. ABS are exchange-traded; MBS are OTC
- b. MBS are created from pooling mortgages, while ABS are created from pooling non-mortgage assets
- c. ABS are government-guaranteed
- d. MBS have no credit risk

Answer and explanation

Answer: B. MBS are created from pooling mortgages, while ABS are created from pooling non-mortgage assets

Rationale: The primary difference between asset-backed securities and mortgage-backed securities is the underlying asset that is being securitised. MBS are created from the pooling of mortgages whereas ABS evolved from MBS and are created from the pooling of non-mortgage assets.

296. [Single choice] [CM-EIP §7.10.1] What is a collateralised loan obligation?

- a. A government-guaranteed loan
- b. A CDO backed by a pool of bonds
- c. A CDO backed by a pool of loans
- d. A loan secured by collateral

Answer and explanation

Answer: C. A CDO backed by a pool of loans

Rationale: CDOs that are backed by a pool of bonds are also known as collateralised bond obligations, while those that are backed by a pool of loans are known as collateralised loan obligations.

297. [Single choice] [CM-EIP §7.11] What is credit risk?

- a. The risk that the bond is called
- b. The risk that the bond is illiquid
- c. The risk that interest rates rise
- d. The risk that the issuer will be unable to make timely principal and interest payments as agreed

Answer and explanation

Answer: D. The risk that the issuer will be unable to make timely principal and interest payments as agreed

Rationale: Credit risk or default risk refers to the risk that the issuer of the bond will be unable to make timely principal and interest payments as agreed in the terms of issue.

298. [Single choice] [CM-EIP §7.11.1] Which rating indicates selective default on a specific issue while other obligations continue to be met?

- a. SD
- b. R
- c. WR
- d. D

Answer and explanation

Answer: A. SD

Rationale: An SD rating is assigned when S&P believes that the obligor has selectively defaulted on a specific issue or class of obligations, but it will continue to meet its payment obligations on other issues or classes of obligations in a timely manner. D indicates a general default.

299. [Single choice] [CM-EIP §7.11.1] What is the highest short-term issuer credit rating category in the S&P scale?

- a. AAA
- b. A-1
- c. A+
- d. B

Answer and explanation

Answer: B. A-1

Rationale: An obligor rated A-1 has strong capacity to meet its financial commitments and is rated in the highest category by S&P. Within this category, certain obligors are designated with a plus sign, indicating extremely strong capacity.

300. [Calculation] [CM-EIP §7.11.2] A 10-year Treasury yields 2% and a 10-year corporate bond yields 4%. What is the credit spread?

- a. 2 basis points
- b. 20 basis points
- c. 200 basis points
- d. 2,000 basis points

Answer and explanation

Answer: C. 200 basis points

Rationale: The spread is $4\% - 2\% = 2\% = 200$ basis points. A credit spread is the rate difference, usually quoted in basis points, between two bonds of the same maturity but of differing credit quality.

301. [Single choice] [CM-EIP §7.11.2] What happens to credit spreads during a crisis?

- a. They are unchanged
- b. They become negative
- c. They narrow
- d. They widen against the benchmark

Answer and explanation

Answer: D. They widen against the benchmark

Rationale: Credit spreads will widen against the benchmark during crises and narrow when market conditions, and investors' risk appetites, improve. As the issuer's default risk increases, its bond price falls and its bond spread widens.

302. [**Single choice**] [**CM-EIP §7.11.3**] **Why might an Asian company choose not to be rated?**

- a. Because it believes the main investors who would buy its bonds know its business and history
- b. Because unrated bonds pay lower yields
- c. Because ratings are illegal in Asia
- d. Because rating agencies refuse to rate Asian companies

Answer and explanation

Answer: A. Because it believes the main investors who would buy its bonds know its business and history

Rationale: Some companies in Asia have chosen not to be rated by rating agencies. They do not see a need for it because they believe the main investors who would buy their bonds know their business and history. An unrated company may not necessarily be of poor credit.

303. [**Multiple response**] [**CM-EIP §7.11.4**] **Which of the following can mitigate poor credit quality of a bond issue? (Select all that apply)**

- a. Creating a dedicated sinking fund
- b. Offering specified assets as collateral
- c. Offering parental guarantees
- d. Removing the trustee

Answer and explanation

Answer: A, B, C

Rationale: Companies can improve credit quality by creating a dedicated sinking fund, offering specified assets as collateral or security, offering parental guarantees, working with a financial institution to repackage the bonds with higher-quality bonds, or obtaining credit insurance or credit default swaps.

304. [**Single choice**] [**CM-EIP §7.11.5**] **In a credit default swap, who pays a lump sum on default?**

- a. The buyer of the CDS
- b. The seller of the CDS
- c. The reference entity
- d. The clearing house

Answer and explanation

Answer: B. The seller of the CDS

Rationale: The buyer of a CDS pays a premium to be insured against a debt default. The seller of a CDS receives regular payments from the buyer to indemnify the buyer with a lump-sum payment in the event of a debt default.

305. [**Single choice**] [**CM-EIP §7.11.5**] **Which institution was bailed out because it had sold multiple CDS on the same mortgage-backed securities?**

- a. Fannie Mae
- b. Lehman Brothers
- c. AIG
- d. Bear Stearns

Answer and explanation

Answer: C. AIG

Rationale: Insurers that sold multiple CDS on the MBS found themselves unable to provide the protection that they sold, exemplified by the bailout of AIG. Lehman Brothers was a purchaser of the securities that went bankrupt.

306. [**Single choice**] [**CM-EIP §7.11.6**] **Which of the following is a red flag for credit analysis?**

- a. An increase in cash flow from operations
- b. A decline in the debt level

- c. Rising sales with stable costs
- d. A non-diversified revenue stream coupled with the loss of major clients

Answer and explanation

Answer: D. A non-diversified revenue stream coupled with the loss of major clients

Rationale: Projected revenue falls include a non-diversified revenue stream coupled with the loss of major clients. Other red flags include cost rises, regulatory restrictions, falling operating cash flow and the departure of key officers.

307. [Single choice] [CM-EIP §7.11.7] At what price range might a distressed bond offer a worthwhile recovery opportunity?

- a. 0.30–0.40 on the dollar
- b. 0.01–0.05 on the dollar
- c. At par
- d. 0.90–0.99 on the dollar

Answer and explanation

Answer: A. 0.30–0.40 on the dollar

Rationale: A distressed bond priced at 0.30–0.40 on the dollar may provide for potential to recover a significant portion of its value when key variables normalise, especially if the challenges relate to a short-term credit squeeze or cyclical downturn.

308. [Single choice] [CM-EIP §7.12.1] What is the relationship between interest rates and bond prices?

- a. Direct
- b. Inverse
- c. Unrelated
- d. Exponential

Answer and explanation

Answer: B. Inverse

Rationale: There is an inverse relationship between interest rates and bond prices. As interest rates rise, the prices of bonds fall, and as rates fall, prices rise. This is also referred to as delta risk.

309. [Single choice] [CM-EIP §7.12.2] For which bonds is reinvestment risk greatest?

- a. Convertible bonds
- b. Zero-coupon bonds with short maturities
- c. Higher coupon bonds with longer holding periods
- d. Floating rate bonds

Answer and explanation

Answer: C. Higher coupon bonds with longer holding periods

Rationale: Reinvestment risk is greater for higher coupon bonds and longer holding periods. A zero-coupon bond has no reinvestment risk if held to maturity because there is nothing to reinvest.

310. [Single choice] [CM-EIP §7.12.4] Which type of inflation does inflation risk concern?

- a. Deflation only
- b. Hyperinflation only
- c. Anticipated inflation
- d. Unanticipated inflation

Answer and explanation

Answer: D. Unanticipated inflation

Rationale: Inflation risk is actually concerned with unanticipated inflation. If inflation is anticipated and has been factored into the price of the bond, then investors are adequately compensated for anticipated inflation.

311. [Single choice] [CM-EIP §7.12.5] What indicates the marketability of a bond?

- a. The size of the spread between the bid and the offer quoted by the dealer
- b. The credit rating
- c. The maturity date
- d. The size of the coupon

Answer and explanation

Answer: A. The size of the spread between the bid and the offer quoted by the dealer

Rationale: An indication of the marketability of a security is the size of the spread between the bid and the offer quoted by the dealer. The greater the spread, the greater the marketability risk.

312. [Single choice] [CM-EIP §7.12.7] What is vega risk?

- a. The risk that interest rates rise
- b. The risk that a change in expected interest rate volatility adversely affects the price of a security with an embedded option
- c. The risk of default
- d. The risk of early redemption

Answer and explanation

Answer: B. The risk that a change in expected interest rate volatility adversely affects the price of a security with an embedded option

Rationale: The price of a bond that has an embedded option depends on the level of interest rates and other factors that influence the value of that embedded option, one of which is the expected volatility of interest rates, also referred to as vega risk.

313. [Calculation] [CM-EIP §7.13] A 10-year bond with par $1,000 and an 8% coupon, with a required yield of 6% and annual coupons, is priced at approximately:

- a. $920.00
- b. $1,000.00
- c. $1,147.20
- d. $1,148.77

Answer and explanation

Answer: C. $1,147.20

Rationale: Discounting eight $80 coupons and the $1,000 principal at 6% gives $1,147.20. With semi-annual coupons the price would be $1,148.77, because the number of periods is doubled and the discount rate halved.

314. [Calculation] [CM-EIP §7.13.1] A zero-coupon bond matures in 15 years with par $1,000, at a required return of 9%. What is its price?

- a. $274.54
- b. $1,000.00
- c. $641.86
- d. $267.00

Answer and explanation

Answer: D. $267.00

Rationale: P = M / (1 + k/2)^(2n) = 1,000 / (1.045)^30 = 1,000 / 3.7453 = $267.00. Even without coupons, the semi-annual rate of 4.5% and the half-year period are used to stay consistent with semi-annual coupon bonds.

315. [Single choice] [CM-EIP §7.14] A 10-year 12% semi-annual coupon bond is priced at 112.46% of par. What is the required yield?

- a. 8%
- b. 10%
- c. 12%
- d. 14%

Answer and explanation

Answer: B. 10%

Rationale: The price-yield table shows that at a required yield of 10% the bond value is 112.46%, at 12% it is 100.00% and at 14% it is 89.41%. A price above par means the coupon exceeds the yield.

316. [Single choice] [CM-EIP §7.15] When does a bond trade at a premium?

- a. When the coupon rate is higher than the market yield
- b. When the coupon rate equals the market yield
- c. When the bond is close to maturity
- d. When the coupon rate is lower than the market yield

Answer and explanation

Answer: A. When the coupon rate is higher than the market yield

Rationale: When the coupon rate is higher than the market yield, the bond will trade at a premium. When the coupon rate is lower than the market yield, the bond will trade at a discount. When they are equal, the bond trades at par.

317. [Single choice] [CM-EIP §7.16.1] Why does a longer maturity increase a bond's price sensitivity to interest rates?

- a. Because longer bonds pay higher coupons
- b. Because longer-term cash flows are more sensitive to changes in interest rates
- c. Because longer bonds are less liquid
- d. Because longer bonds have more credit risk

Answer and explanation

Answer: B. Because longer-term cash flows are more sensitive to changes in interest rates

Rationale: The longer the maturity of a bond, the greater the impact of a change in interest rate on the price of the bond because longer term cash flows are more sensitive to changes in interest rates.

318. [Single choice] [CM-EIP §7.16.2] For two bonds of equal maturity, which is more affected by a rise in market yields?

- a. It depends on the credit rating
- b. The bond with the larger coupon
- c. The bond with the smaller coupon
- d. Both equally

Answer and explanation

Answer: C. The bond with the smaller coupon

Rationale: A bond with smaller coupons generally has a larger percentage of its value attributable to its long-term principal repayment component. Since longer term cash flows are more susceptible to changes in interest rates, bonds with smaller coupons will be more severely affected by a rise in yield.

319. [Single choice] [CM-EIP §7.17.1] For which bonds is duration highest, all else equal?

- a. Higher coupons, lower yields and shorter maturities
- b. Zero coupons with short maturities
- c. Higher coupons, higher yields and shorter maturities
- d. Lower coupons, lower yields and longer maturities

Answer and explanation

Answer: D. Lower coupons, lower yields and longer maturities

Rationale: All else equal, duration is higher for bonds with lower coupons, lower yields and longer maturities.

320. [Single choice] [CM-EIP §7.17.1] Over what size of interest rate change is duration a reliable linear approximation?

- a. Not more than 1% or 100 basis points
- b. Up to 5%
- c. Up to 10%
- d. Any size

Answer and explanation

Answer: A. Not more than 1% or 100 basis points

Rationale: Duration is a linear measure of the approximate price sensitivity of a bond to small interest rate changes, that is not more than 1% or 100 basis points.

321. [Single choice] [CM-EIP §7.17.2] Which bond has the highest convexity?

- a. A 10-year 4% coupon bond
- b. A 10-year zero-coupon bond
- c. A 5-year 8% coupon bond
- d. A 10-year 8% coupon bond

Answer and explanation

Answer: B. A 10-year zero-coupon bond

Rationale: Convexity is directly related to the coupon rate and the bond's remaining term to maturity. The lower the coupon, the higher the convexity. Zero coupon bonds have the highest convexity.

322. [Single choice] [CM-EIP §7.17.3] What should an investor do with portfolio duration when interest rates are expected to fall?

- a. Set duration to zero
- b. Decrease duration to minimise sensitivity
- c. Increase duration to maximise sensitivity
- d. Leave duration unchanged

Answer and explanation

Answer: C. Increase duration to maximise sensitivity

Rationale: With the anticipation of falling interest rates, bond prices are anticipated to rise, and a bond portfolio's duration should be increased to maximise the sensitivity of the portfolio. Conversely, when rising rates are expected, duration should be decreased.

323. [Single choice] [CM-EIP §7.17.3] How does greater convexity affect a bond's price when interest rates rise?

- a. The price is unaffected
- b. The price rises
- c. The price falls faster
- d. The price falls slower

Answer and explanation

Answer: D. The price falls slower

Rationale: When interest rates rise, the price of a more convex bond will not fall as much. When interest rates fall, the price of a more convex bond will rise much faster. Regardless of whether rates are rising or falling, investors should seek bonds with greater convexity.

324. [Single choice] [CM-EIP §7.17.4] **What does PVBP or PV01 measure?**

- a. The change in present value of the entire bond portfolio for a one basis point movement in interest rates
- b. The bond's yield to maturity
- c. The bond's convexity
- d. The bond's price

Answer and explanation

Answer: A. The change in present value of the entire bond portfolio for a one basis point movement in interest rates

Rationale: The PVBP or PV01 is used as a risk management tool for a bond portfolio. It is the change in the present value of the entire bond portfolio for a one basis point movement in interest rates.

325. [Single choice] [CM-EIP §7.18] **What happens to the price of a discount bond as it approaches maturity, all else constant?**

- a. It falls
- b. It rises towards par
- c. It stays constant
- d. It becomes zero

Answer and explanation

Answer: B. It rises towards par

Rationale: Over time, holding all other factors constant, a bond's price must converge to its par value as the maturity date approaches because the bond will be redeemed at par. The price of a discount bond will increase and the price of a premium bond will decline.

326. [Calculation] [CM-EIP §7.19.1] **A 10-year 6% coupon bond with par $1,000 is priced at 115%. What is the current yield?**

- a. 5.00%
- b. 6.90%
- c. 5.22%
- d. 6.00%

Answer and explanation

Answer: C. 5.22%

Rationale: Current yield = Annual interest payment / Bond price = $60 / $1,150 = 5.22%. Current yield considers only the annual interest payment and does not account for any capital gain or loss.

327. [Calculation] [CM-EIP §7.19.2] **A bond priced at $1,148.80 with an 8% semi-annual coupon and 10 years to maturity has a semi-annual YTM of 3.00%. What is the bond-equivalent yield?**

- a. 3.00%
- b. 6.00%
- c. 6.09%
- d. 6.12%

Answer and explanation

Answer: B. 6.00%

Rationale: The market convention is to annualise the semi-annual yield by doubling its value: $3.00\% \times 2 = 6.00\%$, which is the bond-equivalent yield. The effective annual yield, which is the proper way to annualise, is $(1.03)^2 - 1 = 6.09\%$.

328. [Calculation] [CM-EIP §7.19.2] **A bond priced at $1,148.80 with par $1,000 and an 8% annual coupon has 10 years to maturity. Using the approximation formula, what is the YTM?**

- a. 6.00%

- b. 6.09%
- c. 6.96%
- d. 6.06%

Answer and explanation

Answer: D. 6.06%

Rationale: Amortised loss per year = (1,148.80 − 1,000)/10 = $14.88. Approximate YTM = (80 − 14.88) / ((1,148.80 + 1,000)/2) = 65.12 / 1,074.40 = 6.06%.

329. [Calculation] [CM-EIP §7.19.2] A 15-year zero-coupon bond with maturity value $1,000 sells for $300. What is the bond-equivalent yield?

- a. 8.2%
- b. 4.1%
- c. 6.0%
- d. 12.0%

Answer and explanation

Answer: A. 8.2%

Rationale: k/2 = (1,000/300)^(1/30) − 1 = 0.0409 = 4.1% semi-annually. The bond-equivalent yield is 4.1% × 2 = 8.2%. The number of periods is 30 semi-annual periods, double the years.

330. [Single choice] [CM-EIP §7.19.2] Under what two conditions will an investor actually realise the stated YTM?

- a. If the bond is sold before maturity and rates fall
- b. If coupon payments can be consistently reinvested at the YTM and the bond is held to maturity
- c. If the issuer does not call the bond
- d. If the credit rating is upgraded

Answer and explanation

Answer: B. If coupon payments can be consistently reinvested at the YTM and the bond is held to maturity

Rationale: The investor will only realise the YTM stated at the time of purchase if the coupon payments can be consistently reinvested at the YTM and the bond is held to maturity.

331. [Calculation] [CM-EIP §7.19.4] An investor in the 28% marginal tax bracket holds a tax-exempt bond with a YTM of 7%. What is the taxable equivalent yield?

- a. 5.04%
- b. 7.0%
- c. 9.7%
- d. 8.96%

Answer and explanation

Answer: C. 9.7%

Rationale: Taxable equivalent yield = Tax-exempt yield / (1 − marginal tax rate) = 0.07 / (1 − 0.28) = 0.07 / 0.72 = 9.7%.

332. [Single choice] [CM-EIP §7.20.1] What does a declining yield curve indicate?

- a. Inflation is expected to rise
- b. The market is ambivalent
- c. Short-term rates are expected to rise
- d. Generally lower short-term interest rates in the future

Answer and explanation

Answer: D. Generally lower short-term interest rates in the future

Rationale: The declining yield curve is seen when the yields on short-term issues are higher than the yields on longer-maturity issues. This generally indicates lower short-term interest rates in the future.

333. [**Single choice**] [**CM-EIP §7.20.2**] **Which theory argues that the yield curve should always be upward sloping?**

- a. Liquidity preference theory
- b. Market segmentation theory
- c. Purchasing power parity
- d. Expectations theory

Answer and explanation

Answer: A. Liquidity preference theory

Rationale: The theory of liquidity preference argues that the shape of the yield curve should always be upward sloping and that any other shape would be a temporary aberration, because investors require a higher return to hold long-term securities.

334. [**Calculation**] [**CM-EIP §7.20.2**] **A 1-year T-bill yields 6% and a new 1-year T-bill is expected to yield 8% next year. Under the expectations theory, what should a 2-year security yield?**

- a. 6.0%
- b. 7.0%
- c. 8.0%
- d. 14.0%

Answer and explanation

Answer: B. 7.0%

Rationale: The average annual rate of return from rolling 1-year bills is $[(1.06)(1.08)]\hat{}(1/2) - 1 = 0.07$. An investor in a 2-year security will require a yield of 7%, the average of the 6% realised in year one and the 8% expected in year two.

335. [**Single choice**] [**CM-EIP §7.20.2**] **Which theory holds that each maturity's yield is determined solely by supply and demand in that segment?**

- a. Liquidity preference theory
- b. Market segmentation theory
- c. Quantity theory
- d. Expectations theory

Answer and explanation

Answer: B. Market segmentation theory

Rationale: The market segmentation theory presumes that investors have their own maturity preferences, which are dictated by the nature of their liabilities. The yield for each maturity is determined solely by the supply and demand in each segment.

336. [**Calculation**] [**CM-EIP §7.21.1**] **A convertible bond has a conversion ratio of 40 and the share price is $24. What is the conversion value?**

- a. $1,100
- b. $728
- c. $960
- d. $1,000

Answer and explanation

Answer: C. $960

Rationale: Conversion value = Conversion ratio × Market share price = 40 × $24 = $960. The minimum value of a convertible bond is the greater of its conversion value and its investment value.

337. [Calculation] [CM-EIP §7.21.2] **A convertible bond has a par value of $1,000 and a conversion ratio of 40. What is the conversion price?**

- a. $27.50
- b. $24
- c. $40
- d. $25

Answer and explanation

Answer: D. $25

Rationale: Conversion price = Par value of CB / Conversion ratio = $1,000 / 40 = $25.

338. [Calculation] [CM-EIP §7.21.3] **A convertible bond trades at $1,100 with a conversion ratio of 40. What is the conversion parity price?**

- a. $27.50
- b. $25.00
- c. $24.00
- d. $26.00

Answer and explanation

Answer: A. $27.50

Rationale: Conversion parity price = Market price of CB / Conversion ratio = $1,100 / 40 = $27.50. This can be considered a share price breakeven point.

339. [Calculation] [CM-EIP §7.21.4] **A convertible trades at $1,100 with a conversion value of $960. What is the conversion premium in percentage terms?**

- a. 51.0%
- b. 14.6%
- c. 12.7%
- d. 4.6%

Answer and explanation

Answer: B. 14.6%

Rationale: Conversion premium (%) = (Market value of CB − Conversion value) / Conversion value = (1,100 − 960) / 960 = 140 / 960 = 14.6%.

340. [Calculation] [CM-EIP §7.21.5] **A convertible has a conversion premium of $140, bond income of $40 and dividend income of $14. What is the breakeven time?**

- a. 10.00 years
- b. 2.60 years
- c. 5.38 years
- d. 3.50 years

Answer and explanation

Answer: C. 5.38 years

Rationale: Breakeven time = Conversion premium / (Bond income − Dividend income) = 140 / (40 − 14) = 140 / 26 = 5.38 years. The measure does not consider the time value of money.

341. [Calculation] [CM-EIP §7.21.6] **A convertible trades at $1,100 with an investment value of $728. What is the premium over straight value?**

- a. 34%
- b. 66%
- c. 151%
- d. 51%

Answer and explanation

Answer: D. 51%

Rationale: Premium over straight value = (Market price of CB / Investment value) − 1 = 1,100/728 − 1 = 1.51 − 1 = 51%. Despite its use in practice, this measure is flawed because the straight value changes as the discount rate changes.

342. [Single choice] [CM-EIP §7.21.7] How does a convertible bond's yield compare with the underlying share's dividend yield and a plain vanilla bond?

- a. Higher than the dividend yield but lower than a comparable plain vanilla bond
- b. Lower than the dividend yield but higher than a plain vanilla bond
- c. Lower than both
- d. Higher than both

Answer and explanation

Answer: A. Higher than the dividend yield but lower than a comparable plain vanilla bond

Rationale: The yield on a convertible bond is usually higher than the dividend yield of the underlying ordinary share but lower than the yield of a comparable plain vanilla bond.

Chapter 8 — Portfolio Management

343. [Single choice] [CM-EIP §8.2] What does the Efficient Market Hypothesis state?

- a. Prices are always above fair value
- b. Security prices always fully reflect available information
- c. Security prices never reflect available information
- d. Only fundamental analysis can beat the market

Answer and explanation

Answer: B. Security prices always fully reflect available information

Rationale: The EMH states that an efficient market is one where security prices always fully reflect available information. The direct implication is that stocks always trade at their fair values and it is impossible to outperform the overall market consistently on a risk-adjusted basis.

344. [Single choice] [CM-EIP §8.2.1] Under the weak form of the EMH, which technique cannot consistently produce excess returns?

- a. Portfolio diversification
- b. Fundamental analysis
- c. Technical analysis
- d. Insider trading

Answer and explanation

Answer: C. Technical analysis

Rationale: Under the weak form, all past price information is fully reflected in the current stock price, so future prices follow a random walk. This implies that investors will not be able to consistently achieve excess returns using technical analysis.

345. [Single choice] [CM-EIP §8.2.2] Under the semi-strong form of the EMH, which techniques cannot consistently beat the market?

- a. Both fundamental and technical analysis
- b. Technical analysis only
- c. Fundamental analysis only
- d. Neither fundamental nor technical analysis

Answer and explanation

Answer: D. Neither fundamental nor technical analysis

Rationale: Under the semi-strong form, all publicly available information is fully reflected in the current stock price and prices react rapidly to new public information. This implies that neither fundamental analysis nor technical analysis will allow investors to consistently beat the market.

346. [Single choice] [CM-EIP §8.2.3] Why is the strong form of the EMH not supported by the evidence?

- a. Because the history of insider trading suggests inside information does provide an advantage
- b. Because technical analysis always works
- c. Because prices never adjust
- d. Because markets are perfectly efficient

Answer and explanation

Answer: A. Because the history of insider trading suggests inside information does provide an advantage

Rationale: The evidence suggests real-world financial markets are not strong form efficient. The history of insider trading in the US and numerous high-profile cases continue to suggest that having inside information does provide an information advantage to generate abnormal gains.

347. [Single choice] [CM-EIP §8.2.4] What do supporters of market efficiency highlight about fund managers?

- a. That most outperform their benchmarks
- b. That the majority underperform their benchmarks even before fees, and outperformance does not persist
- c. That all managers generate alpha
- d. That fees are irrelevant

Answer and explanation

Answer: B. That the majority underperform their benchmarks even before fees, and outperformance does not persist

Rationale: Supporters of market efficiency highlight that the average fund manager does not outperform the market. Despite all the expertise most fund managers claim to have, the majority underperform their respective benchmarks even if fees are excluded. Even for managers that outperform, the outperformance does not generally persist.

348. [Single choice] [CM-EIP §8.3.1] Which investor should probably adopt a passive investment approach?

- a. One who wants to trade frequently
- b. One with large amounts, time and expertise
- c. One who can only invest small amounts, lacks time and expertise, and does not wish to pay fees
- d. One who believes markets are inefficient

Answer and explanation

Answer: C. One who can only invest small amounts, lacks time and expertise, and does not wish to pay fees

Rationale: Investors who are only able to invest in small amounts, lack the time or expertise to monitor and manage their investments, and do not wish to pay fees should probably adopt a passive investment approach and rely on index funds.

349. [Single choice] [CM-EIP §8.3.2] Which of the following is an active strategy?

- a. Buying a market-cap weighted ETF
- b. Dollar cost averaging into an index
- c. Buying and holding an index fund
- d. Sector rotation

Answer and explanation

Answer: D. Sector rotation

Rationale: Security selection, sector rotation and market timing are examples of active strategies. The objective of active management is to outperform the market on a risk-adjusted basis.

350. [Single choice] [CM-EIP §8.3.2] How many industries does the Industry Classification Benchmark classify the market into?

- a. 5
- b. 8
- c. 10
- d. 12

Answer and explanation

Answer: C. 10

Rationale: Under the Industry Classification Benchmarks, the market is classified into 10 industries, which are further sub-divided into super sectors and sectors. A sector is the smallest subset of an industry.

351. [Single choice] [CM-EIP §8.4.1] What does the efficient frontier represent?

- a. All combinations of assets that generate the highest return at a given level of risk
- b. The portfolio with the lowest risk
- c. The portfolio with the highest return
- d. All possible portfolios, efficient or not

Answer and explanation

Answer: A. All combinations of assets that generate the highest return at a given level of risk

Rationale: The Efficient Frontier represents all possible combinations of the two asset classes that would generate the highest return at a given level of risk. All the different combinations that lie on the Efficient Frontier represent optimal portfolios, but for clients with different risk profiles.

352. [Single choice] [CM-EIP §8.4.1] Why is the efficient frontier a curve rather than a straight line?

- a. Because of transaction costs
- b. Because returns are not perfectly correlated
- c. Because risk is measured in percentages
- d. Because the number of assets is finite

Answer and explanation

Answer: B. Because returns are not perfectly correlated

Rationale: The frontier is a curve instead of a straight line because the returns of the two assets are not perfectly correlated.

353. [Single choice] [CM-EIP §8.4.2] Approximately what proportion of portfolio standard deviation is eliminated by the time there are 10 securities in the portfolio?

- a. One quarter
- b. One third
- c. One half
- d. All of it

Answer and explanation

Answer: C. One half

Rationale: Studies have shown that around half of portfolio standard deviation is eliminated by the time that there are 10 securities in the portfolio. Unfortunately, the benefits of diversification diminish as more securities are added.

354. [Single choice] [CM-EIP §8.4.2] What is the only condition required for adding a second stock to reduce portfolio risk?

- a. The second stock must have a higher return
- b. The second stock must have lower volatility
- c. The two stocks must be in different industries
- d. Correlation between the two stocks must be less than perfectly positive

Answer and explanation

Answer: D. Correlation between the two stocks must be less than perfectly positive

Rationale: It does not matter whether factors affecting the second stock are diametrically opposed to those affecting the first. All that matters is that there is less than perfect positive correlation between the factors driving the two price movements. As long as correlation is not 1, the addition has the potential to diversify risk.

355. [Single choice] [CM-EIP §8.4.3] Which statement about systematic risk is correct?

- a. It cannot be reduced through diversification but can be partly or almost fully offset through hedging
- b. It is the same as firm-specific risk
- c. It is captured by the error term
- d. It can be eliminated by diversification

Answer and explanation

Answer: A. It cannot be reduced through diversification but can be partly or almost fully offset through hedging

Rationale: Systematic risk cannot be reduced through diversification. However, depending on the level of exposure and asset class invested, investors can partly or almost fully offset systematic risk through hedging, for example the use of options and futures.

356. [Single choice] [CM-EIP §8.4.3] What does beta reflect?

- a. Total risk including firm-specific risk
- b. Systematic risk, the sensitivity of the portfolio to market movement
- c. The error term in the risk decomposition
- d. The correlation coefficient

Answer and explanation

Answer: B. Systematic risk, the sensitivity of the portfolio to market movement

Rationale: Systematic risk is reflected in the beta of an investment portfolio. This is the sensitivity of the portfolio to market movement.

357. [Single choice] [CM-EIP §8.4.4] What happens to portfolio risk when two securities with a correlation of +1 are combined?

- a. Risk is eliminated
- b. Risk is reduced but not eliminated
- c. There is no reduction in portfolio risk; it is the weighted average of the individual risks
- d. Risk becomes zero

Answer and explanation

Answer: C. There is no reduction in portfolio risk; it is the weighted average of the individual risks

Rationale: Combining securities that are perfectly positively correlated with each other provides no reduction in portfolio risk. The risk of the resulting portfolio is simply a weighted average of the individual risks of the securities.

358. [Single choice] [CM-EIP §8.4.4] What is the principle behind hedging strategies in terms of correlation?

- a. Combining securities with zero correlation

- b. Combining securities with correlation of 0.5
- c. Combining securities with correlation of +1
- d. Combining two securities with perfect negative correlation, which can eliminate portfolio risk altogether

Answer and explanation

Answer: D. Combining two securities with perfect negative correlation, which can eliminate portfolio risk altogether

Rationale: Combining two securities with perfect negative correlation with each other can eliminate portfolio risk altogether. This is the principle behind hedging strategies.

359. [Single choice] [CM-EIP §8.4.6] What happens to correlations between asset classes during a major financial crisis?

- a. They move towards 1
- b. They remain stable
- c. They become zero
- d. They fall towards −1

Answer and explanation

Answer: A. They move towards 1

Rationale: One concern in relation to diversification is the potential for asset correlations to move towards 1 during major financial crises. Under such conditions, correlation between assets gravitates towards 1, and there is nowhere for funds to hide except in cash and government bonds.

360. [Calculation] [CM-EIP §8.4.7] Security A has an expected return of 18% and a weight of 40%; Security B has 9% and a weight of 60%. What is the portfolio expected return?

- a. 27.0%
- b. 12.6%
- c. 13.5%
- d. 13.3%

Answer and explanation

Answer: B. 12.6%

Rationale: $\text{Rp} = 0.4(0.18) + 0.6(0.09) = 0.072 + 0.054 = 0.126 = 12.6\%$. The expected portfolio return is a weighted average of the constituent returns.

361. [Calculation] [CM-EIP §8.4.7] Security A has a standard deviation of 20% and weight 40%; Security B has 10% and weight 60%; correlation is 0.8. What is portfolio risk?

- a. 2.00%
- b. 14.00%
- c. 13.30%
- d. 10.00%

Answer and explanation

Answer: C. 13.30%

Rationale: $\text{p} = \sqrt{(0.4^2 \times 0.2^2 + 0.6^2 \times 0.1^2 + 2 \times 0.4 \times 0.6 \times 0.8 \times 0.2 \times 0.1)} = \sqrt{(0.0064 + 0.0036 + 0.00768)} = \sqrt{0.01768} = 0.1330 = 13.30\%$.

362. [Calculation] [CM-EIP §8.4.7] Using the same two securities, what would portfolio risk be if the correlation were −1?

- a. 0.00%
- b. 14.00%
- c. 10.00%
- d. 2.00%

Answer and explanation

Answer: D. 2.00%

Rationale: With $= -1$, $p = \sqrt{(0.0064 + 0.0036 - 0.00768)} = \sqrt{0.00232} = 0.0482$, but the weighted difference is $|0.4(0.2) - 0.6(0.1)| = |0.08 - 0.06| = 0.02 = 2\%$. Portfolio risk falls from 14% at $= 1$ to 2% at $= -1$.

363. [Single choice] [CM-EIP §8.4.8] How can foreign exchange risk be removed from an international portfolio?

- a. By hedging the risk using forward exchange contracts
- b. By holding only emerging market securities
- c. By increasing the equity allocation
- d. By avoiding foreign investment entirely

Answer and explanation

Answer: A. By hedging the risk using forward exchange contracts

Rationale: Inherent in international diversification is the risk of fluctuating exchange rates of foreign currency investments. However, one way to remove foreign exchange risk from international portfolio investments is to hedge the risk using forward exchange contracts.

364. [Single choice] [CM-EIP §8.4.8] Does international diversification always improve risk-adjusted returns?

- a. Yes, always
- b. No, adding the wrong securities and markets can lower the risk-adjusted return
- c. Only for institutional investors
- d. Only in developed markets

Answer and explanation

Answer: B. No, adding the wrong securities and markets can lower the risk-adjusted return

Rationale: International diversification does not always lead to higher risk-adjusted returns. It makes sense to add securities from new markets only if the additions have the potential to raise the risk-adjusted return of the portfolio. Sometimes adding the wrong securities and markets can actually lower the risk-adjusted return.

365. [Single choice] [CM-EIP §8.4.9] Which of the following is NOT an assumption of the CAPM?

- a. Investors can borrow or lend at the risk-free rate
- b. There are no taxes or transaction costs
- c. Investors have heterogeneous expectations of risk and return
- d. All investors have the same one-period time horizon

Answer and explanation

Answer: C. Investors have heterogeneous expectations of risk and return

Rationale: CAPM assumes investors have homogeneous expectations of risk, return and covariances of securities. Heterogeneous expectations would mean investors arrive at different optimal portfolios, which breaks the market portfolio argument.

366. [Single choice] [CM-EIP §8.4.9] What is the beta of the market portfolio?

- a. 0
- b. 0.5
- c. 1
- d. 2

Answer and explanation

Answer: C. 1

Rationale: The beta of the market portfolio is 1. A security with beta greater than 1 contains proportionately higher systematic risk than the market and should be compensated with a higher expected return.

367. [Single choice] [CM-EIP §8.4.9] What is the risk-free asset's covariance with any risky asset?

- a. Equal to the market variance
- b. Negative
- c. 1
- d. Zero

Answer and explanation

Answer: D. Zero

Rationale: The risk-free asset has no risk and provides a risk-free rate of return. The risk-free asset's return variance is zero and its return covariance with any risky asset is zero as well.

368. [Single choice] [CM-EIP §8.4.9] What does the capital market line specify?

- a. The equilibrium relation between expected return and total risk for efficiently diversified portfolios
- b. The relation between coupon and yield
- c. The relation between inflation and exchange rates
- d. The equilibrium relation between expected return and systematic risk for all assets

Answer and explanation

Answer: A. The equilibrium relation between expected return and total risk for efficiently diversified portfolios

Rationale: The capital market line specifies the equilibrium relation between expected return and total risk for efficiently diversified portfolios. The security market line specifies the relation between expected return and systematic risk and applies to individual securities as well as portfolios.

369. [Single choice] [CM-EIP §8.4.9] What does the slope of the capital market line represent?

- a. The risk-free rate
- b. The market price of risk for efficient portfolios
- c. The beta of the market
- d. The correlation coefficient

Answer and explanation

Answer: B. The market price of risk for efficient portfolios

Rationale: The slope of the CML, [E(Rm) − Rf]/ m, is the market price of risk for efficient portfolios. It indicates the additional return that the market requires for a 1% increase in the portfolio's standard deviation of returns.

370. [Calculation] [CM-EIP §8.4.9] A portfolio has a beta of 1.20, the risk-free rate is 2.5% and the expected market return is 14%. What is the expected return by CAPM?

- a. 19.3%
- b. 11.5%
- c. 16.3%
- d. 14.0%

Answer and explanation

Answer: C. 16.3%

Rationale: E(r) = rf + (rm − rf) = 2.5 + 1.2(14 − 2.5) = 2.5 + 1.2(11.5) = 2.5 + 13.8 = 16.3%.

371. [Calculation] [CM-EIP §8.4.9] Using the same portfolio, if the actual return is 17%, what is the alpha?

- a. −0.7%

- b. +1.0%
- c. +3.0%
- d. +0.7%

Answer and explanation

Answer: D. +0.7%

Rationale: Alpha = actual return − CAPM predicted return = 17% − 16.3% = +0.7%. At an actual return of 15% the alpha would be −1.3%.

372. [Single choice] [CM-EIP §8.4.9] What does a security lying above the security market line indicate?

- a. It is under-priced, providing above-average expected returns for its beta
- b. It has zero systematic risk
- c. It is fairly priced
- d. It is overpriced

Answer and explanation

Answer: A. It is under-priced, providing above-average expected returns for its beta

Rationale: Securities that lie above the SML provide above average expected returns given their respective betas. Such securities are considered to be under-priced. Securities below the SML are overpriced.

373. [Single choice] [CM-EIP §8.4.9] What does a steep security market line suggest about the market?

- a. The market is overvalued
- b. The market may be excessively risk-averse and therefore cheap
- c. The market is in equilibrium
- d. The risk-free rate is zero

Answer and explanation

Answer: B. The market may be excessively risk-averse and therefore cheap

Rationale: A steep SML indicates that a large incremental return can be earned in the market with a small increase in systematic risk. As such, the market may be excessively risk-averse and therefore cheap. A flat SML suggests the market is overvalued.

374. [Single choice] [CM-EIP §8.4.10] Who developed the Arbitrage Pricing Theory?

- a. Eugene Fama
- b. William Sharpe
- c. Stephen Ross
- d. Harry Markowitz

Answer and explanation

Answer: C. Stephen Ross

Rationale: In response to criticisms of the CAPM, Stephen Ross developed an alternative asset pricing model in the early 1970s, the Arbitrage Pricing Theory. The APT is a multifactor equilibrium pricing model.

375. [Single choice] [CM-EIP §8.4.10] Which of the following does the APT NOT assume, unlike the CAPM?

- a. Markets are perfect
- b. Investors have homogeneous beliefs
- c. Investors are risk-averse utility maximisers
- d. A single-period investment horizon

Answer and explanation

Answer: D. A single-period investment horizon

Rationale: In contrast to the CAPM, the APT does not assume a single-period investment horizon, absence of taxes, borrowing and lending at the risk-free rate, or investors selecting portfolios on the basis of expected return and variance.

376. [Single choice] [CM-EIP §8.4.10] Why is the rate of inflation not an APT risk factor, while unexpected inflation is?

- a. Because the rate of inflation is partially predictable, whereas unexpected inflation is unpredictable by the market as a whole
- b. Because inflation only affects bonds
- c. Because the APT ignores inflation
- d. Because inflation is not an economic factor

Answer and explanation

Answer: A. Because the rate of inflation is partially predictable, whereas unexpected inflation is unpredictable by the market as a whole

Rationale: At the beginning of each period, the risk factor must be unpredictable by the market as a whole. The rate of inflation is not an APT risk factor because it is partially predictable. In contrast, unexpected inflation, the difference between actual and expected inflation, is an APT risk factor.

377. [Single choice] [CM-EIP §8.4.10] How many systematic factors did Chen, Roll and Ross identify?

- a. Three
- b. Four
- c. Five
- d. Two

Answer and explanation

Answer: B. Four

Rationale: Chen, Roll and Ross identified four systematic factors: unanticipated changes in inflation, in industrial production, in the default-risk premium, and in the term structure of interest rates.

378. [Single choice] [CM-EIP §8.5.1] What is the typical range of investment management fees quoted in the guide?

- a. 3.0%–5.0%
- b. 0.1%–0.5%
- c. 1.0%–1.5%
- d. 2.0%–3.0%

Answer and explanation

Answer: C. 1.0%–1.5%

Rationale: There are also investors who do not wish to incur investment management fees, which can range from 1.0% to 1.5%, on average.

379. [Single choice] [CM-EIP §8.5.1] What is the difference between advisory and discretionary portfolio management?

- a. Advisory charges no fees
- b. Discretionary requires client approval for each trade
- c. Advisory is for institutions; discretionary is for retail
- d. In advisory the client remains engaged in decisions; in discretionary the manager decides without permission for each transaction

Answer and explanation

Answer: D. In advisory the client remains engaged in decisions; in discretionary the manager decides without permission for each transaction

Rationale: For investors who wish to remain actively engaged in investment decisions, an advisory portfolio management approach is needed. Under discretionary portfolio management, the fund manager is able to decide on asset allocation and securities selection without having to obtain permission from the client for each transaction.

380. [Single choice] [CM-EIP §8.5.2] What is the first step in the portfolio management process for retail investors?

- a. Setting investment objectives
- b. Monitoring market conditions
- c. Reviewing and adjusting the portfolio
- d. Measuring portfolio performance

Answer and explanation

Answer: A. Setting investment objectives

Rationale: The process is dynamic, continuous and systematic: setting investment objectives, developing and implementing strategies, monitoring market conditions, reviewing and adjusting the portfolio, and measuring investment portfolio performance.

381. [Single choice] [CM-EIP §8.5.2] Why should objectives, constraints and strategy be stated in a written document?

- a. Because regulators require it
- b. Because a well-drafted investment policy provides discipline, minimising whipsaw reactions to temporary price swings
- c. Because it reduces fees
- d. Because it guarantees returns

Answer and explanation

Answer: B. Because a well-drafted investment policy provides discipline, minimising whipsaw reactions to temporary price swings

Rationale: A well-drafted investment policy provides the discipline required for efficient management of the portfolio, minimising whipsaw reactions to temporary price swings or even protracted volatility.

382. [Single choice] [CM-EIP §8.5.3] Who does the investment policy statement belong to?

- a. The custodian
- b. The portfolio manager
- c. The client
- d. The regulator

Answer and explanation

Answer: C. The client

Rationale: The IPS or investment mandate belongs to the client and not to the portfolio manager. It is therefore essential that the investment manager spends sufficient time with the client to define the client's investment goals and acceptable levels of risk.

383. [Single choice] [CM-EIP §8.5.4] In the life cycle approach, what is the appropriate risk posture in the early career stage?

- a. Only cash and money market funds
- b. Balanced income and growth
- c. Capital preservation with low-risk assets
- d. High return, high risk, growth-oriented investments

Answer and explanation

Answer: D. High return, high risk, growth-oriented investments

Rationale: In the first stage, early career, the individual is young with a long time horizon and a growing stream of discretionary income, so he can undertake high return, high risk investments that are growth-oriented.

384. [Single choice] [CM-EIP §8.5.4] What characterises the late career stage of the life cycle approach?

- a. Greater emphasis on income needs and capital preservation, with assets significantly exceeding liabilities
- b. The longest investment horizon
- c. Willingness to take maximum risk
- d. Assets are small relative to liabilities

Answer and explanation

Answer: A. Greater emphasis on income needs and capital preservation, with assets significantly exceeding liabilities

Rationale: At the third stage, late career, the individual is near retirement and there is a greater emphasis on income needs and capital preservation. Assets significantly exceed liabilities, the time horizon is shorter, and the portfolio is typically shifted to significantly lower return, lower risk assets.

385. [Single choice] [CM-EIP §8.5.5] Which instruments should be used to meet an immediate liquidity need?

- a. Real estate and closely held business shares
- b. High quality, short-term debt instruments such as Treasury bills and commercial paper
- c. Long-dated government bonds
- d. Private equity funds

Answer and explanation

Answer: B. High quality, short-term debt instruments such as Treasury bills and commercial paper

Rationale: To meet such liquidity needs, funds should be invested in high quality, short-term debt instruments which can be easily sold, such as Treasury bills and commercial paper. If the assets held have poor marketability, such as real estate, liquidity needs should be monitored carefully.

386. [Single choice] [CM-EIP §8.5.5] Which investment constraint governs a client using CPF money?

- a. The SFA investor classification
- b. No constraint applies
- c. The CPF Investment Scheme, which specifies the types and quantum of permissible investments
- d. Only the client's own preferences

Answer and explanation

Answer: C. The CPF Investment Scheme, which specifies the types and quantum of permissible investments

Rationale: If an individual is using his CPF money to invest, his investment will be governed by the CPF Investment Scheme, which specifies the types and quantum of investments permissible.

387. [Single choice] [CM-EIP §8.5.6] According to Brinson, Hood and Beebower, what proportion of a portfolio's return comes from asset allocation?

- a. Less than 10%
- b. About 50%
- c. About 75%
- d. 91.5% or more

Answer and explanation

Answer: D. 91.5% or more

Rationale: A study by Brinson, Hood and Beebower showed that 91.5% or more of a portfolio's return comes from asset allocation and, therefore, less than 10% can be attributed to security selection.

388. [Single choice] [CM-EIP §8.5.6] What is strategic asset allocation?

- a. The long-run policy asset mix, normally expressed as percentages of total portfolio value
- b. The selection of individual securities
- c. The choice of fund manager
- d. Short-run overweighting and underweighting of asset classes

Answer and explanation

Answer: A. The long-run policy asset mix, normally expressed as percentages of total portfolio value

Rationale: SAA refers to how a portfolio's funds are divided among the various asset classes, given the client's risk profile and the manager's long-term forecasts. SAA constitutes the policy or long-run asset mix, normally expressed in terms of the percentage of total portfolio value invested in each asset class.

389. [Single choice] [CM-EIP §8.5.6] What is the most common valuation approach to tactical asset allocation?

- a. The cyclical consideration approach
- b. The risk premium or spread approach, also called the equity-bond yield gap
- c. The technical analysis approach
- d. The dividend discount approach

Answer and explanation

Answer: B. The risk premium or spread approach, also called the equity-bond yield gap

Rationale: The most common valuation approach is the risk premium approach or spread approach. If the expected return on equities relative to bonds is above average, a higher allocation is made to equities. This is also referred to as the equity-bond yield gap.

390. [Calculation] [CM-EIP §8.6.1] A portfolio is worth $30 million at the start of the quarter and $35 million at the end, with no cash flows. What is the return?

- a. 20.0%
- b. 14.3%
- c. 16.7%
- d. 5.0%

Answer and explanation

Answer: C. 16.7%

Rationale: Rp = (VE − VB) / VB = (35 − 30) / 30 = 5/30 = 16.7%.

391. [Calculation] [CM-EIP §8.6.1] A portfolio is worth $50 million at the start. The client deposits $5 million just before the end of the quarter and the end value is $53 million. What is the return?

- a. +6.0%
- b. −3.6%
- c. +16.0%
- d. −4.0%

Answer and explanation

Answer: D. −4.0%

Rationale: The deposit must be deducted from the ending value: Rp = [(53 − 5) − 50] / 50 = −2/50 = −4.0%.

392. [Calculation] [CM-EIP §8.6.1] Using the same figures, if the $5 million deposit was received at the beginning of the quarter, what is the return?

- a. −3.6%
- b. +6.0%
- c. −2.0%

- d. −4.0%

Answer and explanation

Answer: A. −3.6%

Rationale: The beginning value is adjusted: Rp = [53 − (50 + 5)] / (50 + 5) = −2/55 = −3.6%.

393. [Single choice] [CM-EIP §8.6.1] Which return measure is equivalent to the internal rate of return?

- a. Time-weighted rate of return
- b. Dollar-weighted rate of return
- c. Modified Dietz return
- d. Current yield

Answer and explanation

Answer: B. Dollar-weighted rate of return

Rationale: The DWR is equivalent to the internal rate of return. It equates the present value of all cashflows, including the ending market value, with the beginning market value of the portfolio.

394. [Calculation] [CM-EIP §8.6.1] A portfolio starts at $50 million, receives a $5 million deposit mid-quarter and ends at $53 million. The semi-quarterly IRR is −1.92%. What is the quarterly return?

- a. −1.92%
- b. −3.84%
- c. −3.8%
- d. −3.81%

Answer and explanation

Answer: C. −3.8%

Rationale: The quarterly return is $(1 - 0.0192)^2 - 1 = 0.9808^2 - 1 = 0.96197 - 1 = -3.8\%$. In each half of the quarter the loss was 1.92%.

395. [Calculation] [CM-EIP §8.6.1] A portfolio is worth $50m at the start, $49m just before a $5m deposit, $54m immediately after, and $53m at the end. What is the time-weighted return?

- a. −3.81%
- b. −3.8%
- c. −2.0%
- d. −1.85%

Answer and explanation

Answer: A. −3.81%

Rationale: First half: (49 − 50)/50 = −2.0%. Second half: (53 − 54)/54 = −1.85%. Linked: [(1 − 0.02)(1 − 0.0185)] − 1 = −3.81%.

396. [Calculation] [CM-EIP §8.6.1] A portfolio grows from $100 to $250 in five years. What is the compound annual growth rate?

- a. 15.0%
- b. 20.1%
- c. 30.0%
- d. 50.0%

Answer and explanation

Answer: B. 20.1%

Rationale: 250 = 100(1 + r)^5, so (1 + r)^5 = 2.5 and r = 2.5^(1/5) − 1 = 0.201 = 20.1%.

397. [Single choice] [CM-EIP §8.6.1] Why is the dollar-weighted return inappropriate for evaluating a portfolio manager?

- a. Because it ignores income
- b. Because it cannot be annualised
- c. Because it is too difficult to compute
- d. Because it is strongly influenced by the size and timing of cash flows, over which the manager has no control

Answer and explanation

Answer: D. Because it is strongly influenced by the size and timing of cash flows, over which the manager has no control

Rationale: The DWR is regarded as inappropriate for the purpose of evaluating the performance of a portfolio manager because the DWR is strongly influenced by the size and timing of the cash flows, over which the investment manager has no control.

398. [Calculation] [CM-EIP §8.6.1] A portfolio starts a 30-day month at $10m, receives $100,000 on day 12 and $200,000 on day 18, and ends at $11m. What is the modified Dietz return?

- a. 6.90%
- b. 7.00%
- c. 6.67%
- d. 10.00%

Answer and explanation

Answer: A. 6.90%

Rationale: WCF = 100,000 × (30 − 12)/30 + 200,000 × (30 − 18)/30 = 60,000 + 80,000 = 140,000. Rp = (11,000,000 − 10,000,000 − 300,000) / (10,000,000 + 140,000) = 700,000 / 10,140,000 = 6.90%.

399. [Single choice] [CM-EIP §8.6.1] What is the main disadvantage of the modified Dietz method?

- a. It requires valuation at each cash flow date
- b. It provides a less accurate estimate of the true TWR and can be affected by large cash flows or high volatility
- c. It cannot handle withdrawals
- d. It ignores the ending value

Answer and explanation

Answer: B. It provides a less accurate estimate of the true TWR and can be affected by large cash flows or high volatility

Rationale: The main advantage of the modified Dietz method is that it does not require portfolio valuation on the date of each cash flow. Its main disadvantage is that it provides a less accurate estimate of the true TWR, and can be affected by cash flows, especially if these are large relative to the size of the portfolio or occur during periods of high market volatility.

400. [Single choice] [CM-EIP §8.6.1] When were the revised Global Investment Performance Standards approved?

- a. 1 January 2000
- b. 4 February 2010
- c. 4 February 2005
- d. 1 April 2006

Answer and explanation

Answer: C. 4 February 2005

Rationale: On 4 February 2005, the Chartered Financial Analysts Institute Board of Governors approved the revised GIPS, which created a single global standard of investment performance reporting and increased minimum standards worldwide.

401. [Single choice] [CM-EIP §8.6.1] Under GIPS, how often must portfolios be valued?

- a. Quarterly
- b. Annually
- c. Daily
- d. At least monthly

Answer and explanation

Answer: D. At least monthly

Rationale: GIPS require that portfolios must be valued, marked to market, at least monthly, and returns have to be geometrically linked, that is CAGR must be used for annualized returns.

402. [Calculation] [CM-EIP §8.6.2] Portfolio A returns 10% with a standard deviation of 15%; the risk-free rate is 6%. What is the Sharpe ratio?

- a. 0.27
- b. 0.50
- c. 0.39
- d. 0.67

Answer and explanation

Answer: A. 0.27

Rationale: Sharpe ratio = (Rp − Rf) / p = (0.10 − 0.06) / 0.15 = 0.04/0.15 = 0.27.

403. [Calculation] [CM-EIP §8.6.2] Portfolio B returns 16% with a standard deviation of 20%; the risk-free rate is 6%. What is the Sharpe ratio?

- a. 0.27
- b. 0.50
- c. 0.39
- d. 0.80

Answer and explanation

Answer: B. 0.50

Rationale: Sharpe ratio = (0.16 − 0.06) / 0.20 = 0.10/0.20 = 0.50. Portfolio B has the highest excess return per unit of total risk among the three portfolios compared.

404. [Calculation] [CM-EIP §8.6.2] Portfolio A returns 14% with a beta of 0.9; the risk-free rate is 6%. What is the Treynor ratio?

- a. 8.9%
- b. 10.9%
- c. 14.0%
- d. 15.6%

Answer and explanation

Answer: A. 8.9%

Rationale: Treynor ratio = (Rp − Rf) / p = (0.14 − 0.06) / 0.9 = 0.08/0.9 = 0.089 = 8.9%.

405. [Single choice] [CM-EIP §8.6.2] When do the Sharpe and Treynor measures give identical performance rankings?

- a. When betas are all equal to 1
- b. Never
- c. When portfolios are perfectly diversified, so correlation with the market is 1.0

- d. When the risk-free rate is zero

Answer and explanation

Answer: C. When portfolios are perfectly diversified, so correlation with the market is 1.0

Rationale: If portfolios are perfectly diversified, that is the correlation coefficient between a portfolio's return and the market return is 1.0, the two measures will provide identical portfolio performance rankings, because total risk measured by standard deviation is only systematic risk.

406. [Single choice] [CM-EIP §8.6.2] If a portfolio is inadequately diversified, how do its Treynor and Sharpe rankings compare?

- a. Its Sharpe ranking is likely to be higher than its Treynor ranking
- b. They will be identical
- c. Neither can be computed
- d. Its Treynor ranking is likely to be higher than its Sharpe ranking

Answer and explanation

Answer: D. Its Treynor ranking is likely to be higher than its Sharpe ranking

Rationale: If a portfolio is inadequately diversified, its performance ranking using the Treynor ratio is likely to be higher than its ranking using the Sharpe measure, because the Treynor ratio ignores non-systematic risks, which are present in an inadequately diversified portfolio.

407. [Calculation] [CM-EIP §8.6.2] A portfolio has a beta of 1.2, the risk-free rate is 3% and the market risk premium is 6%. The actual return is 9.5%. What is Jensen's alpha?

- a. -0.7%
- b. $+0.7\%$
- c. -2.9%
- d. $+2.9\%$

Answer and explanation

Answer: A. -0.7%

Rationale: CAPM required return = $3\% + 1.2(6\%) = 10.2\%$. Jensen's alpha = $9.5\% - 10.2\% = -0.7\%$.

408. [Single choice] [CM-EIP §8.6.2] What does the information ratio measure?

- a. Excess return per unit of total risk
- b. Alpha per unit of excess risk
- c. Excess return per unit of downside volatility
- d. Return per unit of beta

Answer and explanation

Answer: B. Alpha per unit of excess risk

Rationale: The information or appraisal ratio looks at the amount of excess return relative to the excess risk. When comparing two portfolios with positive alpha, the appraisal ratio compares the alpha per unit of excess risk. A higher information ratio suggests the manager is generating more outperformance per unit of excess risk taken.

409. [Single choice] [CM-EIP §8.6.2] Why is the Sortino ratio particularly useful for hedge funds?

- a. Because it uses beta
- b. Because hedge funds have no volatility
- c. Because it focuses on downside risk, which suits positively skewed and non-normally distributed returns
- d. Because it ignores the risk-free rate

Answer and explanation

Answer: C. Because it focuses on downside risk, which suits positively skewed and non-normally distributed returns

Rationale: When fund managers protect downside risk with derivatives and enjoy the effects of leveraging to take advantage of positive volatility, they can create positively skewed portfolios. It would be unfair to penalize them for high positive deviation, so the Sortino ratio focuses on downside risk. This is particularly useful in comparing the performance of hedge funds.

410. [Single choice] [CM-EIP §8.6.3] What is a bogey portfolio?

- a. A portfolio with zero risk
- b. The market portfolio
- c. A portfolio that has underperformed
- d. A benchmark designed to measure returns if a passive strategy were adopted

Answer and explanation

Answer: D. A benchmark designed to measure returns if a passive strategy were adopted

Rationale: Part of the analysis process involves identifying a performance benchmark, called the bogey, for comparing the portfolio's results. The bogey is designed to measure returns if a passive strategy were adopted.

411. [Calculation] [CM-EIP §8.6.3] A managed portfolio returns 5.55% and the bogey returns 4.10%. What is the excess return?

- a. 1.45%
- b. 1.10%
- c. 0.35%
- d. 9.65%

Answer and explanation

Answer: A. 1.45%

Rationale: Excess return = 5.55% − 4.10% = 1.45%. This is then decomposed into asset allocation and selection effects.

412. [Calculation] [CM-EIP §8.6.3] The managed portfolio holds 70% equities against a benchmark weight of 60%, and the benchmark equity return is 6%. What is the asset allocation contribution from equities?

- a. 0.10%
- b. 0.60%
- c. 1.05%
- d. 0.35%

Answer and explanation

Answer: B. 0.60%

Rationale: Asset allocation contribution = (Actual weight − Benchmark weight) × Benchmark return = $(0.70 - 0.60) \times 6\% = 0.10 \times 6\% = 0.60\%$.

413. [Calculation] [CM-EIP §8.6.3] The managed portfolio holds 70% equities with an actual return of 7.50% against a benchmark return of 6.00%. What is the selection contribution from equities?

- a. 1.50%
- b. 0.35%
- c. 1.05%
- d. 0.60%

Answer and explanation

Answer: C. 1.05%

Rationale: Selection contribution = Actual weight × (Actual return − Benchmark return) = 0.70 × (7.50% − 6.00%) = 0.70 × 1.50% = 1.05%.

414. [Multiple response] [CM-EIP §8.6.3] Which of the following are characteristics of a good benchmark? (Select all that apply)

- a. Unambiguous, with the asset class and composition clearly stated
- b. Investable as a passively managed portfolio
- c. Measurable, with returns obtainable from the market
- d. Chosen by the client without the manager's agreement

Answer and explanation

Answer: A, B, C

Rationale: Useful benchmarks should be unambiguous, investable, measurable, consistent with the investment manager's investment philosophy, and bought in by the investment manager as an acceptable benchmark.

415. [Single choice] [CM-EIP §8.7] What is a retrocession in the context of fund sales?

- a. A tax rebate
- b. A discount on the offer price
- c. A refund of management fees to investors
- d. A payment to third-party vendors for promoting the fund

Answer and explanation

Answer: D. A payment to third-party vendors for promoting the fund

Rationale: The sales force should ensure there is no misrepresentation about the reliance on third-party vendors, who receive payments such as retrocessions for promoting the fund. Mis-selling can result when there is poor communication and a lack of enforcement of required practices.

Chapter 9 — Exchange Traded Funds (ETFs), Unit Trusts, Real Estate Investment Trusts (REITs) and Insurance

416. [Single choice] [CM-EIP §9.1] What are the two key features of a Variable Capital Company?

- a. It can freely issue and redeem shares and pay dividends from capital, and can adopt an umbrella structure with sub-funds
- b. It must be listed on an exchange and cannot have sub-funds
- c. It is taxed as a partnership and must disclose its shareholder register
- d. It can only be used for closed-end funds and cannot pay dividends

Answer and explanation

Answer: A. It can freely issue and redeem shares and pay dividends from capital, and can adopt an umbrella structure with sub-funds

Rationale: A VCC can freely issue and redeem shares and pay dividends using its capital. A VCC can also adopt an umbrella structure with multiple sub-funds, which increases operating efficiency.

417. [Single choice] [CM-EIP §9.2] How does an ETF differ from a typical mutual fund in terms of trading?

- a. Both trade identically
- b. ETF units can be bought and sold during trading hours like any listed security; mutual funds can only be bought or sold at the end of the day
- c. ETFs can only be bought at the end of the day
- d. Mutual funds trade continuously

Answer and explanation

Answer: B. ETF units can be bought and sold during trading hours like any listed security; mutual funds can only be bought or sold at the end of the day

Rationale: ETF units can be bought and sold by investors during trading hours like any security listed on the Exchange. This is unlike mutual funds, which can only be bought or sold at the end of the day with the mutual fund manager.

418. [Calculation] [CM-EIP §9.2] The SPDR STI ETF is priced at about 1/100th of the STI level. If the STI is at 1,700, what is the price of one unit?

- a. $0.17
- b. $1.70
- c. $17.00
- d. $170.00

Answer and explanation

Answer: C. $17.00

Rationale: Pricing is about 1/100th of the STI level. If the STI is at 1,700, each unit of STI ETF 100 will be priced at $17.00.

419. [Single choice] [CM-EIP §9.2] In what size can units of the STI ETF be created for or redeemed by participating dealers?

- a. 1,000,000 units
- b. 1,000 units
- c. 10,000 units
- d. 100,000 units

Answer and explanation

Answer: D. 100,000 units

Rationale: Unlike a typical unit trust that creates units for investors when they purchase units, units of the STI ETF can only be created for, or redeemed by, participating dealers in sizes of 100,000 units.

420. [Single choice] [CM-EIP §9.2] What happens when an ETF's traded price is above its net asset value?

- a. An arbitrageur sells the ETF shares and buys the underlying portfolio of securities
- b. An arbitrageur buys the ETF shares and sells the underlying portfolio
- c. Nothing happens
- d. The ETF is delisted

Answer and explanation

Answer: A. An arbitrageur sells the ETF shares and buys the underlying portfolio of securities

Rationale: When the traded price of an ETF is higher than its NAV, an arbitrageur will sell the ETF shares and buy the underlying portfolio of securities. If the traded price falls below the NAV, the arbitrageur will sell the underlying portfolio and buy the ETF shares.

421. [Single choice] [CM-EIP §9.2.1] What is representative sampling in a physical ETF?

- a. Holding all index constituents in exact proportions
- b. Investing in a subset of the index comprising the more dominant stocks, used when constituents are not liquid
- c. Using swaps to replicate the index
- d. Holding only cash

Answer and explanation

Answer: B. Investing in a subset of the index comprising the more dominant stocks, used when constituents are not liquid

Rationale: Representative sampling is where the ETF invests in a subset of the index, comprising the more dominant stocks. This method is used when the stocks that make up the index are not liquid. Full replication purchases all of the listed stocks in the exact proportions.

422. [Single choice] [CM-EIP §9.2.1] What is the main advantage and disadvantage of synthetic replication?

- a. Advantage: physical ownership. Disadvantage: illiquidity
- b. Advantage: lower cost. Disadvantage: higher tracking error
- c. Advantage: reduced tracking error. Disadvantage: counterparty risk
- d. Advantage: no counterparty risk. Disadvantage: higher tracking error

Answer and explanation

Answer: C. Advantage: reduced tracking error. Disadvantage: counterparty risk

Rationale: One advantage of synthetic replication is that it reduces the error of the ETF in tracking the index. A disadvantage is counterparty risk, as the counterparty to the derivative contract might default on its obligations.

423. [Single choice] [CM-EIP §9.2.2] What is cash equitisation?

- a. Hedging a portfolio with cash
- b. Investing only in money market funds
- c. Converting equities to cash
- d. Converting cash to equities through investing in ETFs, to minimise the cash drag on a portfolio

Answer and explanation

Answer: D. Converting cash to equities through investing in ETFs, to minimise the cash drag on a portfolio

Rationale: Cash equitisation is converting cash to equities through investing in ETFs, to minimise the cash drag on a portfolio. Cash drag is the underperformance of a portfolio from holding idle cash.

424. [Single choice] [CM-EIP §9.2.3] What collateral might an ETF receive for lending out shares?

- a. Cash or other securities equivalent to about 105% of the stocks' value
- b. A guarantee from the exchange
- c. A government bond at 50% of value
- d. Nothing

Answer and explanation

Answer: A. Cash or other securities equivalent to about 105% of the stocks' value

Rationale: An ETF can transfer the stocks to another investor over the lending period in exchange for collateral such as cash or other securities equivalent to, for example, 105% of the stocks' value. If the borrower cannot fulfil its obligations, the ETF faces a loss equal to the value of the lent stocks.

425. [Single choice] [CM-EIP §9.2.4] How do ETF annual management fees compare with actively managed unit trusts?

- a. ETFs charge more
- b. ETFs normally charge less than 1%, compared with 1–2% for actively managed unit trusts
- c. Both charge the same
- d. ETFs charge no fees at all

Answer and explanation

Answer: B. ETFs normally charge less than 1%, compared with 1–2% for actively managed unit trusts

Rationale: The annual management fees for ETFs are normally less than 1% compared to the 1–2% that is charged by actively managed unit trusts. In addition, there are no initial charges or redemption fees when trading ETFs.

426. [Single choice] [CM-EIP §9.2.5] In which type of ETF is tracking error greater?

- a. Neither has tracking error
- b. Synthetic ETFs
- c. Physical ETFs

- d. Both equally

Answer and explanation

Answer: C. Physical ETFs

Rationale: Tracking error is greater in physical ETFs than in synthetic ETFs, principally because synthetic ETFs can achieve more precise tracking using swaps and derivatives.

427. [Single choice] [CM-EIP §9.2.5] Which tax risk applies to non-US investors holding US-established ETFs?

- a. No tax applies
- b. Capital gains tax
- c. Singapore withholding tax
- d. US withholding taxes on dividends and other distributions

Answer and explanation

Answer: D. US withholding taxes on dividends and other distributions

Rationale: Non-US investors holding US-established ETFs may be subject to US withholding taxes on dividends and other distributions.

428. [Single choice] [CM-EIP §9.2.6] How does a smart beta ETF differ from a standard ETF?

- a. It selects and weights securities on positive performance criteria rather than market capitalisation
- b. It only holds bonds
- c. It has no index
- d. It is actively managed by a stock picker

Answer and explanation

Answer: A. It selects and weights securities on positive performance criteria rather than market capitalisation

Rationale: In contrast to a standard ETF that typically follows an index's market capitalization weighting, a smart beta ETF will select and weight securities based on some positive performance criteria, such as improvements to cashflow, dividends or revenue/earnings ratios.

429. [Single choice] [CM-EIP §9.2.7] Why is holding a long volatility ETF position over time described as potentially deadly?

- a. Because volatility always rises
- b. Because volatility trends downward, and a 1–2% fall each day produces massive losses over time
- c. Because the ETF is delisted
- d. Because the ETF pays no dividends

Answer and explanation

Answer: B. Because volatility trends downward, and a 1–2% fall each day produces massive losses over time

Rationale: Holding on to a long volatility position to wait for the upward spike can be deadly as volatility trends downward, and a 1–2% fall each day will soon lead to massive losses over time. High volatility generally does not last, and the beneficiaries tend to be those that shorted volatility.

430. [Single choice] [CM-EIP §9.2.8] What leverage multiple do leveraged ETFs typically use?

- a. 20x or more
- b. 0.5x or 1x
- c. 2x or 3x
- d. 5x or 10x

Answer and explanation

Answer: C. 2x or 3x

Rationale: Such ETFs are usually leveraged 2x or 3x the index value and there are also those that track the inverse movement of the underlying index. Leveraged ETFs are often issued in pairs, long and short a particular index.

431. [Single choice] [CM-EIP §9.2.8] Why do issuers of leveraged ETFs often carry out reverse splits?

- a. To reduce the leverage
- b. To change the index tracked
- c. To raise new capital
- d. To reset prices at higher levels after sharp movements

Answer and explanation

Answer: D. To reset prices at higher levels after sharp movements

Rationale: Due to the sharp movements of leveraged ETFs, issuers of leveraged ETFs often carry out reverse splits to reset prices at higher levels. Hence regular price resetting takes place for many of the popular ETFs.

432. [Single choice] [CM-EIP §9.3] What is a unit trust in the United States called?

- a. A mutual fund
- b. An investment fund
- c. A unit trust
- d. A Sicav

Answer and explanation

Answer: A. A mutual fund

Rationale: In the US, a CIS is known as a mutual fund and may not be issued in the form of a trust. In the UK the term unit trust is actively used; in Germany it is an investment fund; in Switzerland and Luxembourg the terms are Sicaf and Sicav.

433. [Single choice] [CM-EIP §9.3.1] How many places of domicile can a fund have?

- a. Unlimited
- b. One
- c. Two
- d. As many as it is registered in

Answer and explanation

Answer: B. One

Rationale: The domicile of a fund is where the fund is based and set up. A fund can be registered in multiple jurisdictions, but it can only have one place of domicile.

434. [Single choice] [CM-EIP §9.3.3] How is the price of a closed-end unit trust determined?

- a. By the MAS
- b. By net asset value only
- c. By market demand and supply as well as the performance of the fund
- d. By the trustee

Answer and explanation

Answer: C. By market demand and supply as well as the performance of the fund

Rationale: For an open-end trust the price is determined by net asset value, the performance of the fund net of management fees. For a closed-end trust the price is determined by market demand and supply as well as the performance of the fund.

435. [Calculation] [CM-EIP §9.3.3] A unit trust's investments are worth $10 million and there are 5 million units issued. What is the value of each unit?

- a. $0.50

- b. $2.00
- c. $5.00
- d. $10.00

Answer and explanation

Answer: B. $2.00

Rationale: Each unit represents the same proportion of the value of the assets held by the trust. $10 million / 5 million units = $2 per unit.

436. [Single choice] [CM-EIP §9.3.4] Who holds a unit trust's cash, shares and other investment assets?

- a. The distributor
- b. The custodian bank's shareholders
- c. The fund manager
- d. The trustee

Answer and explanation

Answer: D. The trustee

Rationale: The managers neither hold the investments nor issue or redeem units. This is done by the trustee, who holds the trust's cash, shares and other investment assets.

437. [Single choice] [CM-EIP §9.3.5] What is the typical initial investment outlay for a unit trust quoted in the guide?

- a. $1,000 to $5,000
- b. $10,000 to $50,000
- c. $100,000 or more
- d. $100 to $500

Answer and explanation

Answer: A. $1,000 to $5,000

Rationale: The initial investment outlay, for example $1,000 to $5,000, is affordable to individual investors. For this amount, the average investor can invest in a portfolio of securities that would otherwise be out of his reach.

438. [Single choice] [CM-EIP §9.3.6] Which is a disadvantage of investing in unit trusts?

- a. No diversification
- b. Sales charges on entry and management fees paid regardless of performance
- c. No professional management
- d. No liquidity

Answer and explanation

Answer: B. Sales charges on entry and management fees paid regardless of performance

Rationale: Relatively high sales charges are incurred when an investor buys into a unit trust, and unit trusts pay annual management fees to the fund manager regardless of the fund's performance.

439. [Single choice] [CM-EIP §9.3.7] How do capital guaranteed funds differ from capital preservation funds?

- a. Capital guaranteed funds cannot be sold
- b. They are identical
- c. Capital guaranteed funds have a guarantor and guarantee only if held to maturity; capital preservation funds have no guarantor and use less onerous language
- d. Capital preservation funds have a guarantor

Answer and explanation

Answer: C. Capital guaranteed funds have a guarantor and guarantee only if held to maturity; capital preservation funds have no guarantor and use less onerous language

Rationale: Capital guaranteed funds invest most capital in low-risk assets to preserve capital at maturity, and the guarantee usually applies only if held to maturity. Capital preservation funds aim to provide some level of capital preservation, the language used is less onerous for the issuer, and the fund does not require a guarantor.

440. [Single choice] [CM-EIP §9.3.7] Which fund is described as the best-known example of a quantitative fund?

- a. Fidelity Magellan
- b. Vanguard 500
- c. Berkshire Hathaway
- d. Long-Term Capital Management

Answer and explanation

Answer: D. Long-Term Capital Management

Rationale: Quant funds use price and market modelling based on complex mathematical algorithms. Quite probably the best known example is Long-Term Capital Management, whose board of directors included Myron S. Scholes and Robert C. Merton.

441. [Single choice] [CM-EIP §9.3.7] Why is the name hedge fund described as a misnomer?

- a. Because hedge funds are not necessarily hedged and may take directional bets, being net long or short
- b. Because hedge funds only invest in bonds
- c. Because hedge funds are always market neutral
- d. Because hedge funds never use derivatives

Answer and explanation

Answer: A. Because hedge funds are not necessarily hedged and may take directional bets, being net long or short

Rationale: The name hedge fund can be a misnomer as hedge funds are not necessarily hedged. For example, a hedge fund manager may take directional bets on the markets, that is they can be net long or short.

442. [Single choice] [CM-EIP §9.3.8] Which of the following appears on a monthly fund fact sheet?

- a. The trust deed
- b. Top holdings, country allocation, sector allocation and risk statistics
- c. The auditor's report
- d. The board of directors

Answer and explanation

Answer: B. Top holdings, country allocation, sector allocation and risk statistics

Rationale: Summary information on the monthly fund fact sheets includes fund objective, key information on fees, performance data and ratios, benchmark comparison, top holdings, country allocation, risk statistics and sector allocation.

443. [Single choice] [CM-EIP §9.3.10] What is the typical range of a front-end load?

- a. 10% to 20% of NAV
- b. 0.1% to 0.5% of NAV
- c. 1.5% to 5% of NAV
- d. 5% to 10% of NAV

Answer and explanation

Answer: C. 1.5% to 5% of NAV

Rationale: The load often serves as a commission for the services of the distributors of the unit trust and the front-end load is usually 1.5 to 5% of the NAV.

444. [Single choice] [CM-EIP §9.3.10] What is the typical range of a unit trust annual management fee?

- a. 2.0% to 5.0% of NAV
- b. 5.0% to 10.0% of NAV
- c. 0.1% to 0.15% of NAV
- d. 0.5% to 2.0% of NAV

Answer and explanation

Answer: D. 0.5% to 2.0% of NAV

Rationale: All unit trusts charge annual management fees to compensate their professional managers. This fee typically ranges from 0.5% to 2.0% of the NAV and is accrued daily and debited directly from the unit trust's asset value.

445. [Single choice] [CM-EIP §9.3.10] What is the purpose of a high watermark in a performance fee arrangement?

- a. To ensure the manager generates a certain level of profit before the bonus can be taken, so it does not unfairly benefit from a fund recovering after a fall
- b. To set the minimum investment
- c. To determine the offer price
- d. To cap the manager's total fee

Answer and explanation

Answer: A. To ensure the manager generates a certain level of profit before the bonus can be taken, so it does not unfairly benefit from a fund recovering after a fall

Rationale: There are measures to ensure that a fund manager does not unfairly benefit from a fund that recovers after first falling sharply. A certain performance level, the high watermark, is set to ensure that the fund manager generates a certain level of profits for the fund before the performance bonus can be taken out of the NAV.

446. [Single choice] [CM-EIP §9.3.11] What benchmark might a fund with an absolute return target use?

- a. A broad equity index
- b. An interbank rate such as LIBOR plus 1%
- c. A bond index
- d. No benchmark at all

Answer and explanation

Answer: B. An interbank rate such as LIBOR plus 1%

Rationale: For traditional funds with a relative return target, the benchmark is usually an index. For funds with absolute return targets, an interbank rate, for example LIBOR plus 1%, might be the benchmark target to beat.

447. [Calculation] [CM-EIP §9.3.12] A fund has securities of $100m, other assets of $5m, liabilities including management fees of $3m, no leverage, and 10m units outstanding. What is the NAV per unit?

- a. $9.70
- b. $10.00
- c. $10.20
- d. $10.50

Answer and explanation

Answer: C. $10.20

Rationale: NAV = (Fund securities + Other fund assets − Liabilities − Debt used for leveraging) / Total fund units outstanding = (100 + 5 − 3 − 0) / 10 = 102/10 = $10.20.

448. [Single choice] [CM-EIP §9.3.13] Which expenses are added when calculating the offer price of a unit trust?

- a. Only the trustee fee
- b. Only the audit fee
- c. Expenses related to the sale of securities
- d. Expenses related to the purchase of securities, such as stockbrokers' commission and clearing fees

Answer and explanation

Answer: D. Expenses related to the purchase of securities, such as stockbrokers' commission and clearing fees

Rationale: The offer price is the price at which an investor can buy units. The steps are: determine NAV per unit, add expenses related to purchase of securities such as stockbrokers' commission and clearing fees, add cash held by the trust and accrued income, and add the manager's initial charge.

449. [Single choice] [CM-EIP §9.3.14] What is the typical maximum single-security holding in many funds, and how high may some managers allow it to rise?

- a. 5%, rising to 8–9%
- b. 10%, rising to 20%
- c. 20%, rising to 50%
- d. 1%, rising to 2%

Answer and explanation

Answer: A. 5%, rising to 8–9%

Rationale: Many funds do not allow a single security to constitute more than 5% of total asset holdings, but some managers may allow the fund's exposure to a particular security to rise as high as 8–9% in an attempt to outperform the market.

450. [Single choice] [CM-EIP §9.3.17] Which unit trusts are the exception to the general rule that unit trusts suit medium to long-term investors?

- a. Money market funds
- b. Leveraged ETFs and highly volatile commodity ETFs, which are meant more for traders
- c. Bond funds
- d. Balanced funds

Answer and explanation

Answer: B. Leveraged ETFs and highly volatile commodity ETFs, which are meant more for traders

Rationale: Generally, unit trusts are better suited for investors with medium to long-term investment horizons, the exception being certain leveraged ETFs and highly volatile commodity ETFs, which are meant more for traders.

451. [Calculation] [CM-EIP §9.3.17] A SGD 10,000 legal expense is billed to a fund. What proportion of the fund is it for an SGD 8 million fund and an SGD 1 million fund?

- a. 0.0125% and 0.1%
- b. 0.5% and 5%
- c. 0.125% and 1%
- d. 1% and 0.125%

Answer and explanation

Answer: C. 0.125% and 1%

Rationale: A SGD 10,000 legal expense billed to an SGD 8 million fund is 0.125% of the fund, but billed to an SGD 1 million fund it is 1% of the fund. A small fund size can therefore materially affect performance.

452. [Single choice] [CM-EIP §9.3.18] Why do retrocessions create a conflict of interest?

- a. Because they increase the fund's NAV
- b. Because they are illegal everywhere
- c. Because they are paid by the investor
- d. Because the distributor recommending the fund receives a fee from the fund house, compromising the independence of the advice

Answer and explanation

Answer: D. Because the distributor recommending the fund receives a fee from the fund house, compromising the independence of the advice

Rationale: Retrocessions are fees received by banks and other financial institutions for distributing third-party products to clients. When recommendations of funds are made to new clients, there could be a conflict of interest and the independence of the investment advisors are compromised. Retrocession fees are also a cost offset against fund performance, negatively affecting the NAV.

453. [Single choice] [CM-EIP §9.4.1] What minimum proportion of a REIT must be invested in income-producing real estate?

- a. 45%
- b. 75%
- c. 90%
- d. 100%

Answer and explanation

Answer: B. 75%

Rationale: A REIT is a unit trust that is predominantly invested in real estate and real estate-related assets. Specifically, the Code on CIS requires a minimum of at least 75% of the fund to be invested in income-producing real estate.

454. [Single choice] [CM-EIP §9.4.1] How often must REIT properties be valued under the Code on CIS?

- a. Once a year
- b. Every five years
- c. Daily
- d. Monthly

Answer and explanation

Answer: A. Once a year

Rationale: Unlike equities and bonds in unit trusts where current market values can readily be obtained from the stock exchanges or bond dealers, the Code on CIS only requires the properties on a REIT to be valued once a year.

455. [Single choice] [CM-EIP §9.4.2] What tax advantage do REITs have over direct property investment?

- a. REITs pay no tax at all
- b. REITs distributing at least 90% of taxable income enjoy favourable tax treatment, with no corresponding advantage for direct investment
- c. Direct property is tax-exempt
- d. Both are equally taxed

Answer and explanation

Answer: B. REITs distributing at least 90% of taxable income enjoy favourable tax treatment, with no corresponding advantage for direct investment

Rationale: Since REITs that distribute at least 90% of their taxable income each year enjoy favourable tax treatment, many of them do; there is no corresponding tax advantage for direct investments.

456. [Single choice] [CM-EIP §9.4.3] What is the single-tier leverage limit for Singapore REITs?

- a. 25%
- b. 35%
- c. 45%
- d. No limit

Answer and explanation

Answer: C. 45%

Rationale: In Singapore, REITs are subject to a single-tier leverage limit of 45%. The need to service the debt used to finance the acquisition of underlying properties implies less income for distribution and greater volatility in the equity value of the REIT.

457. [Single choice] [CM-EIP §9.4.3] Why do REITs face refinancing risk?

- a. Because their properties are illiquid
- b. Because they have no tenants
- c. Because they cannot borrow
- d. Because they distribute at least 90% of income, so cannot easily build cash reserves for debt repayment and must refinance or raise new equity

Answer and explanation

Answer: D. Because they distribute at least 90% of income, so cannot easily build cash reserves for debt repayment and must refinance or raise new equity

Rationale: As REITs typically distribute at least 90% of their income annually to unit holders, it is not easy for them to build up cash reserves for debt repayment, so they typically resort to fresh borrowing or raising new equity. If unable to make these arrangements, a REIT may be forced to liquidate some investments.

458. [Single choice] [CM-EIP §9.5.1] Is a business trust a separate legal entity?

- a. No, unlike a company it is not a separate legal entity
- b. Only if listed
- c. Only if it has a trustee
- d. Yes, like a company

Answer and explanation

Answer: A. No, unlike a company it is not a separate legal entity

Rationale: A business trust is a hybrid structure with characteristics of both companies and trusts but, unlike a company, a business trust is not a separate legal entity.

459. [Single choice] [CM-EIP §9.5.1] What is a trustee-manager in a Singapore business trust?

- a. Two separate entities
- b. A single responsible entity vested with the roles of both trustee and manager of the trust assets
- c. A government regulator
- d. The unitholders' committee

Answer and explanation

Answer: B. A single responsible entity vested with the roles of both trustee and manager of the trust assets

Rationale: In Singapore, business trusts are a single responsible entity organization and the trustee-manager is vested with the roles of both trustee and manager of the trust assets. The trustee-manager has legal ownership of the trust's assets.

460. [Single choice] [CM-EIP §9.5.2] Why is a business trust structure attractive to infrastructure providers?

- a. Because they must distribute 90% of income
- b. Because they pay no tax

- c. Because distributions can be paid out of operating cash flow, which matters when depreciating assets reduce accounting profits despite strong cash flows
- d. Because they have a 45% gearing limit

Answer and explanation

Answer: C. Because distributions can be paid out of operating cash flow, which matters when depreciating assets reduce accounting profits despite strong cash flows

Rationale: The main attractive feature of business trusts is their effectiveness in unlocking value from fixed assets that generate cash flow. Such businesses usually have depreciating assets that reduce accounting profits despite strong positive cashflows, which makes business trust structures attractive because dividends can be paid out of operating cash flow.

461. [Single choice] [CM-EIP §9.5.4] How does the gearing limit of a business trust compare with a REIT?

- a. Business trusts are limited to 45%; REITs have no limit
- b. Both have no limit
- c. Both are 45%
- d. Business trusts have no limit; REITs are limited to 45%

Answer and explanation

Answer: D. Business trusts have no limit; REITs are limited to 45%

Rationale: There is no gearing limit for business trusts but a 45% gearing limit for REITs. However, debt could still become a problem if the business trust uses that freedom for indiscriminate borrowing.

462. [Single choice] [CM-EIP §9.5.4] What distribution requirement applies to a listed REIT compared with a business trust?

- a. A REIT must distribute a minimum of 90% of annual income; a business trust has no mandatory minimum
- b. A business trust must distribute 90%; a REIT has no minimum
- c. Neither has a minimum
- d. Both must distribute 90%

Answer and explanation

Answer: A. A REIT must distribute a minimum of 90% of annual income; a business trust has no mandatory minimum

Rationale: A listed REIT has a mandatory distribution of a minimum of 90% of annual income to qualify for favourable tax treatment. A business trust has no mandatory minimum and can distribute from operating cash flow.

463. [Single choice] [CM-EIP §9.6] Why can a stapled security combining a REIT and a business trust not be marketed as a REIT?

- a. Because it is unlisted
- b. Because the issuer is bound by the regulations specific to each security, and such stapled securities cannot be marketed as REITs
- c. Because it has no income
- d. Because it has no trustee

Answer and explanation

Answer: B. Because the issuer is bound by the regulations specific to each security, and such stapled securities cannot be marketed as REITs

Rationale: In a stapled security, the issuer is bound by the regulations specific to each of the securities being stapled. Such stapled securities cannot be marketed as REITs.

464. [Single choice] [CM-EIP §9.7] What does UCITS provide for funds?

- a. A single currency
- b. A tax exemption
- c. A passport for funds to be sold internationally
- d. A guarantee of returns

Answer and explanation

Answer: C. A passport for funds to be sold internationally

Rationale: The UCITS provides the passport for funds to be sold internationally. The original objectives were to remove the barriers to cross-border distribution and marketing of funds by allowing funds to invest in a wider range of financial instruments.

465. [Single choice] [CM-EIP §9.7] Since when has MAS allowed UCITS funds to be distributed and sold in Singapore?

- a. 1998
- b. 2002
- c. 2006
- d. 2013

Answer and explanation

Answer: C. 2006

Rationale: Since 2006, MAS has allowed UCITS funds to be distributed and sold in Singapore if these meet the local regulatory conditions.

466. [Single choice] [CM-EIP §9.8] What is the main benefit of alternative investments?

- a. They are risk-free
- b. They are always liquid
- c. They always outperform equities
- d. They can further diversify the portfolio using asset classes less correlated with traditional equities and fixed income

Answer and explanation

Answer: D. They can further diversify the portfolio using asset classes less correlated with traditional equities and fixed income

Rationale: The main benefit of investing in alternative investments is the ability to further diversify the portfolio using asset classes that are less correlated with the traditional asset classes of equities and fixed income. Through such diversification, the portfolio's risk-adjusted returns may potentially be increased.

467. [Single choice] [CM-EIP §9.8] Do alternative investments provide a buffer during major financial crises?

- a. Not necessarily, because they are also affected by a widespread massive sell-off of assets
- b. Only hedge funds do
- c. Only private equity does
- d. Yes, always

Answer and explanation

Answer: A. Not necessarily, because they are also affected by a widespread massive sell-off of assets

Rationale: During major financial crises, such as the Great Financial Crisis of 2008, investing in alternative investments do not necessarily provide much of a buffer since such investments are also affected by a widespread massive sell-off of assets. However, under normal market conditions, alternative investments have been found to be less correlated with equity and bonds.

468. [Single choice] [CM-EIP §9.8.2] What do hedge funds capture that conventional funds largely cannot?

- a. Beta only
- b. Alpha, with the freedom to hedge away or reduce beta
- c. Only market return
- d. Only dividends

Answer and explanation

Answer: B. Alpha, with the freedom to hedge away or reduce beta

Rationale: Hedge funds capture alpha, with the freedom to hedge away or reduce beta. Conventional funds capture alpha plus beta, with limited ability to reduce beta risk.

469. [Single choice] [CM-EIP §9.8.3] What is basis risk in a hedge fund context?

- a. The risk of currency movements
- b. The risk that the fund closes
- c. The risk that remains as a result of an imperfect hedge because a complete offsetting position may not exist
- d. The risk of rising interest rates

Answer and explanation

Answer: C. The risk that remains as a result of an imperfect hedge because a complete offsetting position may not exist

Rationale: Basis risk is the risk that remains as a result of an imperfect hedge because a complete offsetting position may not exist. Counterparty risk arises when the other party to a derivative transaction does not make good on its contractual obligation.

470. [Single choice] [CM-EIP §9.8.3] What is a gate provision in a hedge fund?

- a. A minimum investment
- b. A lock-up on new investors
- c. An entry fee
- d. A provision allowing the manager to limit withdrawals on a given withdrawal date to a percentage of the fund's net assets

Answer and explanation

Answer: D. A provision allowing the manager to limit withdrawals on a given withdrawal date to a percentage of the fund's net assets

Rationale: Many hedge funds have lock-up periods during which investors are not allowed to redeem their investment, and some have gate provisions, which allow the manager to limit the withdrawals on a given withdrawal date to a percentage of the fund's net assets.

471. [Single choice] [CM-EIP §9.8.4] What is the rationale for convertible arbitrage?

- a. Convertible securities are usually slower to reprice to fair value than the underlying stock, creating arbitrage opportunities
- b. Convertibles pay no interest
- c. Shares are always overvalued
- d. Convertibles always outperform shares

Answer and explanation

Answer: A. Convertible securities are usually slower to reprice to fair value than the underlying stock, creating arbitrage opportunities

Rationale: This strategy involves the purchase of the convertible securities while shorting the underlying stock. The rationale is that, for a variety of reasons, convertible securities are usually slower in repricing to fair value as compared to the underlying stock, which leads to arbitrage opportunities.

472. [Single choice] [CM-EIP §9.8.5] At what price range can distressed bonds trade, depending on the severity of the company's crisis?

- a. 90–99% of par
- b. 10–99% of par
- c. 1–10% of par
- d. Always at par

Answer and explanation

Answer: B. 10–99% of par

Rationale: Depending on the degree of the crisis facing a particular company, the distressed bonds can be priced at 10% to 99% of par. Hedge funds purchase distressed debt and borrowings that they believe to be undervalued.

473. [Single choice] [CM-EIP §9.8.6] What do global macro strategies take?

- a. Only positions in government bonds
- b. Only long positions
- c. Directional views on financial products and derivatives in anticipation of market reactions to economic or political changes
- d. Only market-neutral positions

Answer and explanation

Answer: C. Directional views on financial products and derivatives in anticipation of market reactions to economic or political changes

Rationale: Global macro strategies take directional views on various financial products and their derivatives in anticipation of the market reaction from significant economic or political changes in one or more countries.

474. [Single choice] [CM-EIP §9.8.7] What do directional quantitative trading strategies mainly follow?

- a. Central bank announcements
- b. Dividend announcements
- c. Fundamental valuation
- d. Technical price trends and momentum

Answer and explanation

Answer: D. Technical price trends and momentum

Rationale: Directional quantitative trading strategies employ a combination of technical analysis, market data and funds flow to make investment decisions. These strategies mainly follow technical pricing trends and momentum.

475. [Single choice] [CM-EIP §9.8.8] What happens to returns when a fund of hedge funds diversifies across strategies?

- a. They typically fall to the average return achieved by hedge funds
- b. They become risk-free
- c. They are unaffected
- d. They rise above the best manager

Answer and explanation

Answer: A. They typically fall to the average return achieved by hedge funds

Rationale: With diversification, returns will typically fall to the average return achieved by hedge funds, which may not fit the expectation that some investors have when investing in hedge funds.

476. [Single choice] [CM-EIP §9.8.10] Which private equity stage provides seed and start-up capital in the early stages of a business?

- a. Growth capital

- b. Venture capital
- c. Buyout funds
- d. Distressed businesses

Answer and explanation

Answer: B. Venture capital

Rationale: Venture capitalists co-invest with an entrepreneur to provide seed and/or start-up capital in the early stages of a business. In return for the higher risk, they expect a higher-than-average return on their investment.

477. [Single choice] [CM-EIP §9.8.10] Which private equity stage invests in mature companies with stable cash flows and established products?

- a. Venture capital
- b. Growth capital
- c. Buyout funds
- d. Special situations

Answer and explanation

Answer: C. Buyout funds

Rationale: Buyout funds invest in mature companies with stable cash flows, established products and developed personnel. The buyout could occur pre-IPO for owners who want to exit before the IPO process.

478. [Single choice] [CM-EIP §9.8.11] How long is the investment period of a private equity fund, and when does the harvest period typically begin?

- a. 10 years; harvest from year 11
- b. 2–3 years; harvest from year 4
- c. 1–2 years; harvest from year 3
- d. 5–6 years; harvest from year 6

Answer and explanation

Answer: D. 5–6 years; harvest from year 6

Rationale: The first 5–6 years represent the investment period. The harvest period typically begins from the 6th year onwards when the fund will start making payouts.

479. [Single choice] [CM-EIP §9.8.11] What internal rate of return is typically expected from private equity?

- a. 10–15%
- b. 25–30%
- c. 2–4%
- d. 5–8%

Answer and explanation

Answer: A. 10–15%

Rationale: The incentive for investing in private equity is that the expected internal rate of return would typically be in the double digits, 10–15%. The downside is that the funds would be locked up for several years.

480. [Single choice] [CM-EIP §9.8.12] Why do agriculture commodity funds suffer long downward cycles?

- a. Because demand always falls
- b. Because successes in agricultural production supply often outstrip demand, causing overhanging supply
- c. Because of government regulation
- d. Because of futures market rules

Answer and explanation

Answer: B. Because successes in agricultural production supply often outstrip demand, causing overhanging supply

Rationale: Agriculture commodity funds suffer from the same problem as their underlying commodities: the successes in agriculture production supply often outstrip demand, causing long downward cycles of overhanging supply.

481. [Single choice] [CM-EIP §9.8.12] What is the main concern when investing in real estate funds?

- a. The fund manager's fees
- b. The level of rental income
- c. The mismatch between unit liquidity and the illiquidity of the underlying assets
- d. The tax treatment

Answer and explanation

Answer: C. The mismatch between unit liquidity and the illiquidity of the underlying assets

Rationale: As with REITs, the main concern when investing in real estate funds is the mismatch between unit liquidity and the illiquidity of the underlying assets. In a property bubble situation, asset prices can fall very sharply once the bubble has burst, leading to redemptions.

482. [Single choice] [CM-EIP §9.9.1] When is the sum assured of a whole life insurance policy payable?

- a. Only if the policy is surrendered
- b. After 10 years
- c. Only at a fixed maturity date
- d. On the death of the life insured, whenever this takes place

Answer and explanation

Answer: D. On the death of the life insured, whenever this takes place

Rationale: In the case of a whole life insurance policy, the sum assured will be payable on the death of the life insured, whenever this takes place. Policy owners pay fixed monthly or annual premiums and beneficiaries receive fixed death benefits.

483. [Single choice] [CM-EIP §9.9.2] When is the sum assured of an endowment policy payable?

- a. On the maturity date or on death, whichever is earlier
- b. Only at maturity
- c. Only on surrender
- d. Only on death

Answer and explanation

Answer: A. On the maturity date or on death, whichever is earlier

Rationale: Endowment insurance policies usually mature after a fixed period of time, for example 10, 15 or 20 years, and the sum assured is payable on the maturity date or on the life insured's death, whichever is earlier.

484. [Single choice] [CM-EIP §9.9.2] Why may the guaranteed cash value of an endowment policy be less than the sum of premiums paid?

- a. Because of tax
- b. Because part of the premiums pays for insurance protection while the rest is invested and subject to investment risk
- c. Because of agent commission only
- d. Because the insurer keeps the difference

Answer and explanation

Answer: B. Because part of the premiums pays for insurance protection while the rest is invested and subject to investment risk

Rationale: The guaranteed cash values received may be less than the sum of the premiums paid, because part of the premiums will pay for insurance protection while the rest is invested and subject to investment risks.

485. [Single choice] [CM-EIP §9.9.3] How are benefits linked to investment performance in a with-profits policy?

- a. Only through the stock market
- b. Directly, through unit values
- c. Indirectly, through annual valuation of the life fund and a reversionary bonus decided by the board
- d. Not at all

Answer and explanation

Answer: C. Indirectly, through annual valuation of the life fund and a reversionary bonus decided by the board

Rationale: Under a with-profits policy, the link between the policy benefits and the investment and operating performance of the insurer is not direct and depends on the annual valuation of the fund assets and liabilities. The allocation of the surplus as a reversionary bonus ultimately depends on the decision of the board of directors.

486. [Single choice] [CM-EIP §9.9.3] How do investment-linked policies differ from with-profits policies?

- a. They guarantee a fixed return
- b. They cannot be surrendered
- c. They have no investment element
- d. Policy values are directly linked to the investment performance of underlying units and fluctuate daily

Answer and explanation

Answer: D. Policy values are directly linked to the investment performance of underlying units and fluctuate daily

Rationale: Investment-linked life insurance policies offer policies with values that are directly linked to the investment performance of the underlying instruments, usually by formally linking the values of the policies to units in a fund. The investment element fluctuates daily according to the performance of those investments.

487. [Single choice] [CM-EIP §9.10.1] How is an immediate annuity purchased?

- a. With a lump sum, a single premium
- b. With a combination of premiums and equity
- c. It cannot be purchased directly
- d. With periodic premiums only

Answer and explanation

Answer: A. With a lump sum, a single premium

Rationale: An immediate annuity is a contract where payments to the annuitant begin as soon as it is purchased. An immediate annuity is always purchased with a lump-sum, a single premium.

488. [Single choice] [CM-EIP §9.10.1] On what does the amount received from a deferred annuity depend?

- a. Only on the contributions
- b. On contributions, interest earned, the annuitant's gender and the annuitant's age when payments begin
- c. Only on the annuitant's age

- d. Only on the insurer's profits

Answer and explanation

Answer: B. On contributions, interest earned, the annuitant's gender and the annuitant's age when payments begin

Rationale: The amount that the annuitant will periodically receive depends on his contributions, the interest earned on the contributions, the annuitant's gender, and the annuitant's age when the payments begin.

489. [Calculation] [CM-EIP §9.10.2] What is the present value of $1,000 paid every year for 5 years at a 3% discount rate?

- a. $4,212.36
- b. $5,309.14
- c. $4,579.71
- d. $5,000.00

Answer and explanation

Answer: C. $4,579.71

Rationale: PV = C × [1 − (1 + i)^−n] / i = 1,000 × [1 − (1.03)^−5] / 0.03 = 1,000 × [1 − 0.86261] / 0.03 = 1,000 × 4.57971 = $4,579.71.

490. [Calculation] [CM-EIP §9.10.2] What is the future value of $1,000 paid every year for 8 years at 5%?

- a. $10,000.00
- b. $8,142.00
- c. $8,000.00
- d. $9,549.11

Answer and explanation

Answer: D. $9,549.11

Rationale: FV = C × [(1 + i)^n − 1] / i = 1,000 × [(1.05)^8 − 1] / 0.05 = 1,000 × [1.47746 − 1] / 0.05 = 1,000 × 9.54911 = $9,549.11.

491. [Single choice] [CM-EIP §9.10.3] What risk does an annuity protect against, and how does it compare with life insurance?

- a. Excessive longevity; the opposite of life insurance, which protects against premature death
- b. Inflation; the same as life insurance
- c. Interest rate risk; unrelated to life insurance
- d. Premature death; the same as life insurance

Answer and explanation

Answer: A. Excessive longevity; the opposite of life insurance, which protects against premature death

Rationale: The purpose of an annuity is to protect against loss of income arising from excessive longevity. This is exactly the opposite of the purpose of a life insurance policy, which furnishes protection against loss of income arising out of premature death.

492. [Single choice] [CM-EIP §9.11] What must a fund meet to be included under CPFIS?

- a. Only a minimum size
- b. CPFIS admission criteria, including being in the top quartile of the global peer group and benchmarking total expense ratios to the CPFIS median
- c. Only a MAS licence
- d. Only a credit rating

Answer and explanation

Answer: B. CPFIS admission criteria, including being in the top quartile of the global peer group and benchmarking total expense ratios to the CPFIS median

Rationale: For funds to be included under CPFIS and thus eligible to take in CPF monies, they must meet the CPFIS funds admission criteria, which include being in the top quartile of global peer group and benchmarking total expense ratios to the median of CPFIS funds in the respective risk class.

493. [Calculation] [CM-EIP §9.11.1] An investor invests $60,000 at $5.00 per share, buying 12,000 units. Using dollar cost averaging over 12 months at an average price of $4.622, how many units are bought?

- a. 11,500
- b. 12,000
- c. 12,980
- d. 13,000

Answer and explanation

Answer: C. 12,980

Rationale: With dollar cost averaging, $60,000 / $4.622 = 12,980 units, against 12,000 units if invested at the start. The average cost of purchase is $60,000 / 12,980 = $4.622, lower than the $5.00 paid for a single lump sum.

494. [Single choice] [CM-EIP §9.11.1] What is the practical disadvantage of executing dollar cost averaging accurately?

- a. It incurs higher management fees
- b. It cannot be used with unit trusts
- c. It requires market timing skill
- d. The investor will likely end up with odd lots

Answer and explanation

Answer: D. The investor will likely end up with odd lots

Rationale: Unfortunately, executing dollar cost averaging accurately will cause the investor to likely end up with odd lots.

495. [Single choice] [CM-EIP §9.11.1] If an investor missed the best 25 days of the S&P 500 between 1970 and 2016, what happened to the annualised return?

- a. It fell from 6.7% to 3.0%
- b. It rose from 3.0% to 6.7%
- c. It was unchanged
- d. It fell to zero

Answer and explanation

Answer: A. It fell from 6.7% to 3.0%

Rationale: If he missed investing in the best 25 days of the S&P, his return falls from 1,910% to just 371%, or from 6.7% to 3.0% annually. This is even lower than the return on a 1-month US T-bill rolled over regularly, which would generate an average yield of 4.9%.

496. [Single choice] [CM-EIP §9.11.1] What proportion of active fund managers have been found to underperform the market each year?

- a. 50–60%
- b. 70–80%
- c. 10–20%
- d. 30–40%

Answer and explanation

Answer: B. 70–80%

Rationale: Most working adults do not have the time, expertise or emotional temperament to be disciplined active managers of their own investment portfolio, and 70–80% of active fund managers have been found to underperform the market each year.

497. [Single choice] [CM-EIP §9.13.1] For taxation purposes, how is a Variable Capital Company treated?

- a. As tax-exempt
- b. As a partnership
- c. As a company and a single tax entity
- d. As a trust

Answer and explanation

Answer: C. As a company and a single tax entity

Rationale: For taxation purposes, a VCC is a company and a single tax entity. A VCC also does not need to publicly disclose its shareholder's register, thus offering investor privacy.

498. [Single choice] [CM-EIP §9.13.2] What are the two axes of the CPF Investment Risk Classification System?

- a. Interest rate risk and currency risk
- b. Market risk and operational risk
- c. Credit risk and liquidity risk
- d. Equity risk and focus risk

Answer and explanation

Answer: D. Equity risk and focus risk

Rationale: The Risk Classification System splits the investment risk associated with a unit trust into two axes: equity risk and focus risk. Equity risk relates to the proportion of riskier investments; focus risk reflects how focused the investments are in one region, country, industry or company.

499. [Single choice] [CM-EIP §9.13.2] Which unit trust would have higher focus risk?

- a. A Japan-only equity fund
- b. A regional Asia fund
- c. A balanced global fund
- d. A global equity fund

Answer and explanation

Answer: A. A Japan-only equity fund

Rationale: A unit trust that invests in shares of companies from around the world would have lower focus risk, that is more broadly diversified, than a unit trust that invests in shares of Japanese corporations only. The Japan-only unit trust would have higher focus risk because it is less diversified.

Chapter 10 — Warrants

500. [Single choice] [CM-EIP §10.1.1] What is a warrant?

- a. An obligation to buy securities at a fixed price
- b. A right, but not an obligation, to buy or sell a specified number of securities at a fixed price within a specified period
- c. A share of ownership in a company
- d. A bond convertible into cash

Answer and explanation

Answer: B. A right, but not an obligation, to buy or sell a specified number of securities at a fixed price within a specified period

Rationale: A warrant is a derivative instrument that gives the holder the right, but not the obligation, to buy or sell a specified number of securities, usually the issuer's common share, at a fixed price within a specified time period.

501. [Single choice] [CM-EIP §10.1.1] Why are warrants issued by corporations as a sweetener?

- a. To avoid regulation
- b. To increase the coupon rate
- c. To obtain financing at a lower interest rate, reducing financing cost
- d. To reduce the number of shares

Answer and explanation

Answer: C. To obtain financing at a lower interest rate, reducing financing cost

Rationale: Warrants are generally issued by corporations as a sweetener, attached to an offering of bonds or rights issues. This allows the issuer to obtain financing at a lower interest rate, hence reduce its financing cost.

502. [Single choice] [CM-EIP §10.1.1] How does exercising a warrant differ from exercising an equity call option?

- a. Both cause dilution
- b. Neither causes dilution
- c. Warrants cause no dilution; options do
- d. Warrants cause dilution because new shares are issued; options cause no dilution

Answer and explanation

Answer: D. Warrants cause dilution because new shares are issued; options cause no dilution

Rationale: When the warrants are exercised, new shares will be issued, increasing the number of shares outstanding and resulting in a dilution of the issuer's earnings among its shareholders. This differs from equity call options, which do not cause earnings dilution because no new shares are issued.

503. [Single choice] [CM-EIP §10.1.2] Which of the following is a difference between company warrants and structured warrants?

- a. Company warrants are call only; structured warrants can be calls or puts
- b. Company warrants are cash-settled; structured warrants deliver shares
- c. Company warrants have shorter maturities
- d. Company warrants can be puts; structured warrants are calls only

Answer and explanation

Answer: A. Company warrants are call only; structured warrants can be calls or puts

Rationale: Company-issued warrants are one direction only, that is call warrants. Structured warrants can be both call and put warrants. Company warrants are settled with delivery of shares; structured warrants are cash-settled.

504. [Single choice] [CM-EIP §10.1.2] In the structured warrant code CapitalaMBeCW181009, what does the "e" represent?

- a. The issuer
- b. The exercise style, European
- c. The expiry date
- d. The warrant type

Answer and explanation

Answer: B. The exercise style, European

Rationale: In the code, Capitala is the underlying asset, MB is the issuer Macquarie Bank, e represents a European style warrant, CW represents a Call Warrant, 181009 is the expiry date in YYMMDD format, and COWW is the warrant code.

505. [Single choice] [CM-EIP §10.1.2] What is the typical maturity of a structured warrant compared with a company warrant?

- a. Both 6–12 months

- b. Both measured in years
- c. Structured 6–12 months; company measured in years
- d. Structured measured in years; company 6–12 months

Answer and explanation

Answer: C. Structured 6–12 months; company measured in years

Rationale: Company warrants have a maturity typically measured in years. Structured warrants have a shorter-term maturity of usually 6–12 months.

506. [Single choice] [CM-EIP §10.1.3] How is the exercise price typically set when warrants are issued?

- a. Equal to the prevailing share price
- b. At par value
- c. Below the prevailing share price
- d. Above the prevailing share price

Answer and explanation

Answer: D. Above the prevailing share price

Rationale: Typically, when warrants are issued, the exercise price or strike price is set above the prevailing share price.

507. [Single choice] [CM-EIP §10.1.3] What does the conversion ratio of a warrant state?

- a. The number of warrants required for the conversion to one share
- b. The ratio of premium to intrinsic value
- c. The gearing ratio
- d. The number of shares outstanding

Answer and explanation

Answer: A. The number of warrants required for the conversion to one share

Rationale: The conversion ratio states the number of warrants required for the conversion to one share.

508. [Calculation] [CM-EIP §10.2.1] A warrant has a share price of $0.48, an exercise price of $0.43 and a conversion ratio of 1. What is its intrinsic value?

- a. $0.00
- b. $0.05
- c. $0.43
- d. $0.48

Answer and explanation

Answer: B. $0.05

Rationale: Intrinsic value = (SP − EP) / n = (0.480 − 0.430) / 1 = $0.05. When the share price is below the exercise price the warrant is out-of-the-money and intrinsic value is zero.

509. [Calculation] [CM-EIP §10.2.2] A warrant trades at $0.030 with an exercise price of $0.430 and a conversion ratio of 1. What is the conversion price?

- a. $0.40
- b. $0.43
- c. $0.46
- d. $0.49

Answer and explanation

Answer: C. $0.46

Rationale: Conversion price = n × WP + EP = 1 × 0.030 + 0.430 = $0.46. This is the total price the investor has to pay to convert warrants into one underlying share.

510. [Calculation] [CM-EIP §10.2.3] A warrant trades at $0.030, the exercise price is $0.430 and the share price is $0.26. What is the warrant premium in percentage terms?

- a. 16.54%
- b. 76.92%
- c. 6.25%
- d. 11.54%

Answer and explanation

Answer: B. 76.92%

Rationale: Warrant premium ($) = nWP + EP − SP = 0.030 + 0.430 − 0.26 = $0.20. Premium (%) = 0.20 / 0.26 × 100 = 76.92%.

511. [Calculation] [CM-EIP §10.2.3] A warrant trades at $0.080, the exercise price is $0.430 and the share price is $0.480. What is the warrant premium in percentage terms?

- a. 30.00%
- b. 76.92%
- c. 6.25%
- d. 16.67%

Answer and explanation

Answer: C. 6.25%

Rationale: Warrant premium ($) = 0.080 + 0.430 − 0.480 = $0.030. Premium (%) = 0.030 / 0.480 × 100 = 6.25%. The premium falls sharply as the warrant moves in-the-money.

512. [Single choice] [CM-EIP §10.2.3] In the DF Int'l example, the share price rose 85% while the warrant price rose 167%. What does this illustrate?

- a. That the warrant was overpriced
- b. That the conversion ratio changed
- c. That warrants are less risky than shares
- d. The leverage effect, offering greater percentage gains and losses than the underlying share

Answer and explanation

Answer: D. The leverage effect, offering greater percentage gains and losses than the underlying share

Rationale: Investors are interested in warrants mainly because of the leverage effect. Leverage offers the potential for greater percentage gains and losses than the underlying share for a given change in the price of the underlying share.

513. [Calculation] [CM-EIP §10.2.4] A share price is $0.265 and the warrant price is $0.030 with a conversion ratio of 1. What is the gearing ratio?

- a. 8.8x
- b. 0.11x
- c. 88x
- d. 1.9x

Answer and explanation

Answer: A. 8.8x

Rationale: Gearing ratio = SP / (n × WP) = 0.265 / 0.030 = 8.8x approximately. Gearing gives investors the ability to participate in the performance of the underlying share at a lower price.

514. [Single choice] [CM-EIP §10.2.5] For a warrant with a conversion ratio of 1, what is the maximum value?

- a. The intrinsic value
- b. The underlying share price
- c. The exercise price

- d. The warrant premium

Answer and explanation

Answer: B. The underlying share price

Rationale: For a warrant with a conversion ratio of 1, the maximum value of a warrant is the underlying share price. Most warrant prices do not trade at their maximum value and prices fall as the warrants approach expiration.

515. [Single choice] [CM-EIP §10.2.5] For a warrant with a conversion ratio of 1, what is the minimum value?

- a. The exercise price
- b. Zero always
- c. The intrinsic value, which is SP − EP if SP > EP and zero otherwise
- d. The share price

Answer and explanation

Answer: C. The intrinsic value, which is SP − EP if SP > EP and zero otherwise

Rationale: For a warrant with a conversion ratio of 1, the minimum value of a warrant is the intrinsic value of the warrant: SP − EP if SP > EP, and zero if SP is less than or equal to EP.

516. [Single choice] [CM-EIP §10.2.5] Why does the premium of the warrant price over intrinsic value decrease as the share price rises well above the exercise price?

- a. Because the warrant expires
- b. Because the conversion ratio changes
- c. Because the exercise price falls
- d. Because the leverage potential declines, with a given share price increase constituting an increasingly smaller percentage of the share price

Answer and explanation

Answer: D. Because the leverage potential declines, with a given share price increase constituting an increasingly smaller percentage of the share price

Rationale: As the price of the underlying share increases beyond a point, the premium of the warrant price over the intrinsic value decreases. This is because the leverage potential of the warrant declines at some point, with a given share price increase constituting an increasingly smaller percentage of the share price.

517. [Single choice] [CM-EIP §10.3] How does an increase in the remaining life of a warrant affect its price?

- a. It increases the price
- b. It has no effect
- c. It makes the price zero
- d. It reduces the price

Answer and explanation

Answer: A. It increases the price

Rationale: Other things being equal, the longer the time before the warrants expire, the greater the warrant price. A warrant that is out-of-the-money may become attractive later as the share price appreciates. On the day of expiration, warrants are worth their intrinsic value, after which they are worthless.

518. [Single choice] [CM-EIP §10.3] How does an increase in the exercise price affect a warrant's price?

- a. It increases the price
- b. It reduces the price
- c. It has no effect
- d. It doubles the price

Answer and explanation

Answer: B. It reduces the price

Rationale: The lower the exercise price, the greater the warrant price. The ability to convert warrants into a share by paying a lower exercise price increases the intrinsic value of the warrant and hence the warrant's price.

519. [Single choice] [CM-EIP §10.3] Why does higher volatility of the underlying share increase a warrant's price?

- a. Because volatility extends the maturity
- b. Because volatility increases the exercise price
- c. Because warrants benefit from price upside but limit price downside
- d. Because volatility reduces the conversion ratio

Answer and explanation

Answer: C. Because warrants benefit from price upside but limit price downside

Rationale: The more volatile the price of the underlying share, the higher the warrant price, because warrants benefit from price upside but limit price downside. Volatility can change over time.

520. [Single choice] [CM-EIP §10.3] Why do dividend declarations reduce a call warrant's price?

- a. Because dividends reduce volatility
- b. Because dividends shorten the warrant's life
- c. Because dividends increase the exercise price
- d. Because warrant holders are not entitled to dividends, so call warrants lose out

Answer and explanation

Answer: D. Because warrant holders are not entitled to dividends, so call warrants lose out

Rationale: As warrant holders are not entitled to dividends, the declaration of dividends by the company will reduce the call warrant's price slightly, since call warrants lose out.

521. [Single choice] [CM-EIP §10.3] Why do higher interest rates increase a call warrant's price?

- a. Because of interest savings from the lower capital needed to buy warrants compared with an equivalent long stock position
- b. Because volatility rises
- c. Because dividends fall
- d. Because the exercise price falls

Answer and explanation

Answer: A. Because of interest savings from the lower capital needed to buy warrants compared with an equivalent long stock position

Rationale: The higher the interest rate, the higher the call warrant price due to interest savings from the lower capital needed to buy warrants compared to an equivalent long stock position. For completeness, the higher the interest rate, the lower the put warrant price.

522. [Single choice] [CM-EIP §10.3] Why is there a negative relation between the size of a warrant issue and the warrant price?

- a. Because larger issues are less liquid
- b. Because more warrants outstanding means larger potential earnings dilution if the warrants are deep in-the-money
- c. Because larger issues have shorter maturities
- d. Because larger issues have lower exercise prices

Answer and explanation

Answer: B. Because more warrants outstanding means larger potential earnings dilution if the warrants are deep in-the-money

Rationale: The more warrants a company has outstanding, the larger the potential earnings dilution of the existing shares. If the warrants were deep in-the-money, the conversion of the warrants would increase the number of outstanding shares and reduce the value of each share. Thus there is a negative relation between the size of the warrant issue and the warrant price.

523. [Single choice] [CM-EIP §10.4] How are company warrants settled compared with structured warrants?

- a. Both deliver shares
- b. Both are cash-settled
- c. Company warrants are settled with delivery of shares; structured warrants are cash-settled
- d. Company warrants are cash-settled; structured warrants deliver shares

Answer and explanation

Answer: C. Company warrants are settled with delivery of shares; structured warrants are cash-settled

Rationale: Company warrants are settled with a delivery of shares. Structured warrants are cash-settled, that is no delivery of shares.

Chapter 11 — Technical Analysis and Quantitative Analysis

524. [Single choice] [CM-EIP §11.1.2] What does technical analysis study?

- a. The economy and industry structure
- b. Financial statements
- c. A company's business profitability
- d. Historical market data such as price changes and trading volume, to forecast price fluctuations

Answer and explanation

Answer: D. Historical market data such as price changes and trading volume, to forecast price fluctuations

Rationale: Technical analysis refers to the methodology of forecasting fluctuations in securities prices through an examination of historical market data. It is sometimes called market or internal analysis because it studies demand and supply conditions based on market data such as price changes and trading volume.

525. [Single choice] [CM-EIP §11.1.2] What is the key difference between technical and fundamental analysts regarding irrational factors?

- a. Technical analysts expect irrational factors to persist; fundamental analysts expect only a temporary effect
- b. Fundamental analysts expect irrational factors to persist; technical analysts expect only a temporary effect
- c. Both expect them to persist
- d. Neither considers them

Answer and explanation

Answer: A. Technical analysts expect irrational factors to persist; fundamental analysts expect only a temporary effect

Rationale: A key difference in opinion relates to the influence of irrational factors: technical analysts expect irrational factors to persist while fundamental analysts expect only a temporary effect, with rational beliefs prevailing in the long run.

526. [Single choice] [CM-EIP §11.1.2] Why do technical analysts believe prices exhibit trends that persist over long periods?

- a. Because markets are perfectly efficient

- b. Because new information is priced by the market over a period of time rather than at one point, so prices adjust gradually
- c. Because all investors have the same information
- d. Because prices are random

Answer and explanation

Answer: B. Because new information is priced by the market over a period of time rather than at one point, so prices adjust gradually

Rationale: Technical analysts believe that new information affecting supply and demand is priced by the market over a period of time rather than at one point in time, given that there are different sources of information or because some investors have earlier access. As the speed of adjustment is gradual, prices are expected to exhibit trends that persist over long periods.

527. [Single choice] [CM-EIP §11.1.3] What presents the greatest challenge to technical analysis?

- a. Behavioural finance
- b. Fundamental analysis
- c. The Efficient Market Hypothesis
- d. The random walk theory alone

Answer and explanation

Answer: C. The Efficient Market Hypothesis

Rationale: The Efficient Market Hypothesis presents the greatest challenge to technical analysis. Studies that tested weak-form market efficiency generally support the notion that prices adjust rapidly to new information, leaving little opportunity for technical analysts.

528. [Single choice] [CM-EIP §11.1.3] Why might a popular technical trading rule eventually stop working?

- a. Because regulators ban it
- b. Because charts are unavailable
- c. Because the rule is mathematically wrong
- d. Because many participants act on it at the same time, so some act first and the resulting competition negates the rule's value

Answer and explanation

Answer: D. Because many participants act on it at the same time, so some act first and the resulting competition negates the rule's value

Rationale: If a popular technical analysis trading rule is followed by many market participants, some participants will act first, causing prices to change quicker than expected, so that only those who act earliest will benefit. The popularity and the resulting competition will eventually negate the value of the technical trading rule.

529. [Single choice] [CM-EIP §11.2.1] Why is a moving average more useful than the daily price for detecting trends?

- a. Because it includes past prices, making it more stable and smoothing out erratic changes
- b. Because it predicts the future exactly
- c. Because it ignores volume
- d. Because it is more volatile

Answer and explanation

Answer: A. Because it includes past prices, making it more stable and smoothing out erratic changes

Rationale: The moving average is more stable than the daily price because of the inclusion of past prices, making it a more useful measure for detecting trends. If the current market price starts to diverge directionally from the moving average, this may indicate a trend reversal.

530. [Single choice] [CM-EIP §11.2.2] What does the advance-decline line measure?

- a. The average price of all shares
- b. On a cumulative daily basis, the net difference between the number of shares with price advances and those with price declines
- c. The trading volume of the index
- d. The volatility of the index

Answer and explanation

Answer: B. On a cumulative daily basis, the net difference between the number of shares with price advances and those with price declines

Rationale: The A/D line measures, on a cumulative daily basis, the net difference between the number of shares that have price advances and those that have price declines. The net advance or decline is calculated by subtracting the number of declines from the number of advances, and the line plots a running total over time.

531. [Single choice] [CM-EIP §11.2.2] What does it mean if the index is rising while the advance-decline line is falling?

- a. Volume is increasing
- b. The market is technically strong
- c. Most shares are not participating in the rising market, indicating a market peak
- d. The market is oversold

Answer and explanation

Answer: C. Most shares are not participating in the rising market, indicating a market peak

Rationale: In a rising market, any divergence between the index and A/D line is of concern because it means that most shares are not participating in the rising market, indicating a market peak. Conversely, if the A/D line is rising and the index is declining, it signals a reversal of the falling trend.

532. [Single choice] [CM-EIP §11.2.3] What are the most commonly used MACD parameters?

- a. 20, 50, 200
- b. 14, 30, 7
- c. 5, 10, 20
- d. 12, 26, 9

Answer and explanation

Answer: D. 12, 26, 9

Rationale: The most commonly used values are 12, 26 and 9 days, that is MACD (12,26,9). The MACD series is the difference of EMAs with characteristic times a and b, and the average series is an EMA of the MACD series with characteristic time c.

533. [Single choice] [CM-EIP §11.2.3] What signal is given when the MACD crosses and falls below the signal line?

- a. A bearish signal indicating it may be time to sell
- b. A neutral signal
- c. No signal
- d. A bullish signal

Answer and explanation

Answer: A. A bearish signal indicating it may be time to sell

Rationale: When the MACD crosses and falls below the signal line, it is a bearish signal that indicates that it may be time to sell. Conversely, when the MACD crosses and rises above the signal line, the indicator gives a bullish signal.

534. [Single choice] [CM-EIP §11.2.5] What moving average and multiplier are typically used to construct Bollinger Bands?

- a. A 50-day SMA with a multiplier of 1
- b. A 20-day SMA with a multiplier of 2
- c. A 200-day SMA with a multiplier of 3
- d. A 10-day EMA with a multiplier of 2

Answer and explanation

Answer: B. A 20-day SMA with a multiplier of 2

Rationale: The first step is to compute the simple moving average, typically using a 20-day SMA. Then multiply the standard deviation by two, a commonly used multiplier, and both add and subtract that amount from each point along the SMA.

535. [Single choice] [CM-EIP §11.2.5] What does it mean when prices move close to the upper Bollinger Band?

- a. Volatility has fallen to zero
- b. The market is oversold
- c. The market is more overbought and may be primed for a trend reversal or price correction
- d. The market is fairly valued

Answer and explanation

Answer: C. The market is more overbought and may be primed for a trend reversal or price correction

Rationale: Many traders believe the closer the prices move to the upper band, the more overbought the market, and the closer the prices move to the lower band, the more oversold the market. An overbought or oversold market may be primed for a trend reversal or price correction.

536. [Single choice] [CM-EIP §11.3] What defines an uptrend?

- a. Current highs below previous highs and current lows below previous lows
- b. Current highs above previous highs but current lows below previous lows
- c. A flat price line
- d. Current highs above previous highs and current lows above previous lows

Answer and explanation

Answer: D. Current highs above previous highs and current lows above previous lows

Rationale: An uptrend is in place when current highs are above previous highs and current lows are also above previous lows. A downtrend occurs when current lows are lower than previous lows and current peaks are also lower than previous peaks.

537. [Single choice] [CM-EIP §11.3] What is a support level?

- a. A price range where a substantial increase in demand is expected
- b. The highest price ever reached
- c. The average price over a year
- d. A price range where buyers refuse to pay higher prices

Answer and explanation

Answer: A. A price range where a substantial increase in demand is expected

Rationale: A support level is a price range where a substantial increase in demand is expected. At this level, prices are relatively attractive for buyers who had missed the opportunity to buy earlier.

538. [Single choice] [CM-EIP §11.3] What is the market overhang?

- a. Excess demand at a support level
- b. The supply created by sellers attracted by higher prices to take profits at a resistance level
- c. The volume traded in a day
- d. The gap between bid and offer

Answer and explanation

Answer: B. The supply created by sellers attracted by higher prices to take profits at a resistance level

Rationale: As prices increase, a resistance level develops where buyers refuse to pay higher prices. At the same time, sellers may be attracted by higher prices to take profits on their holdings. This supply is called the market overhang.

539. [Single choice] [CM-EIP §11.3] What is a breakout below a support level?

- a. A continuation of the uptrend
- b. A bullish signal
- c. A bearish signal indicating a reversal of the uptrend
- d. A neutral signal

Answer and explanation

Answer: C. A bearish signal indicating a reversal of the uptrend

Rationale: A breakout above the resistance level is a bullish signal, whereas a breakout below the support level is a bearish signal. A penetration below a support line would be a sell signal as it indicates a reversal of the uptrend.

540. [Single choice] [CM-EIP §11.4] What does a price decline accompanied by heavy volume indicate?

- a. A neutral signal
- b. A buying opportunity
- c. A bullish signal
- d. A very bearish signal reflecting a strong and widespread desire to sell

Answer and explanation

Answer: D. A very bearish signal reflecting a strong and widespread desire to sell

Rationale: A price decline with heavy volume is very bearish because it reflects a strong and widespread desire to sell the security.

541. [Single choice] [CM-EIP §11.4] What might a rally developing on contracting volume signal?

- a. A possible trend reversal
- b. Increased buying interest
- c. Higher volatility
- d. A strong continuation of the rally

Answer and explanation

Answer: A. A possible trend reversal

Rationale: In a sustained bull market, if prices correct downward on declining volume, this may indicate a lull followed by a continuation of the rally. On the other hand, a rally that develops on contracting volume could signal a possible trend reversal.

542. [Single choice] [CM-EIP §11.5.1] What are the three movements in the Dow Theory?

- a. Opening, intraday and closing
- b. Primary, secondary and daily fluctuations
- c. Bull, bear and flat
- d. Short, medium and long term

Answer and explanation

Answer: B. Primary, secondary and daily fluctuations

Rationale: The Dow Theory postulates three movements in the market: the primary movement, the secondary movement, and the daily fluctuations. The primary movement is the market trend, lasting from several months to several years.

543. [Single choice] [CM-EIP §11.5.1] What proportion of the previous advance does the secondary movement typically retrace?

- a. The full amount
- b. One-tenth to one-quarter
- c. One-third to two-thirds
- d. Three-quarters to nine-tenths

Answer and explanation

Answer: C. One-third to two-thirds

Rationale: The secondary movement of the market is shorter in duration than the primary movement and opposite in direction. It usually lasts from several weeks to months, and usually retraces one-third to two-thirds of the previous advance or decline.

544. [Single choice] [CM-EIP §11.5.1] When is a new primary trend confirmed under the Dow Theory?

- a. When volume increases
- b. When the RSI crosses 70
- c. When the DJIA alone signals it
- d. When both the DJIA and DJTA give the same signals

Answer and explanation

Answer: D. When both the DJIA and DJTA give the same signals

Rationale: A new primary trend will only be confirmed when both the DJIA and DJTA give the same signals. When the two averages diverge from each other, the Dow Theory assumes that the prior trend is maintained.

545. [Single choice] [CM-EIP §11.5.1] Which of the following is a weakness of the Dow Theory?

- a. It is slow in identifying a major trend and may be whiplashed
- b. It is too fast to signal
- c. It identifies which shares to buy
- d. It uses too many shares

Answer and explanation

Answer: A. It is slow in identifying a major trend and may be whiplashed

Rationale: The Dow Theory is slow in identifying a major trend and may miss an important price move or be whiplashed, that is be caught by a short-term price reversal even though the long-term trend is intact but not yet visible. It also does not help identify which shares to buy or sell.

546. [Single choice] [CM-EIP §11.5.2] How are trend changes identified using a moving average line?

- a. By a change in the direction of the moving average
- b. By a crossover between the moving average and the index itself
- c. By the slope of the moving average
- d. By the volume traded

Answer and explanation

Answer: B. By a crossover between the moving average and the index itself

Rationale: Trend changes are identified not by a change in the direction of the moving average, but by a crossover between the moving average and the index itself.

547. [Single choice] [CM-EIP §11.5.2] Which moving average is normally used to identify long-term trends?

- a. The 30-day
- b. The 50-day

- c. The 200-day
- d. The 20-day

Answer and explanation

Answer: C. The 200-day

Rationale: Popular moving averages used in technical analysis are the 200-day and 50-day moving averages. The 200-day moving average is normally used to identify long-term trends while a 20- to 30-day moving average is used to identify short- and intermediate-term trends.

548. [Single choice] [CM-EIP §11.5.3] What does an RSI reading above 70 indicate?

- a. A neutral market
- b. A trend reversal has occurred
- c. An oversold condition
- d. An overbought condition

Answer and explanation

Answer: D. An overbought condition

Rationale: The index ranges from zero to 100, and an RSI above 70 is considered overbought. Conversely, an RSI below 30 would signal an oversold condition.

549. [Single choice] [CM-EIP §11.5.3] When is the RSI less useful?

- a. When trends are steady and consistent, such as constant demand for a growth stock keeping its RSI persistently high
- b. In a falling market
- c. When volume is high
- d. During surprises and shocks

Answer and explanation

Answer: A. When trends are steady and consistent, such as constant demand for a growth stock keeping its RSI persistently high

Rationale: The RSI is less useful when trends are steady and consistent. Where constant demand for a growth stock causes its RSI to be persistently high, the RSI level should not be viewed as an indication to sell; rather one should look for an opportunity to accumulate the stock when the RSI is not as high.

550. [Single choice] [CM-EIP §11.6] What is the difference between reversal and continuation patterns?

- a. Reversal patterns indicate a pause; continuation patterns indicate a reversal
- b. Reversal patterns indicate an important reversal in trend; continuation patterns suggest the market is only taking a pause before the existing trend resumes
- c. Both indicate reversals
- d. Both indicate continuations

Answer and explanation

Answer: B. Reversal patterns indicate an important reversal in trend; continuation patterns suggest the market is only taking a pause before the existing trend resumes

Rationale: Reversal patterns indicate that an important reversal in trend is occurring. The continuation patterns suggest that the market is only taking a pause, possibly to correct a short term overbought or oversold condition, after which the existing trend is expected to resume.

551. [Single choice] [CM-EIP §11.6.1] By how much must the price close below the neckline for a head and shoulders breakout to be confirmed?

- a. 10% of the share price
- b. 1% of the share price
- c. 3% of the share price
- d. 5% of the share price

Answer and explanation

Answer: C. 3% of the share price

Rationale: A breakout is confirmed if the price closes below the neckline by 3% of the share price. Once penetrated, the potential magnitude of decline is measured by the distance between the head and the neckline, projected downward from the neckline.

552. [Single choice] [CM-EIP §11.6.1] What does an inverse head and shoulders pattern indicate?

- a. A continuation of a downtrend
- b. No change in trend
- c. A market peak
- d. A market bottom

Answer and explanation

Answer: D. A market bottom

Rationale: While the regular upright head and shoulders formation indicates a market peak, the inverse head and shoulders formation indicates a market bottom.

553. [Single choice] [CM-EIP §11.6.2] What is the main feature of a double top?

- a. The second top forms with distinctly lower volume than the first
- b. Both tops form on equal volume
- c. The pattern has no volume characteristic
- d. The second top forms with distinctly higher volume

Answer and explanation

Answer: A. The second top forms with distinctly lower volume than the first

Rationale: A double top is a reversal pattern indicating a market top. It consists of two tops separated by a valley in prices. The main feature of a double top is that the second top is formed with distinctly lower volume than the first.

554. [Single choice] [CM-EIP §11.6.2] How does volume behave in a double bottom formation?

- a. The first bottom is dull and the second is heavy
- b. The first bottom is accompanied by heavy volume, the second is dull, and volume increases significantly on the breakout
- c. Volume is constant throughout
- d. Volume falls on the breakout

Answer and explanation

Answer: B. The first bottom is accompanied by heavy volume, the second is dull, and volume increases significantly on the breakout

Rationale: The formation of the first bottom is normally accompanied by heavy volume while the second bottom is usually dull with little trading volume. Volume increases significantly on the breakout.

555. [Single choice] [CM-EIP §11.6.3] What is assumed during the formation of a consolidation rectangle?

- a. That the price will break upwards
- b. That the trend will reverse
- c. That the prevailing trend will continue until proven otherwise
- d. That volume will fall to zero

Answer and explanation

Answer: C. That the prevailing trend will continue until proven otherwise

Rationale: During the formation of the rectangle, it is difficult to predict which way the price will ultimately break. Therefore, it is assumed that the prevailing trend will continue until proven otherwise.

556. [Single choice] [CM-EIP §11.6.4] Which triangle is a bullish indicator?

- a. Descending triangle
- b. All triangles
- c. Symmetrical triangle
- d. Ascending triangle

Answer and explanation

Answer: D. Ascending triangle

Rationale: Ascending triangles are bullish indicators, formed by price consolidations bounded by an upward sloping support line and an almost horizontal resistance line. The ascending triangle indicates that buyers are more aggressive than sellers.

557. [Single choice] [CM-EIP §11.6.4] How many points of contact form a triangle?

- a. Four
- b. Five
- c. Two
- d. Three

Answer and explanation

Answer: A. Four

Rationale: Triangles are formed with four points of contact, two support and two resistance points.

558. [Single choice] [CM-EIP §11.6.5] What is the difference between a flag and a pennant?

- a. A flag is formed by converging lines; a pennant by parallel lines
- b. A flag is formed by parallel lines; a pennant by converging lines
- c. A flag is a reversal pattern; a pennant is a continuation pattern
- d. There is no difference

Answer and explanation

Answer: B. A flag is formed by parallel lines; a pennant by converging lines

Rationale: A pennant is formed after a sharp rise or fall in price with lower volume during its formation. The difference between a pennant and a flag is that the pennant is formed by a set of converging, instead of parallel, lines.

559. [Single choice] [CM-EIP §11.6.5] How long can a flag take to form?

- a. One to two years
- b. One day
- c. As few as 5 days to as many as 3–5 weeks
- d. Three to six months

Answer and explanation

Answer: C. As few as 5 days to as many as 3–5 weeks

Rationale: Flags can take as few as 5 days to as many as 3–5 weeks to form. Essentially, they represent a period of controlled profit taking in a rising market.

560. [Single choice] [CM-EIP §11.6.6] What does a rising wedge usually indicate?

- a. A market bottom
- b. A neutral market
- c. A continuation of an uptrend
- d. A bear market rally, with prices usually falling very sharply after completion

Answer and explanation

Answer: D. A bear market rally, with prices usually falling very sharply after completion

Rationale: In a rising wedge, both boundary lines slant upwards. A rising wedge usually develops on a falling price trend. Following its completion, prices usually fall very sharply, especially if volume is high. Rising wedges are characteristic of bear market rallies.

561. [Single choice] [CM-EIP §11.6.6] What does a falling wedge represent?

- a. A temporary interruption of a rising trend
- b. A market peak
- c. A reversal to a downtrend
- d. A bear market rally

Answer and explanation

Answer: A. A temporary interruption of a rising trend

Rationale: A falling wedge is bounded by converging lines that slant downwards. It represents a temporary interruption of a rising trend. When prices move out of a falling wedge, they are likely to drift sideways before they begin to rise.

562. [Single choice] [CM-EIP §11.6.7] Where does a breakaway gap appear?

- a. In the middle of a move
- b. At the beginning of a move, when price breaks out of a price pattern
- c. At the end of a move
- d. Only in weekly charts

Answer and explanation

Answer: B. At the beginning of a move, when price breaks out of a price pattern

Rationale: A breakaway gap appears when a price breaks out of a price pattern. Generally, breakaway gaps develop at the beginning of a move, runaway gaps in the middle of a move, and exhaustion gaps at the end of a move.

563. [Single choice] [CM-EIP §11.6.7] What is an exhaustion gap?

- a. A gap caused by a dividend
- b. The first gap in a series
- c. The last in a series of runaway gaps, representing the end of a rapid rise or fall
- d. A gap that is never filled

Answer and explanation

Answer: C. The last in a series of runaway gaps, representing the end of a rapid rise or fall

Rationale: Exhaustion gaps represent the end of a rapid rise or fall and are the last in a series of runaway gaps. It is not easy to ascertain whether a gap is another runaway gap or the exhaustion gap.

564. [Single choice] [CM-EIP §11.6.7] Should a trading strategy rely on a gap being filled?

- a. Yes, within one week
- b. Only for daily gaps
- c. Yes, gaps are always filled
- d. No, although the general principle is that gaps will eventually be covered, this is not always the case

Answer and explanation

Answer: D. No, although the general principle is that gaps will eventually be covered, this is not always the case

Rationale: Although the general principle is that gaps will eventually be covered, this is not always the case. Accordingly, trading strategies should not be implemented solely on the assumption that a gap will be filled in the immediate future.

565. [Single choice] [CM-EIP §11.7] What pattern does the Elliott Wave Theory describe?

- a. A 5-wave advance and a 3-wave retreat
- b. A single continuous wave
- c. Random price movements
- d. A 3-wave advance and a 5-wave retreat

Answer and explanation

Answer: A. A 5-wave advance and a 3-wave retreat

Rationale: In his study of equity prices in 1938, R.N. Elliott came up with the Elliott Wave Theory, which states that prices move in cycles of waves and sub-sets of waves and exhibit the Golden Ratio, 1.618. Each cycle comprises a 5-wave advance and a 3-wave retreat.

566. [Single choice] [CM-EIP §11.7] What is the Golden Ratio exhibited by Elliott Wave patterns?

- a. 1.000
- b. 1.618
- c. 2.000
- d. 3.142

Answer and explanation

Answer: B. 1.618

Rationale: Elliott Wave charts exhibit patterns related to Fibonacci's series, where each number is obtained by adding the two preceding consecutive numbers. As the series progresses, the ratio of consecutive numbers approaches the Golden Ratio of 1.618.

567. [Single choice] [CM-EIP §11.8] What does quantitative analysis involve?

- a. Reading financial statements
- b. The abstraction of market reality into relations incorporated into a financial model
- c. Chart reading
- d. Interviewing management

Answer and explanation

Answer: B. The abstraction of market reality into relations incorporated into a financial model

Rationale: Quantitative analysis essentially involves the abstraction of market reality, which comprises the economy, financial markets and its environment, into relations that are incorporated into a financial model.

568. [Single choice] [CM-EIP §11.8] What do quants examine beyond the first order of price change?

- a. Only the bid-offer spread
- b. Only the delta
- c. Second and third order derivatives of price changes
- d. Only volume

Answer and explanation

Answer: C. Second and third order derivatives of price changes

Rationale: While less sophisticated traders might just look at price changes, quants will decompose these changes so that specific risk profiles can be created to benefit from these changes. Instead of just looking at the first order of price change, that is delta, quants will look at price changes in their second and third order derivatives.

569. [Single choice] [CM-EIP §11.8] Which hedge fund is cited as a cautionary example of quantitative analysis failing?

- a. Bridgewater Associates
- b. Renaissance Technologies

- c. Two Sigma
- d. Long Term Capital Management

Answer and explanation

Answer: D. Long Term Capital Management

Rationale: Long Term Capital Management was a hedge fund that relied extensively on quantitative analysis. Despite the elegance and seeming mathematical robustness of their quantitative strategies, it still incurred huge losses.

Chapter 12 — Case Studies

570. [Single choice] [CM-EIP §12.1] An entity has S$5 million worth of net assets. Is it an accredited investor?

- a. No, because an entity needs net assets exceeding S$10 million
- b. Yes, because any entity with over S$1 million qualifies
- c. No, because only individuals can be accredited investors
- d. Yes, because net assets exceed S$2 million

Answer and explanation

Answer: A. No, because an entity needs net assets exceeding S$10 million

Rationale: A corporation or other entity qualifies as an accredited investor with net assets exceeding $S10 million, as determined by its most recent audited balance sheet. S$5 million is below that threshold.

571. [Single choice] [CM-EIP §12.1] Which return objective is likely to be unique to an individual investor rather than an institutional investor?

- a. Potential for higher yields
- b. Regular savings to fund children's education
- c. Current income to meet recurring liquidity needs
- d. Long-term capital gain to beat inflation

Answer and explanation

Answer: B. Regular savings to fund children's education

Rationale: Current income, long-term capital gain and higher yields can be return objectives of both individual and institutional investors. Regular savings to fund children's education is likely to be unique to an individual investor.

572. [Single choice] [CM-EIP §12.1] Which statement about the difference between accredited and retail investors is INCORRECT?

- a. A retail investor is likely to be more risk averse than an accredited investor
- b. An accredited investor can withstand a higher level of monetary risk
- c. A retail investor has a longer investment time horizon than an accredited investor
- d. An accredited investor can access a wider range of investment products

Answer and explanation

Answer: C. A retail investor has a longer investment time horizon than an accredited investor

Rationale: Investment time horizon is dependent on the needs of the investor, not the type of investor. The other three statements are correct.

573. [Calculation] [CM-EIP §12.2] Returns are 12% with probability 0.3, 8% with probability 0.5 and −6% with probability 0.2. What is the expected return?

- a. 5.30%
- b. 6.40%
- c. 8.45%
- d. 9.28%

Answer and explanation

Answer: B. 6.40%

Rationale: E(R) = 0.3(0.12) + 0.5(0.08) + 0.2(−0.06) = 0.036 + 0.040 − 0.012 = 0.0640 = 6.40%.

574. [Calculation] [CM-EIP §12.2] Using the same returns with an expected return of 6.40%, what is the coefficient of variation?

- a. 0.47
- b. 0.68
- c. 1.01
- d. 1.33

Answer and explanation

Answer: C. 1.01

Rationale: Variance = $0.3(0.12 - 0.064)^2 + 0.5(0.08 - 0.064)^2 + 0.2(-0.06 - 0.064)^2 = 0.004144$. Standard deviation = $\sqrt{0.004144} = 6.4374\%$. CV = 6.4374% / 6.40% = 1.01.

575. [Calculation] [CM-EIP §12.2] An investor sets aside $200,000 for 30 years at an expected return of 6.4% with 2% inflation. What is the future value in today's dollars?

- a. $709,264.72
- b. $1,610,176.77
- c. $2,279,904.66
- d. $2,865,831.06

Answer and explanation

Answer: A. $709,264.72

Rationale: Real rate = (1.064/1.02) − 1 = 4.31%. Future value in real terms = 200,000 × (1.0431)^30 = $709,264.72.

576. [Single choice] [CM-EIP §12.3] Which of the following suggests a potential economic downturn?

- a. Average duration of unemployment is increasing
- b. The 10-year T-bond spread over the Fed Funds rate is increasing
- c. Disposable personal incomes start to rise
- d. New orders by suppliers start to decline

Answer and explanation

Answer: D. New orders by suppliers start to decline

Rationale: Suppliers normally cut orders when they expect demand to drop. Average duration of unemployment is a lagging indicator, an increasing bond spread bodes well for economic health, and rising disposable personal income is a coincident indicator moving in the wrong direction for a downturn.

577. [Single choice] [CM-EIP §12.3] A slowdown in the Chinese economy occurs with rising interest rates. Which asset classes are likely to do well?

- a. Defensive stocks and investment grade bonds
- b. Cyclical stocks and high-yield bonds
- c. Growth stocks and precious metals
- d. Cash and real estate

Answer and explanation

Answer: A. Defensive stocks and investment grade bonds

Rationale: A slowdown coupled with rising interest rates is characteristic of a late bull to early bear phase. Defensive stocks and investment grade bonds are likely to do well. Real estate is typically leveraged and risky when rates rise, cyclical stocks perform worse in a downturn, and high-yield issuers may default.

578. [Single choice] [CM-EIP §12.3] Uninformed fund managers reduced positions out of fear because news spread that many were doing likewise, and the selling triggered computer-driven strategies. What is the root cause of the price fall?

- a. Algorithmic trading
- b. Herd instinct
- c. Loss aversion
- d. Cognitive bias

Answer and explanation

Answer: B. Herd instinct

Rationale: The managers simply followed the crowd without exercising independent decision-making ability and doing the due diligence, which is herd instinct. Loss aversion and algorithmic trading may have exacerbated the fall but did not provide the original impetus.

579. [Calculation] [CM-EIP §12.4] An investor buys IDR 100 billion of bonds at SGD/IDR 10,076.68, earning 6% while paying 1.5% on an SGD loan. If the exchange rate stays flat, what is the annual gain?

- a. SGD 9,923,903.51
- b. SGD 148,858.55
- c. SGD 446,575.66
- d. SGD 595,434.21

Answer and explanation

Answer: C. SGD 446,575.66

Rationale: The SGD-equivalent loan is 100bn / 10,076.68 = SGD 9,923,903.51. The annual gain is $(0.06 - 0.015) \times 9{,}923{,}903.51$ = SGD 446,575.66.

580. [Calculation] [CM-EIP §12.4] Using the same carry trade, if the IDR weakens 10% against the SGD over the year, what is the total gain?

- a. −SGD 942,771.98
- b. SGD 605,357.18
- c. SGD 942,771.98
- d. −SGD 605,357.18

Answer and explanation

Answer: D. −SGD 605,357.18

Rationale: The new rate is 10,076.68/0.9 = 11,196.31. Bond proceeds = 100bn/11,196.31 = SGD 8,931,514.04. Interest income = 6bn/11,196.31 = SGD 535,890.84. Interest expense = $0.015 \times 9{,}923{,}903.51$ = SGD 148,858.55. Total = (8,931,514.04 − 9,923,903.51) + (535,890.84 − 148,858.55) = −SGD 605,357.18.

581. [Calculation] [CM-EIP §12.4] If SGD/USD is 0.7 and USD/IDR is 13,086.60, how does the carry trade gain compare with the flat-rate scenario?

- a. Higher gain
- b. Insufficient information
- c. Lower gain
- d. Same gain

Answer and explanation

Answer: A. Higher gain

Rationale: The SGD/IDR cross rate is $0.7 \times 13{,}086.60 = 9{,}160.62$, which is lower than 10,076.68, so the IDR has appreciated against the SGD. The IDR-denominated investment and income appreciate in value, so the gain is higher than SGD 446,575.66.

582. [Calculation] [CM-EIP §12.5] A stock trades at USD 40 with next-year EPS of USD 1.08. What is the prospective P/E, and why is it high?

- a. 37.04, because the market has overvalued the stock
- b. 37.04, because the market is factoring in growth potential
- c. 35.71, because the market has overvalued the stock
- d. 35.71, because the market is factoring in growth potential

Answer and explanation

Answer: B. 37.04, because the market is factoring in growth potential

Rationale: Prospective P/E = 40 / 1.08 = 37.04x. Since P/E ratios can be high because of strong expected future growth, the high prospective P/E is likely because investors are factoring in the high double-digit long-term growth potential.

583. [Calculation] [CM-EIP §12.5] EPS is 1.12 in FY18A and 5.81 in FY23F. What is the 5-year compounded annualised EPS growth rate?

- a. 31.57%
- b. 38.99%
- c. 40.01%
- d. 52.30%

Answer and explanation

Answer: B. 38.99%

Rationale: 5-year EPS growth = $[(5.81/1.12)\hat{}(1/5) - 1] \times 100\% = [(5.1875)\hat{}0.2 - 1] \times 100\% = 38.99\%$.

584. [Calculation] [CM-EIP §12.5] A stock has a prospective P/E of 37.04 and a 5-year EPS growth rate of 38.99%. What is the PEG ratio and the recommendation?

- a. 0.92, buy
- b. 0.92, sell
- c. 0.95, buy
- d. 0.95, sell

Answer and explanation

Answer: C. 0.95, buy

Rationale: PEG = 37.04 / 38.99 = 0.95. As the PEG ratio is less than 1, the stock is slightly undervalued even after accounting for its strong expected future growth, so the investor can consider buying.

585. [Single choice] [CM-EIP §12.6] Free cash flows to a levered firm should be discounted using:

- a. The cost of ordinary equity only
- b. The cost of preferred equity only
- c. The cost of debt only
- d. The weighted average cost of capital

Answer and explanation

Answer: D. The weighted average cost of capital

Rationale: The appropriate discount rate should take into account capital from all sources for which a return is required. The cost of ordinary equity alone would be correct only if the firm were all-equity funded, which it is not because it is levered.

586. [Single choice] [CM-EIP §12.6] Which of the following is NOT part of estimating free cash flows to the firm?

- a. Analysing historical ratios to make interest expense projections
- b. Interviewing company management on revenue growth potential
- c. Consulting industry specialists on expected profitability margins

- d. Estimating future fixed asset investments

Answer and explanation

Answer: A. Analysing historical ratios to make interest expense projections

Rationale: Revenue growth and profitability margins are needed to derive EBIT, and capital expenditure is another key input. Interest expense is excluded from the FCFF computation.

587. [Multiple response] [CM-EIP §12.6] Enterprise value can be calculated using which of the following? (Select all that apply)

- a. The sum of the present values of the FCFFs for the projection period plus the terminal value
- b. The sum of the present values of the FCFFs for the projection period plus the present value of the terminal value
- c. The sum of the FCFFs for the projection period plus the present value of the terminal value
- d. The sum of the estimated market values of debt, common stock and preferred stock

Answer and explanation

Answer: B, D

Rationale: Enterprise value can be calculated either as the worth today of the cash flows the firm generates, or as the value today of the claims on those cash flows. Options (a) and (c) fail to discount the terminal value and the FCFFs respectively.

588. [Single choice] [CM-EIP §12.7] A 10-year 4% bond has 6 years left and a YTM of 6.0%. Why is it priced below face value?

- a. Because it is callable
- b. Because its YTM is higher than its coupon rate
- c. Because its YTM is lower than its coupon rate
- d. Because it is unrated

Answer and explanation

Answer: B. Because its YTM is higher than its coupon rate

Rationale: A bond has to be priced at a discount in order for the YTM to be higher than the coupon rate. Here $6.0\% > 4\%$, so the bond trades at a discount.

589. [Single choice] [CM-EIP §12.7] An investor insists on holding a bond to maturity. What should he be most concerned with?

- a. That interest rates are decreasing
- b. The coupon rate of the bond
- c. That the bond is callable
- d. That interest rates are increasing

Answer and explanation

Answer: C. That the bond is callable

Rationale: A callable bond can be redeemed before maturity, and if that happens the investor is exposed to reinvestment risk from having to reinvest the proceeds at lower yields. Holding to maturity means interest rate changes are of little concern because the price converges to par.

590. [Single choice] [CM-EIP §12.7] Interest rates are expected to rise. Which bond price is likely to decrease the most in percentage terms?

- a. The 10-year 6% callable bond with 5 years left
- b. The 8% bond with 4 years left
- c. All will decrease equally
- d. The 10-year 4% bond with 6 years left

Answer and explanation

Answer: D. The 10-year 4% bond with 6 years left

Rationale: The bond with the longest maturity and lowest coupon is most sensitive to interest rate changes. The 4% bond with 6 years left has both the longest remaining maturity and the lowest coupon, so its price is most sensitive.

591. [Multiple response] [CM-EIP §12.8] An investor does not believe superior returns can be achieved through fundamental analysis. Which forms of market efficiency must he necessarily believe in? (Select all that apply)

- a. Weak form
- b. Semi-weak form
- c. Semi-strong form
- d. Strong form

Answer and explanation

Answer: A, C

Rationale: Conducting fundamental analysis to identify a mispriced stock assumes all publicly available information is not already reflected in the price, so he must believe the market is semi-strong form efficient. Because publicly available information includes historical price and volume information, he must also believe it is weak form efficient. Semi-weak form is not a standard classification.

592. [Calculation] [CM-EIP §12.8] A share has a beta of 1.2 and a past return of 9.5%. The risk-free rate is 3% and the market risk premium is 6%. What is Jensen's alpha?

- a. -2.9%
- b. -0.7%
- c. 0.7%
- d. 2.9%

Answer and explanation

Answer: B. -0.7%

Rationale: CAPM required return $= 0.03 + 1.2(0.06) = 10.2\%$. Jensen's alpha $= 9.5\% - 10.2\% = -0.7\%$.

593. [Single choice] [CM-EIP §12.8] A retiree needs $100,000 from a $500,000 actively managed portfolio in three months, is willing to risk only $100,000 in full, and needs income for 20 years. Which allocation best fits?

- a. 20% money market, 10% T-bills, 50% long-term government bonds, 20% developed equities
- b. 20% money market, 10% T-bills, 10% long-term government bonds, 10% high yield, 50% developed equities
- c. 10% money market, 80% long-term government bonds, 10% developed equities
- d. 20% money market, 10% T-bills, 10% long-term government bonds, 40% high yield, 20% developed equities

Answer and explanation

Answer: A. 20% money market, 10% T-bills, 50% long-term government bonds, 20% developed equities

Rationale: He needs 20% of the portfolio for the gift shortly, so 20% in money market funds is appropriate. His risk budget is $100,000 of $500,000, so a 60% allocation to high-yield bonds and equities is too risky. He needs long-term government bond coupons to fund annual expenses, so a 50% allocation to long-term government bonds is appropriate.

594. [Calculation] [CM-EIP §12.9] A warrant has a share price of $2.50, an exercise price of $1.50 and a conversion ratio of 1. What is its intrinsic value?

- a. $0
- b. $1
- c. $1.50
- d. $2.50

Answer and explanation

Answer: B. $1

Rationale: Since the warrant is in-the-money, intrinsic value = $(2.50 - 1.50)/1 = \$1$.

595. [Calculation] [CM-EIP §12.9] A warrant trades at $1.30 with an exercise price of $1.50 and a conversion ratio of 1. If the share price rises to $4.00, what is the minimum percentage gain on the warrant?

- a. Impossible to determine
- b. 92.31%
- c. 207.69%
- d. 307.69%

Answer and explanation

Answer: B. 92.31%

Rationale: At a share price of $4.00 the intrinsic value is $(4.00 - 1.50)/1 = \$2.50$, which is the minimum value of the warrant. Minimum percentage gain = $2.50/1.30 - 1 = 92.31\%$. Only the future minimum price is needed, not the future warrant price.

596. [Single choice] [CM-EIP §12.9] An investor wants to sell USD 50,000 for SGD. Quote A is USD/SGD 1.2500–1.2550 and Quote B is USD/SGD 1.2475–1.2515. Which rate is best?

- a. 1.2475
- b. 1.2515
- c. 1.2500
- d. 1.2550

Answer and explanation

Answer: C. 1.2500

Rationale: For each quote the first rate is the price at which the market maker will buy USD and the second is the price at which it will sell USD. Since the investor wants to sell USD, the first rate of each quote applies, and the highest of those is best: 1.2500.

597. [Multiple response] [CM-EIP §12.10] Which features are common to unit trusts, business trusts and REITs? (Select all that apply)

- a. They are established under a trust deed
- b. They are exchange-traded
- c. They allow for diversification
- d. They are professionally managed

Answer and explanation

Answer: A, C, D

Rationale: All three are established under a trust deed, allow diversification by pooling and investing funds, and have professionally managed assets. Option (b) is incorrect because while all REITs are listed, unit trusts are typically not listed and some business trusts are unlisted.

598. [Multiple response] [CM-EIP §12.10] What are the main differences between business trusts and REITs? (Select all that apply)

- a. The trustee and manager are separate in a REIT but not in a business trust
- b. Business trusts do not face investment restrictions, unlike REITs
- c. Business trusts are unlisted whereas REITs are listed
- d. There is a 75% gearing limit on REITs but only a 45% limit on business trusts

Answer and explanation

Answer: A, B

Rationale: A business trust is a single responsible entity organisation with a trustee-manager, unlike a REIT where trustee and manager are separate. Business trusts can actively engage in any business area

whereas REITs must invest only in real estate for rental income. Most business trusts are listed, and there is no gearing limit for business trusts but a 45% limit for REITs.

599. [Single choice] [CM-EIP §12.10] An investor wants a steady stream of income rather than trading gains. Which scheme best suits him?

- a. A closed-end fund investing for capital appreciation
- b. A unit trust described as an aggressive fund
- c. A business trust comprising golf course assets
- d. A REIT focused on heartland retail malls

Answer and explanation

Answer: D. A REIT focused on heartland retail malls

Rationale: An aggressive unit trust generates returns from capital gains and is unlikely to provide steady income. Golf course cash flows are highly sensitive to economic fluctuations and business trusts have no mandatory minimum distribution. A REIT typically distributes at least 90% of annual income, which is likely to be more stable for heartland retail malls.

Appendix A — Review Questions

600. [Single choice] [CM-EIP Appendix A] An investment manager discovers the discount rate used in an NPV calculation was underestimated. What happens when the error is corrected?

- a. NPV falls and IRR is unchanged
- b. Both NPV and IRR fall
- c. NPV rises and IRR falls
- d. Both NPV and IRR rise

Answer and explanation

Answer: A. NPV falls and IRR is unchanged

Rationale: NPV discounts the cash flows at the given rate, so raising the rate lowers NPV. IRR is the rate at which NPV equals zero and is a property of the cash flows alone, so it does not change.

601. [Single choice] [CM-EIP Appendix A] A portfolio has an expected return of 6% and a standard deviation of 12%, with normally distributed returns. The probability of a loss exceeding 6% is 16%. How often should such a loss be expected?

- a. Once every 5–6 years
- b. Once every 6–7 years
- c. Once every 7–8 years
- d. Once every 8–9 years

Answer and explanation

Answer: B. Once every 6–7 years

Rationale: A 16% probability is about 1 in 6.25, so the loss should be expected roughly once every 6–7 years.

602. [Multiple response] [CM-EIP Appendix A] Which statements about value at risk are TRUE? (Select all that apply)

- a. It is not applicable to bonds, shares and currencies
- b. It measures and quantifies the minimum possible loss over a specific time within a confidence level
- c. It can be calculated by Monte Carlo simulation or the historical method
- d. It can be used when determinants of future returns differ from those of past returns

Answer and explanation

Answer: B, C

Rationale: VaR quantifies the loss at a stated confidence level and can be computed historically or by Monte Carlo simulation. It applies to bonds, shares and currencies, and its historical form assumes past determinants persist.

603. [Multiple response] [CM-EIP Appendix A] Which of the following are forms of operational risk? (Select all that apply)

- a. A flash crash caused by a massive erroneous sell order
- b. Poor system infrastructure leading to technical disruptions
- c. Human errors resulting in erroneous trades
- d. Inability to access funding due to a credit downgrade

Answer and explanation

Answer: A, B, C

Rationale: Execution errors, system failures and human error are operational risks. Inability to access funding due to a credit downgrade is funding liquidity risk, not operational risk.

604. [Single choice] [CM-EIP Appendix A] Which of the following is a leading indicator?

- a. The average prime rate charged by banks
- b. Average duration of unemployment
- c. The 10-year T-bond spread over the Fed Funds rate
- d. Nonfarm payroll employment

Answer and explanation

Answer: C. The 10-year T-bond spread over the Fed Funds rate

Rationale: The 10-year spread is one of the ten Composite Leading Economic Index components. Average duration of unemployment and the average prime rate are lagging indicators; nonfarm payrolls are coincident.

605. [Single choice] [CM-EIP Appendix A] Spot USD/SGD is 1.2520 and the 30-day forward points are −10. What is the 30-day forward rate?

- a. 1.2530
- b. 1.2520
- c. 1.2500
- d. 1.2510

Answer and explanation

Answer: D. 1.2510

Rationale: Negative forward points are subtracted: $1.2520 - 0.0010 = 1.2510$, a forward discount.

606. [Single choice] [CM-EIP Appendix A] A company has current assets of $500,000 and current liabilities of $250,000. What is its current ratio?

- a. 2.0
- b. 1.5
- c. 2.5
- d. 0.5

Answer and explanation

Answer: A. 2.0

Rationale: Current ratio = Current assets / Current liabilities = 500,000 / 250,000 = 2.0.

607. [Single choice] [CM-EIP Appendix A] Which index weighting scheme automatically adjusts for bonus and rights issues?

- a. Geometric average
- b. Value-weighted
- c. Price-weighted

- d. Equal-weighted

Answer and explanation

Answer: B. Value-weighted

Rationale: Capital changes typically do not affect market capitalisation, so a value-weighted index adjusts automatically. A price-weighted index requires a divisor adjustment.

608. [Single choice] [CM-EIP Appendix A] Which of the following is NOT covered by the Singapore Deposit Insurance Scheme?

- a. A Singapore dollar savings account
- b. A Singapore dollar fixed deposit
- c. A foreign currency deposit
- d. Monies under the CPF Investment Scheme

Answer and explanation

Answer: C. A foreign currency deposit

Rationale: The scheme covers Singapore dollar-denominated deposits only. Foreign currency deposits, structured deposits and investment products are excluded.

609. [Single choice] [CM-EIP Appendix A] Which factor reduces a company warrant's price?

- a. An increase in the remaining life of the warrant
- b. An increase in the underlying share price
- c. An increase in volatility
- d. An increase in dividend expectations

Answer and explanation

Answer: D. An increase in dividend expectations

Rationale: Warrant holders do not receive dividends, so higher dividend expectations reduce the call warrant's price. Rising share price, rising volatility and longer remaining life all increase the price.

Appendix B — Formulae Sheet

610. [Single choice] [CM-EIP Appendix B] Which formula gives the coefficient of variation?

- a. Standard deviation of returns / Expected rate of return
- b. Expected rate of return / Standard deviation of returns
- c. Variance / Expected rate of return
- d. Standard deviation × Expected rate of return

Answer and explanation

Answer: A. Standard deviation of returns / Expected rate of return

Rationale: The coefficient of variation measures risk per unit of expected return: CV = Standard Deviation of Returns / Expected Rate of Return.

611. [Single choice] [CM-EIP Appendix B] Which formula gives the taxable equivalent yield?

- a. Tax-exempt yield × (1 − marginal tax rate)
- b. Tax-exempt yield / (1 − marginal tax rate)
- c. Tax-exempt yield / marginal tax rate
- d. Tax-exempt yield + marginal tax rate

Answer and explanation

Answer: B. Tax-exempt yield / (1 − marginal tax rate)

Rationale: Taxable Equivalent Yield = Tax Exempt Yield / (1 − Marginal Tax Rate). An investor in the 28% bracket with a 7% tax-exempt yield has a TEY of 0.07/0.72 = 9.7%.

612. [Single choice] [CM-EIP Appendix B] Which formula gives the approximate yield-to-maturity?

- a. Annual interest payment / current market price
- b. Annual interest payment / par value
- c. (Annual interest payment ± amortised gain or loss) / ((Current market price + Par value)/2)
- d. (Annual interest payment ± amortised gain or loss) / Current market price

Answer and explanation

Answer: C. (Annual interest payment ± amortised gain or loss) / ((Current market price + Par value)/2)

Rationale: The approximation relates the net annual effective cash flow to the average amount invested in the bond during the holding period, so the denominator is the average of current market price and par value.

613. [Single choice] [CM-EIP Appendix B] Which formula gives the Sortino ratio?

- a. p / p
- b. (Rp − Rf) / p
- c. (Rp − Rf) / p
- d. (Rp − Rf) / negative

Answer and explanation

Answer: D. (Rp − Rf) / negative

Rationale: The Sortino ratio is (Rp − Rf) / negative, where negative is the average downside volatility. The Sharpe ratio uses total standard deviation and the Treynor ratio uses beta.

614. [Single choice] [CM-EIP Appendix B] Which formula gives the gearing ratio of a warrant?

- a. SP / (nWP)
- b. (SP − EP) / n
- c. nWP + EP
- d. nWP / SP

Answer and explanation

Answer: A. SP / (nWP)

Rationale: Gearing ratio = SP / (nWP), where SP is the share price, WP the warrant price and n the conversion ratio. It compares the price of warrants needed to convert to one share relative to the share price.

Appendix C — Essential Readings

615. [Single choice] [CM-EIP Appendix C] How many items are on the essential readings list?

- a. Five
- b. Seven
- c. Ten
- d. Twelve

Answer and explanation

Answer: B. Seven

Rationale: The source list runs to seven items across five publishers, covering general investments texts, technical analysis and portfolio management.

616. [Single choice] [CM-EIP Appendix C] Which listed text is the only one covering technical analysis?

- a. Essentials of Investments
- b. Analysis of Investments and Management of Portfolios
- c. Technical Analysis Plain and Simple
- d. Managing Investment Portfolios: A Dynamic Process

Answer and explanation

Answer: C. Technical Analysis Plain and Simple

Rationale: Technical Analysis Plain and Simple by Michael N. Kahn is the only technical analysis text on the list and covers Chapter 11.

Appendix D — Glossary

617. [Single choice] [CM-EIP Appendix D] What is the definition of a basis point?

- a. 1% or 0.01
- b. 0.001% or 0.00001
- c. 0.1% or 0.001
- d. 0.01% or 0.0001

Answer and explanation

Answer: D. 0.01% or 0.0001

Rationale: A basis point is a measure equal to 0.01% or 0.0001. It is the standard unit for quoting credit spreads and interest rate changes.

618. [Single choice] [CM-EIP Appendix D] What is the definition of a blue chip company?

- a. A company known for quality, reliability and the ability to weather downturns and operate profitably in adverse conditions
- b. A company listed on a secondary board
- c. A company with a share price below $5
- d. A company with high growth and no dividends

Answer and explanation

Answer: A. A company known for quality, reliability and the ability to weather downturns and operate profitably in adverse conditions

Rationale: Blue chip companies are known for quality, reliability and the ability to weather downturns and operate profitably in the face of adverse economic conditions, which helps to contribute to their long record of stable and reliable growth.

619. [Single choice] [CM-EIP Appendix D] What is the definition of a call market?

- a. A market where trading is continuous
- b. A market where trades can be arranged only when the market is called, usually once a day
- c. A market where prices are negotiated by telephone
- d. A market for call options only

Answer and explanation

Answer: B. A market where trades can be arranged only when the market is called, usually once a day

Rationale: A call market is a market in which trades can be arranged only when the market is called, which is usually once a day. Price, but not timing, prioritisation is practised.

Exam method and final revision checklist

620. [Calculation] [CM-EIP Exam method] The CM-EIP paper has 90 questions in 150 minutes. How many seconds per question does that allow, and how many correct answers are needed to pass at 70%?

- a. 100 seconds; 68 correct
- b. 120 seconds; 70 correct
- c. 100 seconds; 63 correct
- d. 90 seconds; 63 correct

Answer and explanation

Answer: C. 100 seconds; 63 correct

Rationale: 150 minutes is 9,000 seconds, and 9,000 / 90 = 100 seconds per question. The pass mark is 70% of 90 = 63 correct answers, leaving room for 27 errors and no more.

621. [Single choice] [CM-EIP Exam method] Why should a candidate never leave a blank answer in the CM-EIP paper?

- a. Because the system rejects incomplete papers
- b. Because blanks are marked as wrong and also reduce the pass mark
- c. Because blanks are penalised twice
- d. Because there is no penalty for a wrong answer, so a guess can only help

Answer and explanation

Answer: D. Because there is no penalty for a wrong answer, so a guess can only help

Rationale: There is no penalty for a wrong answer, so never leave a blank. If time is short, fill the remaining answers with the best judgement and return only if time allows.

622. [Single choice] [CM-EIP Exam method] Which day-count convention applies to Singapore Government Securities in the CM-EIP examination?

- a. 365 days
- b. Actual/actual
- c. 30/360
- d. 360 days

Answer and explanation

Answer: A. 365 days

Rationale: Use 365 days for Singapore Government Securities and 360 days for everything else. Carrying the 360-day convention into an SGS question is one of the most common calculation errors.

623. [Single choice] [CM-EIP Exam method] A candidate is told the formulae sheet is provided in the examination. What skill should the final revision therefore target?

- a. Memorising every formula verbatim
- b. Retrieval — being able to locate any formula within seconds
- c. Deriving each formula from first principles
- d. Ignoring the formulae sheet entirely

Answer and explanation

Answer: B. Retrieval — being able to locate any formula within seconds

Rationale: Because the formulae sheet is provided, the skill is retrieval rather than memorisation. Rehearse Appendix B until any formula can be located within seconds.

624. [Single choice] [CM-EIP Exam method] A candidate carries over the 75% pass mark from FMRP or RES 12B. What is the error?

- a. The CM-EIP pass mark is 80%
- b. None — all three papers share a 75% pass mark
- c. The CM-EIP pass mark is 70%, so the target is 63 of 90, not 68
- d. The CM-EIP pass mark is 65%

Answer and explanation

Answer: C. The CM-EIP pass mark is 70%, so the target is 63 of 90, not 68

Rationale: The CM-EIP pass mark is 70%, not 75%. At 90 questions that is 63 correct answers, not 68. Carrying over the FMRP or RES 12B figure is a listed trap.

625. [Single choice] [CM-EIP Exam method] How are the case studies weighted in the CM-EIP paper?

- a. There are 6 case studies carrying 18 questions

- b. There are no case studies
- c. They form a separate section worth a fifth of the marks
- d. There are 3 case studies carrying 9 questions between them, within the 90-question paper

Answer and explanation

Answer: D. There are 3 case studies carrying 9 questions between them, within the 90-question paper

Rationale: The paper includes 3 case studies of 9 questions in total, not a separate case-study section worth a fifth of the marks. The 6 case studies carrying 18 questions is the CM-SIP structure.

626. [Single choice] [CM-EIP Exam method] A calculation question is consuming time and the candidate has two minutes invested. What is the recommended action?

- a. Mark it, move on, and return on a second pass
- b. Guess immediately and move on
- c. Skip the rest of the section
- d. Continue until it is solved, since the marks are equal

Answer and explanation

Answer: A. Mark it, move on, and return on a second pass

Rationale: Leave the hardest calculation for a second pass. Mark it, move on, and return. Two minutes lost on one question costs two easier questions, and every question carries the same mark.

627. [Single choice] [CM-EIP Exam method] What should a candidate note first when reading a case study?

- a. The names of the parties
- b. The client's objective, horizon, risk tolerance and any constraint
- c. The currency of the transaction
- d. The date of the scenario

Answer and explanation

Answer: B. The client's objective, horizon, risk tolerance and any constraint

Rationale: Read the scenario once and note the client's objective, horizon, risk tolerance and any constraint — the questions usually turn on one of those.

628. [Single choice] [CM-EIP Exam method] How should a multiple-response question be approached?

- a. Select all options if any one is clearly correct
- b. Look for a pattern in the letter combinations
- c. Evaluate each option independently on its own merits
- d. Select only the option that appears most technical

Answer and explanation

Answer: C. Evaluate each option independently on its own merits

Rationale: Evaluate each option independently. Do not look for a pattern in the letters; a statement is true or false on its own merits.

629. [Single choice] [CM-EIP Exam method] Which pair of distinctions is described as deciding the most marks?

- a. Primary versus secondary market, and dealer versus broker
- b. Duration versus convexity, and CML versus SML
- c. REIT versus business trust, and company versus structured warrant
- d. All of the ten listed distinctions are high value

Answer and explanation

Answer: D. All of the ten listed distinctions are high value

Rationale: The guide lists ten distinctions that decide the most marks, including primary versus secondary market, dealer versus broker, duration versus convexity, CML versus SML, DWR versus TWR, REIT versus business trust and company versus structured warrant. All are worth learning as pairs.